

CHAPTER ONE

INTRODUCTION TO OPERATION RESEARCH

1.1. History and Definitions of Operations Research

- OR came in to existence as a discipline during World War II when there was a critical need to manage scarce resources.
- The term “Operations research” was coined as a result of research on military operations during this war.
- Since the war involved strategic and tactical problems which were greatly complicated, to expect adequate solutions from individual or specialists in a single discipline was unrealistic.
- Therefore, group of individuals who collectively were considered specialists in Mathematics, Economics, Statistics and probability theory, engineering, behavioral, and physical science were formed as special unit within the armed forces to deal with strategic and tactical problems of various military operations.
- The objective was the most effective/efficient utilization of most limited military resources by the use of quantitative techniques.
- After the war ended, scientists who had been active in the military OR groups made efforts to apply the operations research approach to civilian problems, related to business, industry, research and development, and even won Nobel prizes when they returned to their peacetime disciplines.

1.2 Defining Operations Research:

The British/Europeans refer to "operational research", the Americans to "Operations research" - but both are often shortened to just "OR" (which is the term we will use). Another term which is used for this field is "Management Science" ("MS"). The Americans sometimes combine the terms OR and MS together and say "OR/MS" or "ORMS". Yet other terms sometimes used are "Industrial Engineering" ("IE"), "Decision Science" ("DS") and “problem solving”. In recent years there has been a move towards a standardization upon a single term for the field, namely the term "OR".

- OR is the application of the methods of science to complex problems in the direction and management of large systems of men, machines, materials and money in industry, business, government and defense.
- OR is a systematic application of quantitative methods, techniques and tools to the analysis of problems involving the operation of systems.
- OR is essentially a collection of mathematical techniques and tools which in conjunction with systems approach, is applied to solve practical decision problems of an economic or engineering nature.
- OR seeks the determination of the optimum course of action of a decision problem under the restriction of limited resources. It is quite often associated almost exclusively with the use of mathematical techniques to model and analyze decision problems.
- OR is the application of a scientific approach to solving management problems in order to help managers make better decisions. As implied by this and other definitions, operations research encompasses a number of mathematically oriented techniques that have either been developed within the field of management science or been adapted from other disciplines, such as natural sciences, mathematics, statistics, and engineering.

1.3 Features of Operations Research Approach

The important features or characteristics of OR approach to any decision and control problems can be summarized as:

- i. Inter-disciplinary approach***
- ii. Methodological Approach***
- iii. Holistic Approach or Systems Orientation***
- iv. Objectivistic Approach***
- v. Decision Making***
- vi. Use of Computers***
- vii. Human factors***

In deriving quantitative solution, we do not consider human factors which doubtlessly play a great role in the problems. So study of the OR is incomplete without a study of human factors.

1.4. Models and Modeling in Operations Research

What do we mean by the term model?

- ✓ An OR model is an abstract representation of an existing problem or situation.
- ✓ It can be in the form of a graph or chart, but mostly, an OR model consists of a set of mathematical relationship. In OR terminology, these are called objective function and constraints.
- ✓ Models attempt to describe the essence of a situation or activity by abstracting from reality.
- ✓ So that the decision- maker can study the relationship among relevant variables in isolation.
- ✓ Models do not, and cannot, represent every aspect of reality because of the innumerable and changing characteristics of the real life problems to be represented.
- ✓ Instead, they are limited approximation of reality. For example, to study the flow of materials through a factory, a scaled diagram on paper showing the factory floor, position of equipment, tools, and workers can be constructed.
- ✓ It would not be necessary to give such details as the color of machines, the height of the workers, or the temperature of the building.

For a model to be effective, it must be representative of those aspects of reality that are being investigated and have a major impact on the decision situation.

A model is constructed to analyze and understand the given system for the purpose of improving its performance. The reliability of the solution obtained from a model depends on the validity of the model in representing the system under study. A model allows the opportunity to examine the behavioral changes of a system without disturbing the on-going operations.

Note: The key to model building lies in abstracting only the relevant variables that affect the criteria of the measures of performance of the given system and expressing the relationship in a suitable form.

I. Classification of OR Models

There are many ways to classify models:

1. Classification based on structure

i. Physical Models

- These models provide a physical appearance of the real object under study either reduced in size or scaled up.
- Physical models are useful only in design problems because they are easy to observe, build, and describe.

Since these models cannot be manipulated and are not very useful for prediction, problems such as portfolio analysis selection, media selection, production scheduling, etc cannot be analyzed by physical models.

1. Iconic models

They are least abstract, physical model and look like the reality. Example toy

2. Analog Models

They are physical models like iconic model. However they are more abstract than the iconic ones. They substitute analogy for important aspects of reality. Examples include charts, clock, thermometer, Barometer, blue prints, graphs and the like.

ii. Symbolic models

These models use symbols (letters, numbers) and functions to represent variables and their relationships to describe the properties of the system.

2. Classification based on function or purpose

Models based on the purpose of their utility include:

I. Descriptive models

Descriptive models simply describe some aspects of a situation, based on observation, survey, questionnaire results or other available data of a situation and do not recommend anything. Example: Organizational chart, plant layout diagram, etc.

II. Predictive Models

- These models indicate “If this occurs, then that follow”. They relate dependent and independent variables and permit trying out, “what if” questions.
- In other words, these models are used to predict outcomes due to a given set of alternatives for the problem. These models do not have an objective function as a part of the model to evaluate decision alternatives.

For example, $S = a + bA + cI$ is a model that describes how the sales (S) of a product changes in advertising expenditures (A) and disposal personal income (I). Here, a, b, and c are parameters whose values must be estimated.

III. Normative (Optimization) models

These models provide the “best” or “Optimal” solution to problems subject to certain limitations on the use of resources. These models provide recommended courses of action.

For example, in mathematical programming, models are formulated for optimizing the given objective function, subject to restrictions on resources in the context of the problem under consideration and non-negativity of variables. These models are also called *prescriptive models*, because they prescribe what the decision maker ought to do.

3. Classification Based on Time Reference

1. Static Models

Static models represent a system at some specified time and do not account for changes over time.

For example, an inventory model can be developed and solved to determine an economic order quantity for the next period assuming that the demand in planning period would remain the same as that for today.

2. Dynamic models

In dynamic models, time is considered as one of the variables and allows the impact of changes due to change in time.

Thus, sequences of interrelated decisions over a period of time are made to select the optimal course of action to optimize the given objective. Dynamic programming is an example of a dynamic model.

4. Classification based on Degree of certainty

1) Deterministic Models

If all the parameters, constants and functional relationships are assumed to be known with certainty when the decision is made, the model is said to be *deterministic*.

Thus, in such a case, the outcome associated with a particular course of action is known. That is, for a specific set of input values, there is a uniquely, determined output which represents the solution of the model under consideration of certainty.

The results of the models assume single value. Linear programming models are examples of deterministic models.

2) Probabilistic (Stochastic) models

- Models in which at least one parameter or decision variable is a random variable are called probabilistic (or stochastic).
- Since at least one decision variable is random, a dependent variable which is the function of independent variable(s) will also be random.
- This means consequences or pay off due to certain changes in the independent variable cannot be produced with certainty.
- It is possible to predict a pattern of values of both the variables by their probability distribution. Insurance against risk of fire, accidents, sickness, etc are examples where the pattern of events is studied in the form of a probability distribution.

5. Classification Based on Method of solution or Quantification

A. Heuristic Model

These models employ some sets of rules which, though perhaps not optimal, do facilitate solutions of problems when applied in a consistent manner.

B. Analytical Models

These models have a specific mathematical structure and thus can be solved by known analytical or mathematical techniques. Any optimization model (which requires maximization or minimization of an objective function) is an analytical model.

C. Simulation Models

These models have a mathematical structure but are not solved by applying mathematical techniques to get a solution. Instead, a simulation model is essentially a computer-assisted experimentation on a mathematical structure of a real-life problem in order to describe and evaluate its behavior under certain assumptions over a period of time.

Advantage of Models

Models in general are used as an aid for analyzing complex problems. However, a model can also serve other purposes as:

i) A model provides economy in representation of the realities of the system.

That is, models help decision makers to visualize a system so that he/she can understand the system's structure or operation in a better way. For example, it is easier to represent a factory layout on paper than to construct it. It is cheaper to try out modifications of such systems by rearrangement on paper.

ii) The problem can be viewed in its entirety, with all the components being considered simultaneously.

iii) Models serve as aids to transmit ideas and visualization among people in the organization.

For example, process chart can help the management to communicate about better work methods to workers.

iv) A model allows us to analyze and experiment in a complex situation to a degree that would be impossible in the actual system and its environment.

For example, the experimental firing of satellite may be costly and require years of preparation.

v) Models simplify the investigation considerably and provide a powerful and flexible for predicting the future state of the process or system.

1.5. Methodology of Operations Research

For effective use of OR techniques, it is essential to follow some steps that are helpful for decision-makers to make better solution. The flow diagram representing the methodology of OR is shown as:

Step 1 Observation and defining a problem

The first step in OR process is the identification of a problem that exists in a system (organization).

Step 2- Formulating a model

Model formulation involves an analysis of the system under study, determining objective of the decision-maker, and alternative course of action, etc, so as to understand and describe, in precise terms, the problem that an organization faces.

The major steps which have to be taken in to consideration for formulating the model are:

- ☞ *Problem Components.*
- ☞ Decision environment
- ☞ Alternative courses of Action
- ☞ Measure of effectiveness
- ☞ Collecting Data and Constructing a Mathematical Model

Certain basic components required in every decision problem model are:

- ☞ Controllable (decision) Variables
- ☞ Uncontrollable variable.
- ☞ Objective function.

A stated objective helps to focus attention on what the problem actually is.

- ☞ Constraints or Limitations
- ☞ Functional relationships
- ☞ Parameters- These are constants in the functional relationships. Parameters can be deterministic or probabilistic in nature.

Step 3.Solving the Mathematical Model

This involves obtaining the numerical values of decision variables. Obtaining these values depends on the specific form or type, of mathematical models.

Step 4. Validating (Testing) the solution

After solving the mathematical model, it is important to review the solution carefully to see that values make sense and that the resulting decisions can be implemented. Some of the reasons for validating the solution are:

- (i) The mathematical model may not have enumerated all the limitations of the problem under consideration.
- (ii) Certain aspect of the problem may have been overlooked, omitted or simplified,
- (iii) The data may have been incorrectly estimated or recorded, perhaps when entered in to the computer.

Step 5 Implementing the solution

The decision-maker has not only to identify good decision alternatives but also to select alternatives that are capable of being implemented.

Step 6 Modifying the Model

For a mathematical model to be useful, the degree to which it actually represents the system or problem being modeled must be established.

Step 7 Establishing control over the solution

The dynamic environment and changes within the environment can have significant implications regarding the continuing validity of models and their solutions. Thus, a control procedure has to be established for detecting significant changes in decision variables of the problem so that suitable adjustments can be made in the solution without having to build a model every time a significant change occurs.

1.6. Basic Operations Research Models

There is no unique set of problems which can be solved by using OR models or techniques. Some OR models or techniques include:

1. Allocation Models

Allocation models are used to allocate resources to activate in such a way that some measure of effectiveness (objective function) is optimized. Mathematical programming is the broad term of OR techniques used to solve allocation problems.

- a) If the measure of effectiveness such as profit, cost, etc., is represented as a linear function of several variables and if limitations of resources (constraints) can be expressed as a system of linear inequalities or equalities, the allocation problem is classified as linear programming problems.
- b) If the objective function of any or all constraints cannot be expressed as a system of linear equalities or inequalities, the allocation problem is classified as a non-linear programming problem.
- c) When the solution values or decision variables for the problem are restricted to being integer values, the problem is classified as an *integer programming*.
- d) The problem having multiple, conflicting and incommensurable objective function (goals) subject to linear constraints is called *goal programming*.
- e) If decision variables in the linear programming problem depend on chance, such a problem is called is called a *stochastic programming problem*.
- f) If resources such as workers, machines or salesmen can be assigned to perform a certain number of activities such as jobs or territories on one- to one basis so as to minimize total time, cost or distance involved in performing a given activity, such problems are classified as *assignment problems*.
- g) Conversely, if the resources can be used for more than one activity, the allocation problem is classified as a *transportation problem*.

2. Inventory Model

Inventory Models deal with the problem of determination of how much to order at a point in time and when to place an order. The main objective is to minimize the sum of three conflicting inventory costs: the Cost of holding or carrying extra inventory, the cost of shortage or delay in delivery of items when it is needed, a cost of ordering or set-up.

3. Competitive (Game Theory) Model

These models are used to characterize the behavior of two or more opponents (called players) who compete for the achievement of conflicting goals.

4. Network Models

These models are applied to the management (planning, controlling and scheduling) of large-scale projects. PERT/CPM Techniques help in identifying potential trouble spots in a project through the identification of the critical path. These techniques improve project coordination and

enable the efficient use of resources. Network methods are also used to determine time-cost trade off, resource allocation and updating of activity time.

5. Decision analysis Model

These models deal with the selection of an optimal course of action given the possible pay offs and their associated probabilities of occurrence. These models are broadly applied to problems involving decision making under risk and uncertainty.

Features of OR solution

A solution that works but is quite expensive compared to the potential savings from its application should not be considered successful. Also a solution that is well within the budget but which does not accomplish the objective is not successful either. The following are features of good solution:

- *Technically appropriate.* The solution should work technically; meet the constraints and operate in the problem environment.
- *Reliable.* The solution must be useful for a reasonable period of time under the conditions for which it was designed.
- *Economically viable.* The economic value should be more than what it costs to develop and should be seen as a wise investment in hiring OR talents.
- *Behaviourally appropriate.* The solution should be behaviourally appropriate and must remain valid for a reasonable period of time within the organization.

Difficulties in problem solving

Sometimes difficulties in problem solving arise due to lack of an appropriate methodology for it and psychological perceptions on the part of the problem solver. The major difficulties in problem solving:

(i) Failure to recognize the existence of a problem	(iii) Failure to use all available information
- Some people tend to personalize problems	- The problem-solver fails to seek out information

- Problems arise in context which people have had no experience.	
ii) Failure to define the correct problem - One situation may contain many intertwined problems. - Obvious problems are often symptoms of much deeper problems. - The inability to identify accurately what is going on can lead to inaccurate problem identification. - Attitudes and beliefs can blind the problem solver to the real causes on undesirable situations.	iv) Failure to recognize or question assumptions. - it is assumed that there is a solution to every problem - Rigid thinking blocks ones view point.

CHAPTER TWO

2. Linear Programming: Application and Model Formulation

2.1. INTRODUCTION

In 1947, George Danzig developed the use of algebra for determining solutions to problems that involved the optimal allocation of scarce resources. In spite of numerous potential applications in business, response to this new technique was low due to substantial computational burden, which is now removed with subsequent advances in computer technology and related software during the last three decades.

The term linear implies that all the mathematical relations used in the problem are linear or straight-line relations, while the term programming refers to the method of determining a particular program or plan of action, i.e., the use of algorithms that is a well defined sequence of steps that will lead to an optimal solution. Taken as a whole, the term linear programming refers to a family of mathematical techniques for determining the optimum allocation of resources and obtaining a particular objective when there are alternative uses of the limited or constrained resources.

The technique of linear programming is applicable to problems in which the total effectiveness can be expressed as linear function of individual allocations and the limitations on resources give rise to linear equation or inequalities of the individual allocations.

The usefulness of this technique is enhanced by the availability of several user-friendly soft wares such as STORM, TORA, QSB+, LINDO, etc. However, there is no general package for building an LP model. Model building is an art of practice.

2.1. LINEAR PROGRAMMING MODELS

Linear programming models are mathematical representations of LP problems. Linear programming models have certain characteristics in common. Knowledge of these characteristics enables us to recognize problems that are amenable to a solution using LP models, and to be able to correctly formulate an LP model. These characteristics can be grouped as *components and assumptions*. The components relate to the structure of a model, where as the assumptions reveal the conditions under which the model is valid.

2.1.1. COMPONENTS OF LP MODELS

There are four major components of LP models including: Objective function, decision variables, constraints and parameters.

➤ *Objective and Objective Function*

The **objective** in problem solving is the criterion by which all decisions are evaluated. It provides the focus for problem solving. In linear programming models, a single, quantifiable objective must be specified by the decision maker. Because we are dealing with optimization, the objective will be either maximization or minimization. Hence, every LP problem will be either maximization or a minimization

problem. Once the objective is specified, it becomes the measure of effectiveness against which alternate solutions are judged. An LP model consists of a mathematical statement of the objective called the **objective function**.

➤ *Decision variables*

They represent unknown quantities to be solved for. The decision maker can control the value of the objective, which is achieved through choices in the levels of decision variables. For example, how much of each product should be produced in order to obtain the greatest profit?

➤ *Constraints*

However, the ability of a decision maker to select values of the decision variables in an LP problem is subject to certain restrictions or limits coming from a variety of sources. The restrictions may reflect availabilities of resources (e.g., raw materials, labor time, etc.), legal or contractual requirements (e.g., product standards, work standards, etc.), technological requirements (e.g., necessary compressive strength or tensile strength) or they may reflect other limits based on forecasts, customer orders, company policies, and so on. In LP model, the restrictions are referred to as constraints. Only solutions that satisfy all constraints in a model are acceptable and are referred to as feasible solutions. The optimal solution will be the one that provides the best value for the objective function.

Generally speaking, a constraint has four elements:

- A right hand side (RHS) quantity that specifies the limit for that constraint. It must be a constant, not a variable.
- An algebraic sign that indicates whether the limit is an upper bound that cannot be exceeded, a lower bound that is the lowest acceptable amount, or an equality that must be met exactly.
- The decision variables to which the constraint applies.
- The impact that one unit of each decision variable will have on the right-hand side quantity of the constraint.

Constraints can be arranged into three groups:

- *System constraints* – involve more than one decision variable,
- *Individual constraints* – involve only one variable, and
- *Non-negativity constraints* – specify that no variable will be allowed to take on a negative value. The non-negativity constraints typically apply in an LP model, whether they are explicitly stated or not.

➤ *Parameters*

The objective function and the constraints consist of symbols that represent the decision variables (e.g., X_1 , X_2 , etc.) and numerical values called parameters. The parameters are fixed values that specify the impact that one unit of each decision variable will have on the objective and on any constraint it pertains to as well as the numerical value of each constraint.

The following simple example illustrates the components of LP models:

x_1 = quantity of product 1
 x_2 = quantity of product 2
 x_3 = quantity of product 3

Decision variables

Maximize $Z = 4x_1 + 7x_2 + 5x_3$ (profit) ← Objective function

Subject to $2x_1 + 3x_2 + 6x_3 \leq 300$ labor hours
 $5x_1 + 4x_3 \leq 200$ lb. raw material A
 $3x_1 + 5x_2 + 2x_3 \leq 360$ lb. raw material B

System constraints

$x_1 = 30$ units
 $x_2 \geq 40$ units

Individual Constraints

$x_1, x_2, \text{ and } x_3 \geq 0$ ← Non-negativity constraints

2.1.2. ASSUMPTIONS OF LP MODELS

Linearly (proportionality)

The linearity requirement is that each decision variable has a linear impact on the objective function and in each constraint in which it appears. In terms of a mathematical model, a function or equation is linear when the variables included are all to the power 1 (not squared, cubed, square root, etc.) and no products (e.g., x_1x_2) appear. On the other hand, the amount of each resource used (supplied) and its contribution to the profit (or cost) in the objective function must be proportional to the value of each decision variable. For example, if production of one unit requires 5 hours of a particular resource, then making 3 units of that product requires 15 hours (3×5) of that resource.

Divisibility (Continuity)

The divisibility requirement pertains to potential values of decision variables. It is assumed that non-integer values are acceptable. However, if the problem concerns, for example, the optimal number of houses to construct, 3.5 do not appear to be acceptable. Instead, that type of problem would seem to require strictly integer solutions. In such cases, integer-programming methods should be used. It should be noted, however, that some obvious integer type situations could be handled under the assumption of divisibility. For instance, suppose 3.5 to be the optimal number of television sets to produce per hour, which is unacceptable, but it would result in 7 sets per two hours, which would then be acceptable.

Certainty

This requirement involves two aspects of LP models. One aspect relates to the model parameters, i.e., the numerical values. It is assumed that these values are known and constant. In practice, production times and other parameters may not be truly constant. Therefore, the model builder must make an assessment as to the degree to which the certainty requirement is met. Large departures almost surely will have a significant effect on the model. The other aspect is the assumption that all relevant constraints have been identified and represented in the model.

Additivity

The value of the objective function and the total amount of each resource used (or supplied), must be equal to the sum of the respective individual contributions (profit or cost) by decision variables. For example, the total profit earned from the sale of two products A and B must be equal to the sum of the profits earned separately from A and B. Similarly, the amount of a resource consumed for producing A and B must be equal to the sum of resources used for A and B respectively.

Non-negativity

It assumes that negative values of variables are unrealistic and, therefore, will not be considered in any potential solutions. Only positive values and zero will be allowed and the non-negativity assumption is inherent in LP models.

2.1.3. Advantages of Linear Programming

Following are certain advantages of linear programming.

- Linear programming helps in attaining the optimum use of scarce productive resources. It also indicates how a decision maker can employ productive resources effectively by selecting and allocating these resources.
- LP techniques improve quality of decisions. The decision making approach of the user of this technique becomes more objective and less subjective.
- Highlighting of bottlenecks in the production process is one of the most advantages of this technique. For example, when a bottleneck occurs, some machines can not meet demand while others remain idle for some of the time.
- LP also helps in re-evaluation of a basic plan for changing conditions. If conditions change when the plan is carried out, they can be determined so as to adjust the remainder of the plan for best results.

2.1.4. Limitations of Linear Programming

In spite of having many advantages and wide area of applications, there are some limitations associated with this technique. These are:

- LP treats all relationships among decision variables as linear. However, generally neither the objective functions nor the constraints in real-life situations concerning business and industrial problems are linearly related to variables.
- While solving an LP model, there is no guarantee that we will get integer valued solutions. For example, in finding out how many men and machines would be required to perform a particular job, a non integer valued solution will be meaningless. Rounding off the solution to the nearest integer will not yield an optimal solution.
- An LP model does not take in to consideration the effect of time and uncertainty. Thus, the LP model should be defined in such a way that any change due to internal as well as external factors can be incorporated.
- It deals with only with a single objective, whereas in real-life situations we may come across conflicting multi-objective problems.

- Parameters appearing in the model are assumed to be constant but, in real life situation they are frequently neither known nor constant.

2.1.5. FORMULATING LP MODELS

Just as it is to define a problem, careful formulation of the model that will be used to solve the problem is important. Linear programming algorithms (solution techniques) are widely used and understood and computer packages are readily available for solving LP problems. Consequently, obtaining solutions is not the real issue, what is very important to note is failure to check that all constraints have been accounted for and have been correctly formulated results in ill-structuring of the model that can easily lead to poor decisions.

Steps in formulating LP models:

- Identify the decision variables.
- Determine the objective function.
- Identify the constraints.
- Determine appropriate values for parameters and determine whether an upper limit, lower limit or equality is called for.
- Use this information to build a model.
- Validate the model.

In many cases, the decision variables are obvious; in others it might require brief discussion with the appropriate manager. However, identifying the constraints and determining appropriate values for the parameters can require considerable time and effort. Potential sources of information include historical records, interviews with managers and staff, and data collection. Validating the model will involve a critical review of the output, perhaps under a variety of inputs, in order to decide if the results are reasonable.

2.2. LINEAR PROGRAMMING APPLICATIONS

There is a wide range of problems that lend themselves to solution by linear programming techniques. This discussion is only meant to give an indication of the LP techniques for managerial decision making and the apparent diversity of situations to which linear programming can be applied. Some of these include production management (product mix, blending problems, production planning, Assembly line balancing...), Marketing management (media selection, traveling sales man problem, physical distribution), Financial management (portfolio selection, profit planning), agricultural application, military applications, personnel management (staffing problem, Determination of equitable salary and etc.

Product Mix

Organizations often produce similar services that use the same resources. For example, labor, material cost, etc. because of limited resources available during any time period a decision must be made concerning how much of each product to produce or make available. Linear programming answers what mix of output (or service) will maximize profit given the availability of scarce resources

Diet problem

It usually involves the mixing of raw materials or other ingredients to obtain an end product that has certain characteristics. For example, what mix of inputs will achieve the desired results in the output for the least cost? Other applications that fall into this category include mixing feed for livestock, mixing pet foods, mixing building materials (concrete, mortar, paint), and so on.

Blending problems

They are very similar to diet problems. Strictly speaking, however, blending problems have additional requirement, i.e. to achieve a mix that have specific consistency. For example, how many quarts of the different juices each with different sugar content proportion must be mixed together to achieve one gallon that has a sugar content of 17 percent?

Portfolio selection

These problems generally involve allocating a fixed dollar amount among a variety of investments, such as bonds, stocks, real states, etc. The goal usually is to maximize income or total return. The problems take on an added dimension when certain other requirements are specified (for example, no more than 40 percent of the portfolio can be invested in bonds).

Dear learner, the section below presents some examples that demonstrate how a linear programming model is formulated. Although these examples are simplistic, they are realistic and represent the type of problem to which linear programming can be applied. By carefully studying each of these examples, the learner can become familiar with the process of formulating linear programming models.

Product Mix

ABC private limited company is engaged in the production of power and traction transformers. Both of these categories of transformers pass through three basic processes: core preparation, core to coil assembly, and vapor phase drying. A power transformer yields a contribution of Birr 50,000 and traction transformer contributes Birr 10,000. The time required in the production of these two products in terms of hours for each of the processes is as follows.

	Power transformer	Traction Transformer
Core preparation	75	15
Core to Coil Assembly	160	30
Vapor Phase Drying	45	10

If the capacities available are 1000, 1500, and 750 machine hours in each processes respectively, formulate the problem as LP?

Solution

Step1. Identify decision variables

Since the products to be produced are power and traction transformers using the available resource to attain the objective set, we consider them as decision variables. This is because the organization's problem here is how many of each product to produce in order to attain the objective, which requires the management decision.

Let X_1 = the no of power transformers to be produced.

X_2 = the no of traction transformer to be produced

Step2. Determine Objective Function

From the problem above, we understand that the problem is maximization problem. Hence,

$$Z_{\max} = 50,000X_1 + 10,000X_2$$

This is because, each unit of X_1 contributes Birr 50,000 and X_2 contributes Birr 10,000 to objective function.

Step3. Identify constraints

Here we have two constraints: structural and non-negativity constraints. The structural constraint is the amount of machine hours available in each process.

Step4. Determining Parameters

Parameters are already identified in the table of the problem.

 **Note:** Step 3 and 4 can be performed simultaneously as:

$$\begin{array}{l} 75X_1 + 15X_2 \leq 1000 \text{ hrs- Core preparation process} \\ 160X_1 + 30X_2 \leq 1500 \text{ hrs- Core to Coil Assembly} \\ 45X_1 + 10X_2 \leq 750 \text{ hrs- Vapor Phase Drying} \end{array} \quad \left. \vphantom{\begin{array}{l} 75X_1 + 15X_2 \leq 1000 \text{ hrs- Core preparation process} \\ 160X_1 + 30X_2 \leq 1500 \text{ hrs- Core to Coil Assembly} \end{array}} \right\} \text{Machine hour constraint}$$
$$X_1, X_2 \geq 0$$

Step5. Building and validating the model

$$Z_{\max} = 50,000X_1 + 10,000X_2$$

Subject to:

$$75X_1 + 15X_2 \leq 1000 \text{ hrs}$$

$$160X_1 + 30X_2 \leq 1500 \text{ hrs}$$

$$45X_1 + 10X_2 \leq 750 \text{ hrs}$$

$$X_1, X_2 \geq 0$$

Investment Application

1. An individual investor has Birr 70,000 to divide among several investments. The alternative investments are municipal bonds with an 8.5% return, certificates of deposits with a 10% return, Treasury bill with a 6.5% return, and income bonds with a 13% return. The amount of time until maturity is the same for each alternative. However, each investment alternative has a different perceived risk to the investor; thus it is advisable to diversify. The investor wants to know how much to invest in each alternative in order to maximize the return. The following guidelines have been established for diversifying the investment and lessening the risk perceived by the investor.
 1. No more than 20% of the total investment should be in an income bonds.
 2. The amount invested in certificates of deposit should not exceed the amount invested in other three alternatives.
 3. At least 30% of the investment should be in treasury bills and certificates of deposits.
 4. The ratio of the amount invested in municipal bonds to the amount invested in treasury bills should not exceed one to three.
 5. The investor wants to invest the entire Birr 70,000

Required: Formulate a LP model for the problem.

Solution

Decision variables

Four decision variables represent the monetary amount invested in each investment alternative. Let X_1 , X_2 , X_3 and X_4 represent investment on municipal bonds, certificates of deposits, Treasury bill, and income bonds respectively.

Objective function

The problem is maximization because from the word problem we know that the objective of the investor is to maximization return from the investment in the four alternatives.

Therefore, the objective function is expressed as:

$$Z_{\max} = 0.085 X_1 + 0.1X_2 + 0.065X_3 + 0.13X_4$$

Where,

Z = total return from all investment

$0.085X_1$ = return from the investment in municipal bonds.

$0.100X_2$ = return from the investment in certificates in deposit

$0.065X_3$ = return from the investment in treasury bills

$0.130X_4$ = return from the investment in income bonds

Model Constraints

In this problem the constraints are the guidelines established for diversifying the total investment. Each guideline is transformed in to mathematical constraint separately.

Guideline 1 above is transformed as:

$$X_4 < 20 \% (70,000) \Rightarrow X_4 < 14,000$$

Guideline 2

$$X_2 \leq X_1 + X_3 + X_4 \Rightarrow X_2 - X_1 - X_3 - X_4 \leq 0$$

Guideline 3

$$X_2 + X_3 \geq 30 \% (70,000) \Rightarrow X_2 + X_3 \geq 21,000$$

Guideline 4

$$X_1 / X_3 \leq 1/3 \Rightarrow 3X_1 \leq X_3 \Rightarrow 3X_1 - X_3 \leq 0$$

Finally, $X_1 + X_2 + X_3 + X_4 = 70,000$, because the investor's money equals to the sum of money invested in all alternatives.

This last constraint differs from $\leq$, and $\geq$ inequalities previously developed in that there is a specific requirement to invest an exact amount. Therefore, the possibility to invest more or less than Birr 70,000 is not considered.

This problem contains all three types of constraints possible in LP problems: $\leq$, $\geq$ and $=$. As this problem demonstrates there is no restriction on mixing these types of constraints.

The complete LPM for this problem can be summarized as:

$$Z_{\max} = 0.085 X_1 + 0.1 X_2 + 0.065 X_3 + 0.13 X_4$$

Subjected to:

$$X_4 \leq 14,000$$

$$X_2 - X_1 - X_3 - X_4 \leq 0$$

$$X_2 + X_3 \geq 21,000$$

$$3X_1 - X_3 \leq 0$$

$$X_1 + X_2 + X_3 + X_4 = 70,000,$$

$$X_1 + X_2 + X_3 + X_4 \geq 0$$

Marketing Application

2. Supermarket store chain has hired an advertising firm to determine the types and amount of advertising it should have for its stores. The three types of advertising available are radio and television commercials, and news papers advertisements. The retail chain desires to know the number of each type of advertisement it should purchase in order to maximize exposure. It is estimated that each ad or commercial will reach the following potential audience and cost the following amount.

Exposure		
Type of Advertisement	(people /ad or commercial)	Cost
Television commercial	20,000	Birr 15,000
Radio commercial	12,000	6,000
<u>News paper advertisement</u>	<u>9,000</u>	<u>4,000</u>

The company must consider the following resource constraints.

1. The budget limit for advertising is Birr 100,000
2. The television station has time available for 4 commercials.
3. The radio station has time available for 10 commercials.
4. The news paper has space available for 7 ads.
5. The advertising agency has time and staff available for producing no more than a total of 15 commercials and/or ads.

Decision variables

Let X_1 = number of television commercials

X_2 = number of radio commercials

X_3 = number of news paper ads

Objective function

It is not only profit which is to be maximized. In this problem the objective to be maximized is audience exposure.

$Z_{\max} = 20,000x_1 + 12,000x_2 + 9,000x_3$, where

Z = total level of audience exposure

$20,000x_1$ = estimated number of people reached by television commercials

$12,000x_2$ = estimated number of people reached by radio commercials

$9,000x_3$ = estimated number of people reached by news paper ads

Model constraints

Budget constraint

$$15,000x_1 + 6,000x_2 + 4,000x_3 \leq \text{Birr } 100,000$$

Capacity constraint

Television commercials and radio commercials are limited to 4 and 10 respectively and news paper ads are limited to 7.

$$x_1 \leq 4 \text{ television commercials}$$

$$x_2 \leq 10 \text{ radio commercials}$$

$$x_3 \leq 7 \text{ news paper ads}$$

Policy constraint

The total number of commercials and ads can not exceed 15

$$x_1 + x_2 + x_3 \leq 15$$

The complete linear programming model for this problem is summarized as:

$$Z_{\max} = 20,000x_1 + 12,000x_2 + 9,000x_3$$

Sub. to:

$$15,000x_1 + 6,000x_2 + 4,000x_3 \leq \text{Birr } 100,000$$

$$x_1 \leq 4$$

$$X_2 \geq 10$$

$$X_3 \geq 7$$

$$X_1 + X_2 + X_3 \leq 15$$

$$X_1, X_2, X_3 \geq 0$$

Chemical mixture

A chemical corporation produces a chemical mixture for the customer in 1000- pound batches. The mixture contains three ingredients- Zinc, mercury and potassium. The mixture must conform to formula specifications (i.e., a recipe) supplied by a customer. The company wants to know the amount of each ingredient to put in the mixture that will meet all the requirements of the mix and minimize total cost.

The customer has supplied the following formula specifications for each batch of mixture.

1. The mixture must contain at least 200 lb of mercury
2. The mixture must contain at least 300 lb of zinc
3. The mixture must contain at least 100 lb of potassium

The cost per pound of mixture is Birr4, of zinc Birr 8 and of potassium Birr 9.

Required: Formulate LPM for the problem

Solution

Decision variables

The model of this problem consists of three decision variables representing the amount of each ingredient in the mixture.

X_1 = number of lb of mercury in a batch

X_2 = number of lb of zinc in a batch

X_3 = number of lb of potassium in a batch

Objective function

$$Z_{\min} = 4x_1 + 8x_2 + 9x_3$$

Constraints

$x_1 \geq 200$ lb... specification 1

$x_2 \geq 300$ lb... specification 2

$x_3 \geq 100$ lb... specification 3

Finally, the sum of all ingredients must equal 1000 pounds.

$$x_1 + x_2 + x_3 = 1000 \text{ lb}$$

The complete LPM of the problem is:

$$Z_{\min} = 4x_1 + 8x_2 + 9x_3$$

Subject to:

$$x_1 \geq 200 \text{ lb}$$

$$x_2 \geq 300 \text{ lb}$$

$$x_3 \geq 100 \text{ lb}$$

$$x_1 + x_2 + x_3 = 1000 \text{ lb}$$

$$x_1, x_2, x_3 \geq 0$$

Activity

The manager of a department store in Adama is attempting to decide on the types and amounts of advertising the store should use. He has invited representatives from the local radio station, television station, and newspaper to make presentations in which they describe their audiences.

- a. The television station representative indicates that a TV commercial, which costs Birr 15,000, would reach 25,000 potential customers. The breakdown of the audience is as follows.

Male	Female
------	--------

Old	5,000	5,000
Young	5,000	10,000

b. The news paper representative claims to be able to provide an audience of 10,000 potential customers at a cost of Birr 4000 per ad. The breakdown of the audience is as follows.

	Male	Female
Old	4,000	3,000
Young	2,000	1,000

c. The radio station representative says that the audience for one of the station's commercials, which costs Birr 6000, is 15,000 customers. The breakdown of the audience is as follows.

	Male	Female
Old	1,500	1,500
Young	4,500	7,500

The store has the following advertising policy:

- Use at least twice as many radio commercials as news paper ads.
- Reach at least 100,000 customers
- Reach at least twice as many young people as old people
- Make sure that at least 30% of the audience is women

Available space limits the number newspaper ads to 7. The store wants to know the optimal number of each type of advertising to purchase to minimize total cost.

Required:

Formulate appropriate linear programming model.

2. An electronic company produces three types of parts for automatic machines. It purchases casting of the parts from the local foundry and then finishes parts on drilling, shaping and polishing machines.

The selling prices of parts A, B, and C respectively are Birr8, 10, 14. All parts made can be sold. Casting for parts A, B and C respectively costs Birr 5, 6 and 10. The shop possesses only one of each type of machine. Costs per hour to run each of the three machines are Birr 20 for drilling, 30 for shaping, and 30 for polishing. The capacities (parts per hour) for each part on each machine are shown in the following table:

Machine	Capacity per hour		
	Part A	Part B	Part C
Drilling	25	40	25
Shaping	25	20	20
Polishing	40	30	40

The management of the shop wants to know how many parts of each type it should produce per hour in order to maximize profit for an hour's run. Formulate this problem as an LP model so as to maximize total profit to the company.

2.3. Graphical Method of solving Linear Programming Problems

2.3.1. Introduction

In the previous section, you have seen how to formulate a given problem as a Linear Programming model. The next step, after the formulation of LP model is to devise effective methods to solve the model and ascertain the optimal solution. Let's start with the Graphical Method of solving LPP and then move on to Simplex Algorithm for solving the Linear Programming model.

Graphical linear programming is a relatively straightforward for determining the optimal solution to certain linear programming problems involving only two decision variables. Although graphic method is limited as a solution approach, it is very useful in the presentation of LP, in that it gives a "picture" of how a solution is derived thus a better understanding of the solution. Moreover, graphical methods provide a visual portrayal of many important concepts.

The graphic solution procedure is one of the methods of solving two variable linear programming problems.

In this method, the two decision variables are considered as ordered pairs (X_1, X_2) , which represent a point in a plane, i.e, X_1 is represented on X-axis and X_2 on Y-axis.

Graphical method has the following advantages:

- It is simple
- It is easy to understand, and
- It saves time.

2.3.2. Steps in graphic solution method:-

Step I Defining the problem. Formulate the problem mathematically. Express it in terms of several mathematical constraints & an objective function. The objective function relates to the optimization aspect, i.e. Maximization or minimization Criterion.

Step II Plot the constraints Graphically. Each inequality in the constraint equation has to be treated as an equation. An arbitrary value is assigned to one variable & the value of the other variable is obtained by solving the equation. In the similar manner, a different arbitrary value is again assigned to the variable & the corresponding value of other variable is easily obtained. These 2 sets of values are now plotted on a graph and connected by a straight line. The same procedure has to be repeated for all the constraints. Hence, the total straight lines would be equal to the total no of equations, each straight line representing one constraint equation.

Step III: *Locate the solution space.* Solution space or the feasible region is the graphical area which satisfies all the constraints at the same time. Such a solution point (x, y) always occurs at the corner Points of the feasible Region the feasible region is determined as follows:

- a) For "greater than" & "greater than or equal to" constraints, the feasible region or the solution space is the area that lays above the constraint lines.
- b) For "Less Than" & "Less than or equal to" constraint, the feasible region or the solution space is the area that lays below the constraint lines.

Step IV Selecting the graphic solution technique. Select the appropriate graphic technique to be used for generating the solution. There are two graphic techniques to find solution;

1. Corner Point Method and

2. Iso-profit (or Iso-cost) method may be used; however, it is easier to generate solution by using the corner point method and we use this method to find solution.

2.3.3. Corner Point Method to find Solution for Graphics Method

- I. Since the solution point (x, y) always occurs at the corner point of the feasible or solution space, identify each of the extreme points or corner points of the feasible region by the method of simultaneous equations.
- II. By putting the value of the corner point's co-ordinates [e.g. (2, 3)] into the objective function, calculate the profit (or the cost) at each of the corner points.
- III. In a Maximize problem, the optimal solution occurs at that corner point which gives the highest profit. In a minimization problem, the optimal solution occurs at that corner point which gives the lowest cost.

Important Theorems in solving LPP

On the process of determining an optimum feasible solution to the linear programming problem, the following four important theorems are used:-

Theorem I: The feasible solution spaces constitute a convex set.

Theorem II: Within the feasible solution space, feasible solution corresponds to the extreme (or Corner) points of the feasible solution space.

Theorem III There is a finite number of basic feasible solutions, with the feasible solution space.

Theorem IV The optimum feasible solution, if it exists, will occur at one, or more, of the

extreme points that are basic feasible solutions.

Note. Convex set is a polygon "Convex" implies that if any two points of the polygon are selected arbitrarily then straight line segment joining these two points lays completely within the polygon. The extreme points of the convex set are the basic solution to the linear programming problem.

Important Terms

Some of the important terms commonly used in linear programming are disclosed as follows:

- a. **Solution:** Values of the decision variable $x_i (i = 1, 2, 3 \dots)$ satisfying the constraints of a general linear programming model is known as the solution to that linear programming model.
- b. **Feasible solution:** Out of the total available solution a solution that also satisfies the non-negativity restrictions of the linear programming problem is called a feasible solution.
- c. **Basic solution:** For a set of simultaneous equations in Q unknowns (P, Q) a solution obtained by setting (P - Q) of the variables equal to zero & solving the remaining P equation in P unknowns is known as a basic solution. The variables which take zero values at any solution are detained as non-basic variables & remaining are known as-basic variables, often called basic solution.
- d. **Basic feasible solution:** A feasible solution to a general linear programming problem which is also basic solution is called a basic feasible solution.
- e. **Optimal feasible solution:** Any basic feasible solution which optimizes (i.e. maximize or minimizes) the objective function of a LP models is known as the optimal feasible solution to that linear programming model.
- f. **Degenerate Solution:** A basic solution to the system of equations is termed as degenerate if one or more of the basic variables become equal to zero.

Example 1

XYZ Ltd. Co. Wishes to purchase a maximum of 3600 units of two types of product, A & B are available in the market. Product A occupies a space of 3 cubic feet & cost Birr 9 whereas B occupies a space of 1 cubic feet & cost Birr 13 per unit. The budgetary constraints of the company do not allow spending more than Birr 39,000. The total availability of space in the company's godown is 6000 cubic feet. Profit margin of both the product A & B is Birr 3 & Birr 4 respectively. Formulate the above problem as a linear programming model and solve it by using graphical method. You are required to ascertain the best possible combination of purchase of A & B so that the total profits are maximized.

Solution

- (a) The problem can be formulated as a linear programming model as follows:-

Let x_1 = Number of units of product A &

x_2 = Number of units of product B

Objective function, *Maximize* $Z = 3x_1 + 4x_2$

Subject to: -

$$x_1 + x_2 \leq 3600 \quad (\text{Maximum unites constraint})$$

$$3x_1 + x_2 \leq 6000 \quad (\text{Storage area constraint})$$

$$9x_1 + 13x_2 \leq 39000 \quad (\text{Budgetary constraint})$$

$$x_1 \& x_2 \geq 0$$

(b) Find the solution by using graphics method

Step I: Treating all the constraints as equality,

The 1st constraint is: $x_1 + x_2 \leq 3600$ *change* this in to $x_1 + x_2 = 3600$ and find the X and Y intercept as follows.

Put $x_1 = 0 \Rightarrow x_2 = 3600 \therefore$ the point is $(0, 3600)$

Put $x_2 = 0 \Rightarrow x_1 = 3600 \therefore$ the point is $(3600, 0)$

Draw the graph with x_1 on x-axis & x_2 on y-axis as shown in the following figure.

The 2nd constraint is: $3x_1 + x_2 \leq 6000$ change this in to $3x_1 + x_2 = 6000$ and find X and Y intercept.

Put $x_1 = 0 \Rightarrow x_2 = 6000 \therefore$ the point is $(0, 6000)$

Put $x_2 = 0 \Rightarrow x_1 = 2000 \therefore$ the point is $(2000, 0)$

Draw the graph with x_1 on x-axis & x_2 on y-axis as shown in the following figure.

The 3rd constraint is: $9x_1 + 13x_2 \leq 39,000$ *change* this in to $9x_1 + 13x_2 = 39,000$ and find the X and Y intercept as follows.

Put $x_1 = 0 \Rightarrow x_2 = 3000 \therefore$ the point is $(0, 3000)$

Put $x_2 = 0 \Rightarrow x_1 = 4333.3 \therefore$ the point is $(4333.3, 0)$

Draw the graph with x_1 on x-axis & x_2 on y-axis as shown in the **Figure 2.1**.

Step II: Determine the set of the points which satisfy the constraint:

$$x_1 + x_2 \leq 3600$$

This can easily be done by verifying whether the origin (0, 0) satisfies the constraint. Here, $0 + 0 < 3600$ hence the solution is true; all the points below the line will satisfy the constraint.

Like it's in the above iteration, determine the set of points which satisfy the constraint $3x_1 + x_2 \leq 6000$. At origin;

$0 + 0 < 6000$. Hence, all the points below the line will satisfy the constraint.

Like it's in the above iteration, determine the set of points which satisfy the constraint. $9x_1 + 13x_2 \leq 39,000$. At origin;

$0 + 0 < 39,000$. Hence, all the points below the line will satisfy the constraint.

The intersection of the above graphic denotes the feasible region for the given problem.

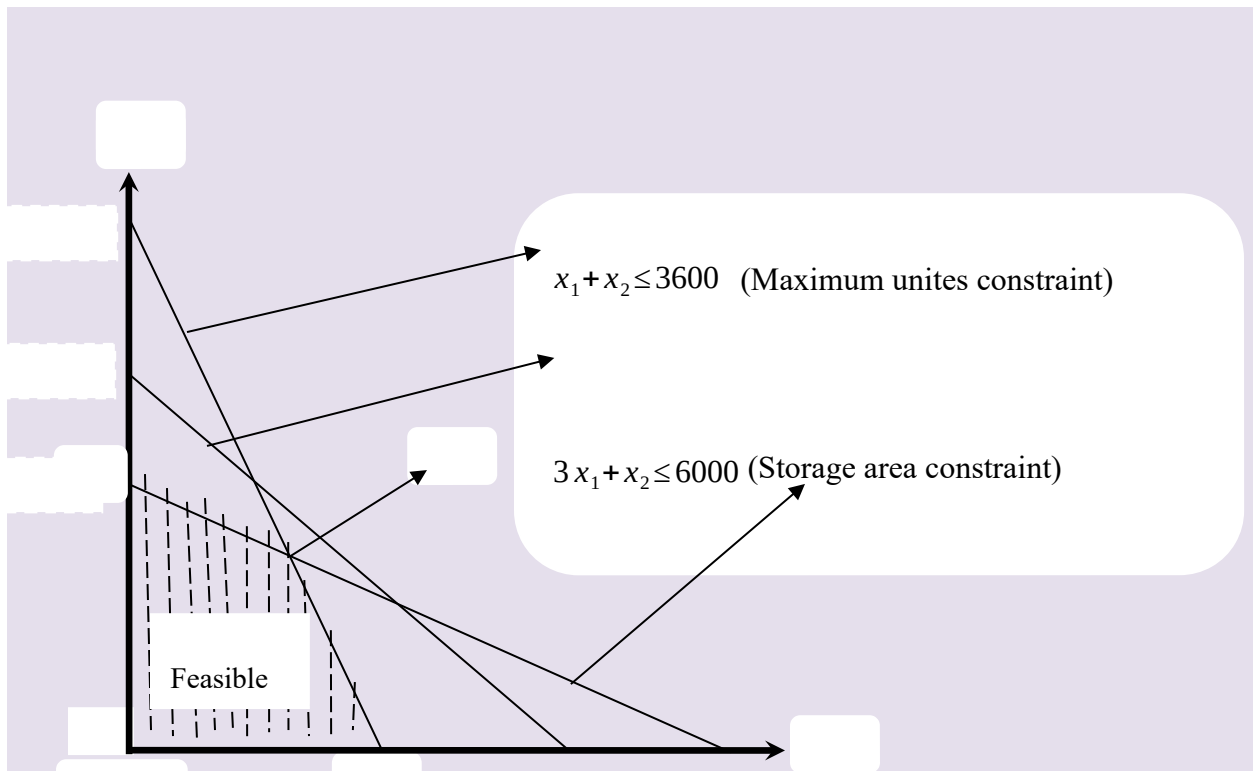


Figure 2.1. Graphical presentation of XYZ Ltd. Co. purchase problem

Step III: Finding feasible region by plotting the graph

Always keep in mind two things: -

- For $\geq$ constraint the feasible region will be the area, which lays above the constraint lines, and for $\leq$ constraints, it will lay below the constraint lines. This would be useful in identifying the feasible region.
- According to a theorem on linear programming, an optimal solution to a problem (if it exists) is found at a corner point of the solution space.

Step IV. Find the optimal solution by the corner point method.

At corner points (O, A, B, C), find the profit value from the objective function. Those points which maximize the profit are the optimal point.

	Corner	Coordinates	Objective function	Value
	O	(0,0)	$Z=0+0$	0
	A	(0,3000)	$Z=0+4 \times 3000$	12,000
→	B_{Optimal} B_{solution}	(1950,1650)	$Z=3 \times 1950 + 4 \times 1650$	12,450
	C	(2000,0)	$3 \times 2000 + 0$	6000

In order to get the value of point B apply simultaneous equation by taking the two intersection lines. Solve the point.

$$3x_1 + x_2 \leq 6000 \text{ (Maximum unit constraint)}$$

$$9x_1 + 13x_2 \leq 39000 \text{ (Budgetary constraint)}$$

($\therefore$ A+B these two lines are intersecting)

$$x_1 + x_2 \leq 3600 \quad \dots 1$$

$$9x_1 + 13x_2 = 39000 \quad \dots 2$$

Multiply equation (1) by -9 on both sides:

$$-9x_1 - 9x_2 = -32,400 \quad \dots 3$$

$$9x_1 + 13x_2 = 39000 \quad \dots 4$$

$$4x_2 = 6,600 \quad \therefore x_2 = 1650$$

Put the Value of x_2 in first equation:

$$\Rightarrow X_1 = 1950$$

At point (1950, 1650) Maximize $Z = 3x_1 + 4x_2$

$$Z = (3 \times 1950) + (4 \times 1650) = 12,450$$

= 12,450 which is the maximum value

Result

The optimal solution is:

No of units of product A = 1950

No of units of product B = 1650

Total profit, = 12,450 which is the maximum

Example 2:

Suppose that a machine shop has two different types of machines; machine 1 and machine 2, which can be used to make a single product. These machines vary in the amount of product produced per hour, in the amount of labor used and in the cost of operation.

Assume that at least a certain amount of product must be produced and that we would like to utilize at least the regular labor force. How much should we utilize each machine in order to utilize total costs and still meet the requirement? The resource used, the cost and the required hour is given in the following table.

Constraints	Resource used		Minimum required hours
	Machine 1	Machine 2	
Product produced/hr	20	15	100
Labor/hr	2	3	15
Operation Cost	Birr25	Birr30	

1. Formulation of the linear programming model

$$\begin{array}{l}
 \text{Min. } Z = 25X_1 + 30X_2 \\
 \text{Subject to:} \\
 20X_1 + 15X_2 \geq 100 \\
 2X_1 + 3X_2 \geq 15 \\
 X_1, X_2 \geq 0
 \end{array}
 \left. \vphantom{\begin{array}{l} \text{Min. } Z = 25X_1 + 30X_2 \\ \text{Subject to:} \\ 20X_1 + 15X_2 \geq 100 \\ 2X_1 + 3X_2 \geq 15 \\ X_1, X_2 \geq 0 \end{array}} \right\} \text{LPP Model}$$

The 1st constraint is: $20x_1 + 15x_2 \geq 100$ change this in to $20x_1 + 15x_2 = 100$ and find X and Y intercept.

$$\text{Put } x_1 = 0 \Rightarrow x_2 = \frac{20}{3} \therefore \text{the point is } (0, 20/3)$$

$$\text{Put } x_2 = 0 \Rightarrow x_1 = 5 \therefore \text{the point is } (5, 0)$$

Draw the graph with x_1 on x-axis & x_2 on y-axis as shown in the following figure.

The 2nd constraint is $2X_1 + 3X_2 \geq 15$ change this in to $2X_1 + 3X_2 = 15$ and find X and Y intercept.

$$\text{Put } x_1 = 0 \Rightarrow x_2 = 5 \therefore \text{the point is } (0, 5)$$

$$\text{Put } x_2 = 0 \Rightarrow x_1 = 7.5 \therefore \text{the point is } (7.5, 0)$$

Draw the graph with x_1 on x-axis & x_2 on y-axis as shown in the following figure.

Determine the set of the points which satisfy the constraint:

$$20x_1 + 15x_2 \geq 100$$

This can easily be done by verifying whether the origin (0, 0) satisfies the constraint. Here,

$0 + 0 > 100$ Hence the solution is false; all the points above the line will satisfy the constraint.

Like it's in the above iteration, determine the set of points which satisfy the constraint

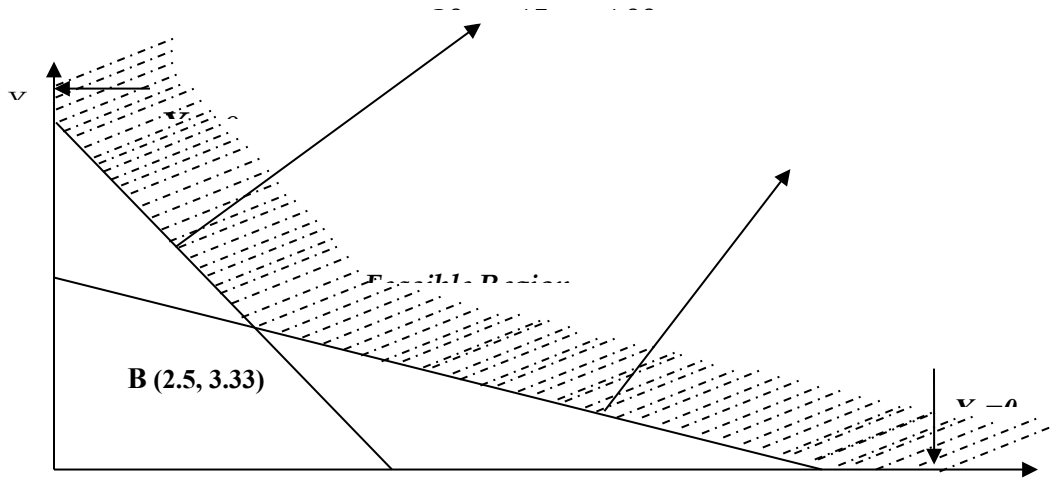
$$2X_1 + 3X_2 \geq 15 \text{ at origin;}$$

$0 + 0 > 1$. Hence, all the points above the line will satisfy the constraint.

Finding feasible region by plotting the graph

Always keep in mind two things: -

- For $\geq$ constraint the feasible region will be the area, which lays above the constraint lines, and for $\leq$ constraints, it will lay below the constraint lines. This would be useful in identifying the feasible region.
- According to a theorem on linear programming, an optimal solution to a problem (if it exists) is found at a corner point of the solution space.



Corner point	Coordinates	Objective function	Value	Feasible solution
<i>Minimize $Z = 25X_1 + 30X_2$</i>				
A	(0, 20/3)	$Z = 0 + 20/3 \times 30$	200	
B	(2.5, 3.33)	$Z = 2.5 \times 25 + 3.33 \times 30$	162.5	✓
C	(7.5, 0)	$Z = 7.5 \times 25 + 0$	187.5	

Use the same steps to find the value of B by using the simultaneous equation as presented previously.

Result

The optimal solution is:

$X_1 = 2.5$, $X_2 = 3.33$ and Minimum cost $Z = 162.5$

2.3.4. Special Cases in Graphics Methods

The following are the major special cases in graphics solution

- **Alternative Optima**
- **Infeasible Solution**
- Unbounded solution

a) Alternative Optimal solution

When the objective function is parallel to a binding constraint; (a constraint that is satisfied in the equality sense by the optimal solution), the objective function will assume the same optimal value at more than one solution point, for this reason they are called **alternative optima**. Example 1 shows that normally there is infinity of such solutions. The example also demonstrates the practical significance of encountering alternative optima.

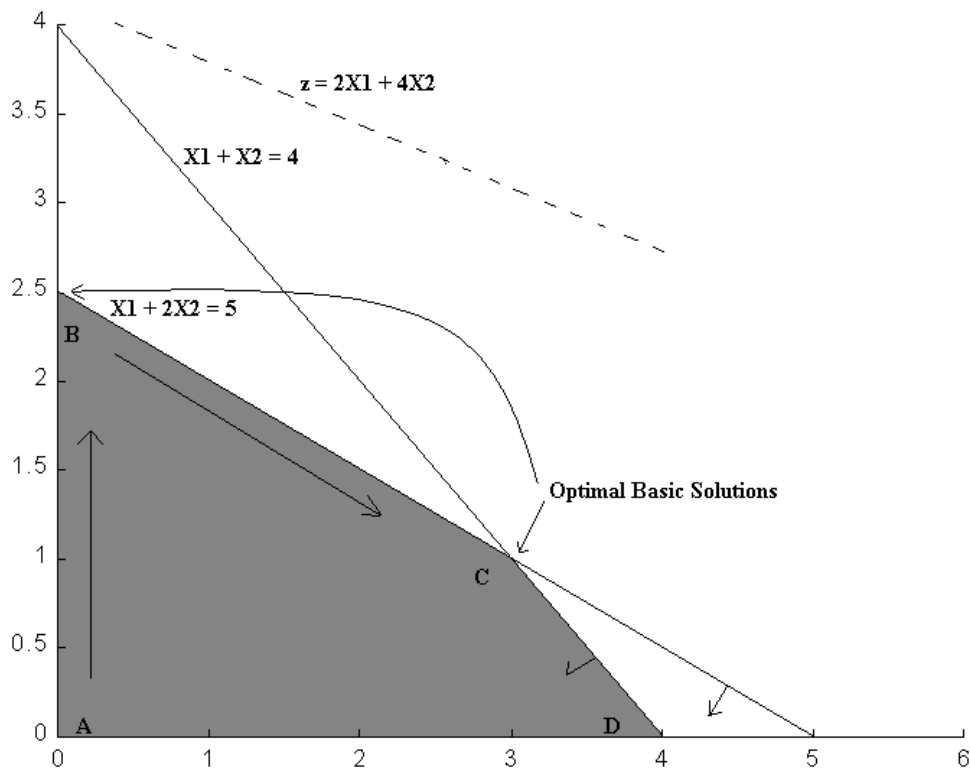
Example1

$$\text{Maximize } z = 2x_1 + 4x_2$$

Subject to:

$$x_1 + 2x_2 \leq 5$$

$$x_1 + x_2 \leq 4 \wedge x_1, x_2 \geq 0$$



The above Figure demonstrates how alternative optima can arise in LP model when the **objective function is parallel to a binding constraint**. Any point on the line segment BC represents an alternative optimum with the same objective value $z = 10$. Mathematically, we can determine all the points (x_1, x_2) on the line segment BC as a nonnegative weighted average of the points B and C. That is, given 0, α , and 1 and

$$B: x_1 = 0, x_2 = 5/2$$

$$C: x_1 = 3, x_2 = 1$$

Then all points on the line segment BC are given by

$$x_1 = \alpha(0) + 3(1-\alpha) = 3 - 3\alpha$$

$$x_2 = \alpha(5/2) + 1(1-\alpha) = 1 + 3\alpha/2$$

Observe that when $\alpha=0$, $(x_1, x_2) = (3, 1)$, which is point C. When $\alpha=1$, $(x_1, x_2) = (0, 5/2)$, which is point B. For values of α between 0 and 1 (x_1, x_2) lays between B and C

Example 2:

The information given below is for the products A and B.

<i>Department</i>	<i>Machine hours per week</i>		<i>Maximum available</i>
	<i>Product A</i>	<i>Product B</i>	<i>per week</i>

Cutting	3	6	900
Assembly	1	1	200
<i>Profit per unit</i>	<u><i>Birr 8</i></u>	<u><i>Birr 16</i></u>	

Assume that the company has a marketing constraint on selling products B and therefore it can sale a maximum of 125 units of this product.

Required:

- Formulate the LPP of this problem?
- Find the optimal solution?

Solution: Let X_1 =The Number of units' f product A produced per week

X_2 = The number of units f product B produced per week

- The LPP Model of the problem is:

$$\text{Max. } Z = 8X_1 + 16X_2$$

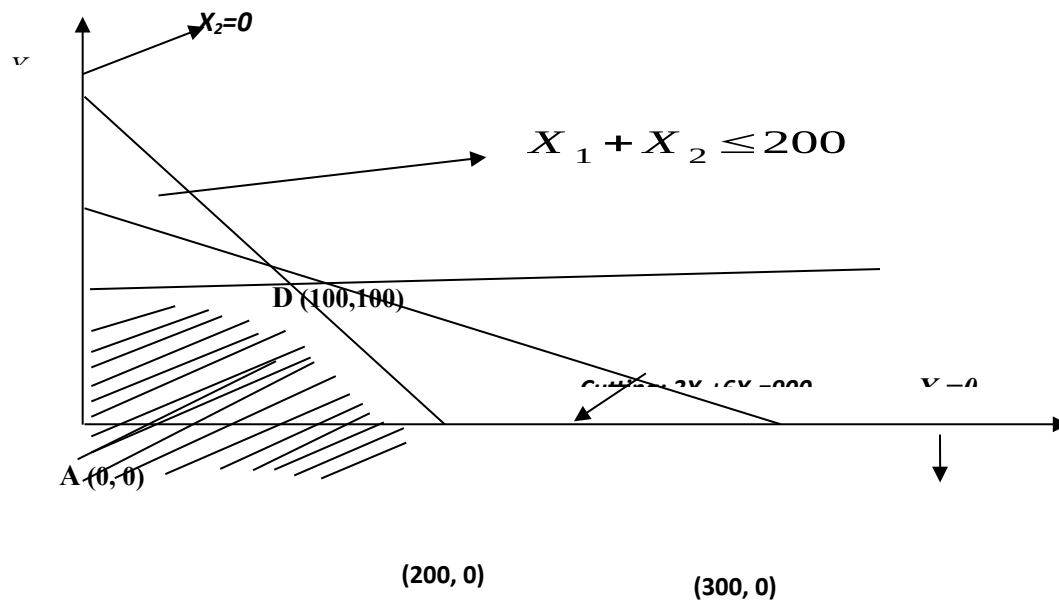
Subject to:

$$3X_1 + 6X_2 \leq 900$$

$$X_1 + X_2 \leq 200$$

$$X_2 \leq 125$$

$$X_1, X_2 \geq 0$$



<u>Corners</u>	<u>Coordinates</u>	<u>Maximize $Z = 8X_1 + 16X_2$</u>
A	(0, 0)	0
B	(0, 125)	2000
C	(50, 125)	2400
D	(100, 100)	2400
E	(200, 100)	1600

Interpretation:

Both **C** and **D** are optimal solutions. Any point on the line segment **CD** will also lead to the same optimal solution.

==> Multiple optimal solutions provide more choices for management to reach their objectives.

b) 2. Infeasible Solution

A solution is called feasible if it satisfies all the constraints and non-negativity condition. However, it is sometimes possible that the constraints may be inconsistent so that there is no feasible solution to the problem. Such a situation is called *infeasibility*.

Example:

$$\text{Maximize } Z = 20X_1 + 30X_2$$

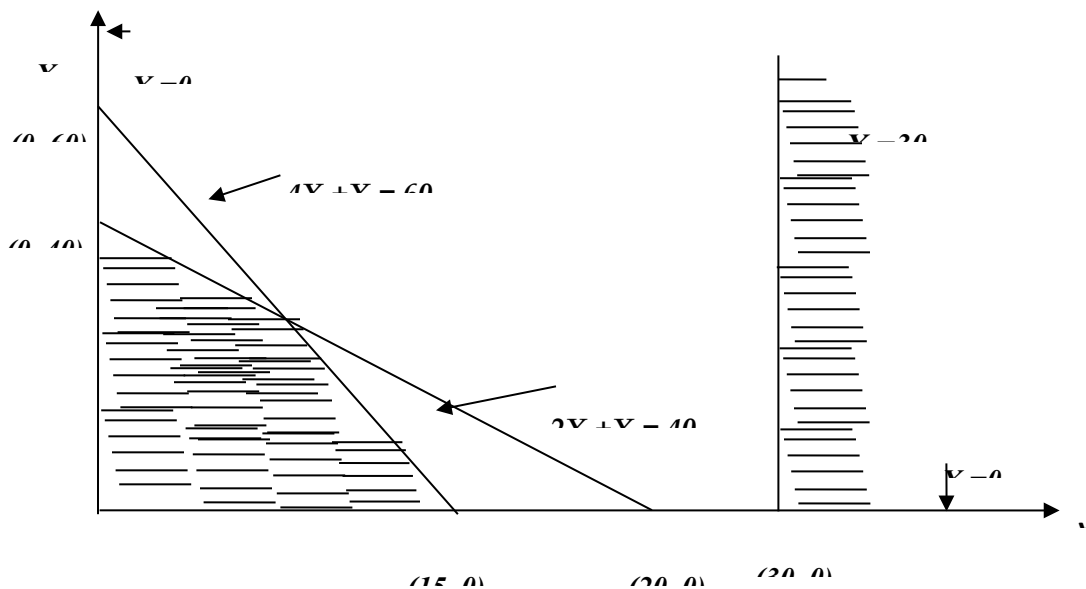
Subject to:

$$2X_1 + X_2 \leq 40$$

$$4X_1 + X_2 \leq 60$$

$$X_1 \geq 30 \quad \text{and } X_1, X_2 \geq 0$$

Solution:



Note:

- In the above graph, there is no common point in the shaded area.
- All constraints cannot be satisfied simultaneously and there is no feasible solution to the problem.

c) Unbounded Solution

When the value of decision variables in LP is permitted to increase infinitely without violating the feasibility condition, then the solution is said to be unbounded. Here, the

objective function value can also be increased infinitely. However, an unbounded feasible region may yield some definite value of the objective function.

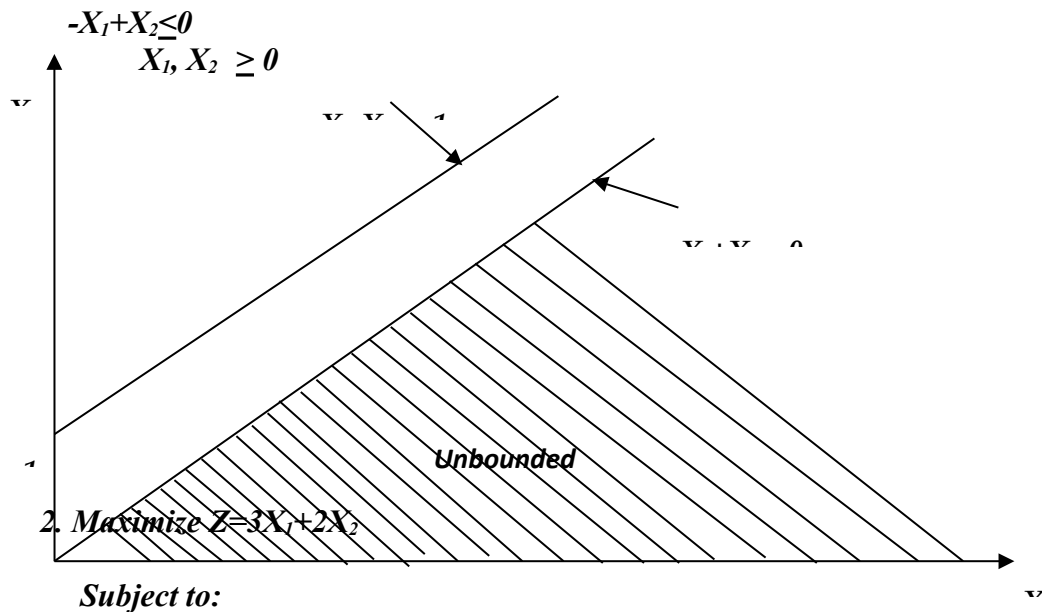
Example:

Use the graphical method to solve the following LPP.

1. Maximize $Z=3X_1+4X_2$

Subject to:

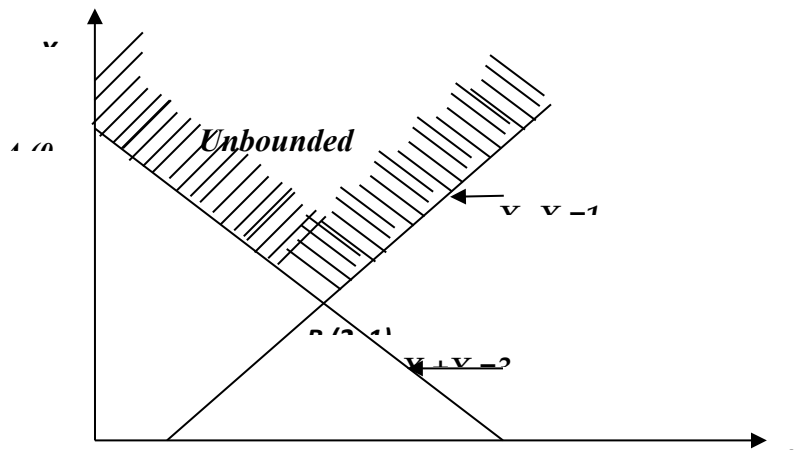
$$X_1 - X_2 \leq -1 \implies -X_1 + X_2 \geq 1 \text{ since the quantity solution is positive}$$



$$X_1 - X_2 \leq 1$$

$$X_1 + X_2 \geq 3$$

$$X_1, X_2 \geq 0$$



Note: here that the two corners of the region are A(0,3) and B(2,1). The value of Maximize $Z(A)=6$ and Maximize $Z(B)=8$. But there exist number of points in the shaded region for which the value of the objective function is more than 8. For example, the point (10, 12) lies in the region and the function value at this point is 70 which is more than 8.

Remark: An unbounded solution does not mean that there is no solution to the given LPP, but implies that *there exists an infinite number of solutions*.

Exercise 1:

Use graphical method to solve the following LPP.

1. Maximize $Z=7/4X_1+3/2X_2$

2. Maximize $Z=X_1+X_2$

Subject to:

Subject to:

$$8X_1+5X_2 \leq 320$$

$$X_1+X_2 \leq 1$$

$$4X_1+5X_2 \leq 20$$

$$-3X_1+X_2 \geq 3$$

$$X_1 \geq 15$$

$$X_1, X_2 \geq 0$$

$$X_2 \geq 10$$

$$X_1, X_2 \geq 0$$

Answer: No feasible solution

Answer: Unbounded solution

3. Maximize $Z=3X_1+2X_2$

4. Maximize $Z=6X_1-4X_2$

Subject to:

Subject to:

$$X_1-X_2 \leq 1$$

$$2X_1+4X_2 \leq 4$$

$$X_1+X_2 \geq 3$$

$$4X_1+8X_2 \geq 16$$

$$X_1, X_2 \geq 2$$

$$X_1, X_2 \geq 0$$

Answer: Unbounded solution

Answer: Infeasible solution

Exercise 2.

I. Solve the following LP problems using the graphical method.

$$1. \text{ Maximize } Z=15X_1-10X_2$$

Subject to:

$$4X_1+6X_2 \leq 360$$

$$3X_1+0X_2 \leq 180$$

$$0X_1+5X_2 \leq 280$$

$$X_1, X_2 \geq 0$$

*Answer: $X_1=60, X_2=20$
and Maximize $Z=1,100$*

$$2. \text{ Maximize } Z=2X_1+X_2$$

Subject to:

$$X_1+2X_2 \leq 10$$

$$X_1+X_2 \leq 6$$

$$X_1-X_2 \leq 2$$

$$X_1-2X_2 \leq 1$$

*Answer: $X_1=4, X_2=2$
and Maximize $Z=10$*

$$3. \text{ Maximize } Z=10X_1+15X_2$$

Subject to:

$$2X_1+X_2 \leq 26$$

$$2X_1+4X_2 \leq 56$$

$$-X_1+X_2 \leq 5$$

$$X_1, X_2 \geq 0$$

Answer: $X_1=4, X_2=2$

and Maximize $Z=230$

$$4. \text{ Min. } Z=3X_1+2X_2$$

Subject to:

$$5X_1+X_2 \geq 10$$

$$X_1+X_2 \geq 6$$

$$X_1+4X_2 \geq 12$$

$$X_1, X_2 \geq 0$$

Answer: $X_1=1, X_2=5$

and Minimize $Z=13$

- II.** A manufacturer produces two different models; X and Y, of the same product. The raw materials r_1 and r_2 are required for production. At least 18 Kg of r_1 and 12 Kg of r_2 must be used daily. Almost at most 34 hours of labor are to be utilized. 2Kg of r_1 are needed for each model X and 1Kg of r_1 for each model Y. For each model of X and Y, 1Kg of r_2 is required. It takes 3 hours to manufacture a model X and 2 hours to manufacture a model Y. The profit realized is \$50 per unit from model X and \$30 per unit from model Y. How many units of each model should be produced to maximize the profit?

Answer: 10 units of model X, 2 units of model Y and the maximum profit is \$ 560.

- III.** A manufacturing firm produces two machine parts P_1 and P_2 using milling and grinding machines. The different machining times required for each part, the machining times available on different machines and the profit on each machine part are as given below:

Manufacturing time

Maximum time

	Required (min)	available per week (minimum)	
Machine	P_1	P_2	
<i>Lathe</i>	<i>10</i>	<i>5</i>	<i>25,000</i>
<i>Milling Machine</i>	<i>4</i>	<i>10</i>	<i>2000</i>
<i>Grinding Machine</i>	<i>1</i>	<i>1.5</i>	<i>450</i>
<i>Profit per unit (\$)</i>	<i><u>\$50</u></i>	<i><u>\$100</u></i>	

Determine the number of pieces of P_1 and P_2 to be manufactured per week to maximize profit.

Answer: $X_1=187.5$, $X_2=125$ and Maximize $Z=21,875$

2.4. THE SIMPLEX METHOD

The graphical method of solving linear programming problems is a simple way to find a solution since the optimum solution is searched among the corner points of the solution space. However, the graphical method is restricted to problems with two decision variables. When the number of variables and the number of constraints increase, it becomes difficult to visualize the solution space. As a result, the graphical method cannot be employed successfully in such cases. In order to avoid this limitation, the simplex method, or iterative or step by step method is efficient method for solving linear programming problems, which was developed by George B. Datzing in 1947.

The simplex method is an algebraic procedure that starts with a feasible solution that is not optimal and systematically moves from one feasible solution to another until an optimal solution is found. In case of the graphical approach, optimal solution occurs at the extreme points where the constraints intersect. Solutions where constraints intersect are called basic solutions, and those satisfying all of the constraints together with non-negativity constraints are called basic feasible solutions.

Constraints are generally expressed using inequalities either in 'less than' or 'greater than' or in mixed form. Thus, constraints are not in standard form, meaning they should be converted into equalities. To convert the inequality constraint into equality, we introduce slack or surplus variables. In economic terminology, slack variables represent unused capacity and surplus variables represent excess amount. The contribution (cost or profit) associated with the slack and surplus variables is zero. An inequality of the 'less than or equal to' type is transformed into equality by introducing a non-negative slack variable, as follows: -

Example

Non Standard form

$$X_1 + 2X_2 \leq 6$$

Standard form

$$X_1 + 2X_2 + S_1 = 6$$

,where X_1 and X_2 are decision variables and S_1 is a slack variable, added to the smaller side of the inequality.

On the other hand, an inequality with 'greater than or equal to' type is changed into equality by subtracting surplus variable as follows: -

Example

Non Standard form

$$X_1 + 2X_2 \geq 6$$

Standard form

$$X_1 + 2X_2 - S_1 = 6$$

,where X_1 & X_2 are decision variables and S_1 is a surplus variable, subtracted from larger side of the inequality.

As stated above, since both the slack and surplus variables are insignificant with no contribution in the objective function, they are represented with coefficient of zero in the objective function.

To find a unique solution, the number of variables should not exceed the number of equations. When the number of variables is more than the number of equations, the number of solutions is unlimited. So as to get a unique solution, we have to set at least $(n-m)$ variables to zero, where n is the number of variables and m is the number of equalities. Those variables that are set to zero are called *non basic variables* indicating they are not in the solution. The variables that are in solution are called *basic variable*.

To demonstrate the simplex method, we will use the microcomputer problem with the following objective function and constraints.

The Microcomputer Problem, which was discussed in graphic approach, can be standardized as:

Max.	$Z = 60X_1 + 50X_2 + 0S_1 + 0S_2 + 0S_3$
Subject to	$4X_1 + 10X_2 + S_1 = 100$
	$2X_1 + X_2 + S_2 = 22$
	$3X_1 + 3X_2 + S_3 = 39$

, where X_1 & X_2 are decision variables and S_1 , S_2 & S_3 are slack variables.

Here, the number of variables (5) is greater than the number of equations (3). Therefore, the decision variables are set to zero and we have

$$X_1 = 0, X_2 = 0, S_1 = 100, S_2 = 22, \text{ and } S_3 = 39.$$

This solution will serve as an initial feasible solution. An initial feasible solution is a first solution used to generate other basic feasible solutions. The initial basic feasible solution is obtained by setting all the decision variables to zero. As a result, the initial basic feasible solution is entirely composed of the slack variables. X_1 & X_2 are non basic variables since they are not in solution. S_1 , S_2 & S_3 are basic variables since they are in solution.

A *tableau* is a system of displaying the basic feasible solution, the constraints of the standard linear programming problem as well as the objective function in a tabular form. A tableau is useful in

summarizing the result of each iteration, i.e. the process of moving from one solution/corner to another solution/corner in order to select the optimal solution.

2.4.1. THE SIMPLEX ALGORITHM

2.4.1.1. Maximization case

The solution steps of the simplex method can be outlined as follows:

Step 1. Formulate the linear programming model of the real world problem, i.e., obtain a mathematical representation of the problem's objective function and constraints.

Step 2. Express the mathematical model of L.P. problem in the standard form by adding slack variables in the left-hand side of the constraints and assign a zero coefficient to these in the objective function.

Thus we can restate the problem in terms of equations:

$$\begin{array}{ll} \text{Maximize} & Z = C_1X_1 + C_2X_2 + \dots + C_nX_n + 0S_1 + 0S_2 + \dots + 0S_m \\ \text{Subject to} & a_{11}X_1 + a_{12}X_2 + \dots + a_{1n}X_n + S_1 = b_1 \\ & a_{21}X_1 + a_{22}X_2 + \dots + a_{2n}X_n + S_2 = b_2 \\ & \cdot \quad \cdot \quad \quad \quad \cdot \quad \cdot \quad \cdot \\ & \cdot \quad \cdot \quad \quad \quad \cdot \quad \cdot \quad \cdot \\ & \cdot \quad \cdot \quad \quad \quad \cdot \quad \cdot \quad \cdot \\ & a_{m1}X_1 + a_{m2}X_2 + \dots + a_{mn}X_n + S_m = b_m \end{array}$$

, where $X_1, X_2, \dots, X_n$ and $S_1, S_2, \dots, S_m$ are non-negative.

Note that the slack variables have been assigned zero coefficients in the objective function. The reason is that these variables typically contribute nothing to the value of the objective function.

Step 3. Design the initial feasible solution. An initial basic feasible solution is obtained by setting the decision variables to zero.

$X_1 = X_2 = \dots = X_n = 0$. Thus, we get $S_1 = b_1, S_2 = b_2, \dots, S_m = b_m$.

Step 4. Set up the initial simplex tableau. For computational efficiency and simplicity, the initial basic feasible solution, the constraints of the standard LPP as well as the objective function can be displayed in a tabular form, called the *simplex tableau* as shown below:

Initial Simplex Tableau

C _j (Contribution Per Unit)		C ₁ C ₂ ... C _n ,, 0 0 ... 0	Quantity Column	Ratio XB/a _{mn}
Basic variables	CB $\rightarrow$ $\downarrow$	X ₁ , X ₂ , ... X _n S ₁ S ₂ ... S _m	b(=X _j)	
S ₁	CB ₁	a ₁₁ a ₁₂ ... a _{1n} 1 0 ... 0	b ₁ =Xb ₁	
S ₂	CB ₂	a ₂₁ a ₂₂ ... a _{2n} 0 1 ... 0	b ₂ =Xb ₂	
S _m	CB _m	a _{m1} a _{m2} ...a _{mn} 0 0 ... 1	b _m =Xb _m	
Z _j =∑CB _i X _j		0 0 ... 0 0 0 ... 0	∑CB _i X _j	
C _j -Z _j (Net contribution per unit)		C ₁ -Z ₁ C ₂ -Z ₂ ... C _n -Z _n		

The interpretation of the data in the above tableau is given as under. Other simplex tableau will have similar interpretations.

- In the first row labeled " C_j ", we write the coefficients of the variables in the objective function. These values will remain the same in subsequent tableaus.
- The second row shows the major column headings.
- In the first column of the second row, under the label "*Basic variables*" (also called *Product mix* column), the basic variables are listed.
- In the second column of the second row, under the label "CB", the coefficients of the current basic variables in the objective function are listed. Thus the coefficients of $S_1, S_2, \dots, S_m$, which are included in the initial feasible solution, are written in the CB column.
- The values listed under the non-basic variables ($X_1, X_2, \dots, X_n$) in the initial simplex tableau consists of the coefficients of the decision variables in the constraint set. They can be interpreted as physical rates of substitution.
- The values listed under the basic variables ($S_1, S_2, \dots, S_m$) in the initial simplex tableau represents the coefficients of the slack variables in the constraints set.
- In the next column (also called *Quantity column*), we write the solution values of the basic variables.
- To find an entry in the Z_j row under a column, we multiply the entries of that column by the corresponding entries of 'CB' column and add the results, i.e., $Z_j = \sum CB_i X_j$. The Z_j row entries will all be equal to zero in the initial simplex tableau. The other Z_j entries represent the decrease in the value of objective function that would result if one of the variables not included in the solution were brought into the solution. The Z_j entry under the "*Quantity Column*" gives the current value of the objective function.
- The last row labeled " $C_j - Z_j$ ", called the *index row* or *net evaluation row*, is used to determine whether or not the current solution is optimal or not. The calculation of $C_j - Z_j$ row simply involves subtracting each Z_j value from the corresponding C_j value for that column, which is written at the top of that column. We observe that $C_j - Z_j$ values are meaningful for the non-basic variables only. This is because for a basic variable, $Z_j = 1 \times C_j = C_j$ so that $C_j - Z_j = C_j - C_j = 0$.

👉 **Note:** The entries in the $C_j - Z_j$ row (i.e., *index row*) represent the net contribution to the objective function that results by introducing one unit of each of the respective column variables. A plus value indicates that a greater contribution can be made by bringing the variable for that column into the solution. A negative value indicates the amount by which contribution would decrease if one unit of the variable for that column were brought into the solution. *Index row* elements are also known as the **shadow prices (or accounting prices)**

Step 5. We test if the current solution is optimum or not. If all the elements or entries in the $C_j - Z_j$ row (i.e., *index row*) are negative or zero, then the current solution is optimum. If there exists some positive number, the current solution can be further improved by removing one basic variable from the basis and replacing it by some non-basic one. *So start trying to improve the current solution in line with the following steps.*

Step 6. Further, iterate towards an optimum solution. To improve the current feasible solution, we replace one current basic variable (called the *departing or leaving variable*) by a new non-basic variable (called the *entering variable*).

- We now determine the **variable to enter** into the solution mix, *the entering variable*. One way of doing this is by identifying the **column with the largest positive value in the $C_j - Z_j$ row** of the simplex table. The column with the largest positive entry in the $C_j - Z_j$ row is called the **key or pivot column**. **The non-basic variable at the top of the key column is the entering variable that will replace a basic variable.**
- Next, we determine the **departing variable** to be replaced in the **basis** solution. This is accomplished by dividing each number in the **quantity column** by the corresponding number in the **key column** selected in identifying the entering variable. We compute the ratio b_1/a_{1j} , b_2/a_{2j} ... b_m/a_{mj} . This is called **replacement ratio**.

Replacement Ratio (RR) =	$\frac{\text{Solution Quantity (Q)}}{\text{Key Column Element}}$
--------------------------	--

- The row with the minimum ratio is the **key row or pivot row**. The corresponding variable in the key row (the departing variable) will leave the basis.
- We identify the **key or pivot element**. This is the number that lies at the intersection of the key column and key row of a given simplex tableau.

Step 7. Evaluate the new solution by constructing a second simplex tableau. After identifying the entering and departing variable, all that remains is to find the new basic feasible solution by constructing a new simplex tableau from the current one.

Now we evaluate or update the new solution in the following way:

- New values for the key row are computed by simply **dividing every element of the key row by the key element** to obtain a unit vector (1) in the key element.
- The new values of the elements in the remaining rows for the new simplex table can be obtained by performing elementary row operations on all rows so that all elements except the key element (1) in the key column are zero, i.e. unit vector.
- New entries in the CB column and XB column are entered in the new table of the current solution.

Elementary row operation for entering variable in new simplex tableau

Improved row for entering variable	=	Old row of leaving variable
		Intersection key element

Elementary row operation for new simplex tableau

New row of existing variable	=	Old row of existing variable	–	Corresponding key element	×	Improved row for entering variable
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- Compute the values of the $C_j - Z_j$ row. If all the numbers in $C_j - Z_j$ row are either negative or zero, an optimum solution has been obtained.

Step 8. If any of the numbers in $C_j - Z_j$ row are positive, repeat the steps (6-7) again until an optimum solution has been obtained.

 **Note: Rules for Ties.** In choosing a key column and a key row, whenever there is a tie between two numbers, the following rules may be followed:

- The *column farthest (extreme) to the left* may be selected if there is a tie between two numbers in the *index row*.
- The *nearest ratio to the top* may be selected whenever there is a tie between two replacement ratios in the ratio column.

Example (Maximization Case)

Finding the initial Feasible Solution

As discussed above, the initial feasible solution is found by setting the decision variables to zero.

$$\text{Max } Z = 60X_1 + 50X_2 + 0S_1 + 0S_2 + 0S_3$$

$$\text{Subject to } 4X_1 + 10X_2 + S_1 = 100$$

$$2X_1 + X_2 + S_2 = 22$$

$$3X_1 + 3X_2 + S_3 = 39$$

$$X_1, X_2 \geq 0$$

Initial Simplex tableau

C_j		60	50	0	0	0	Quantity
Basic V.		X_1	X_2	S_1	S_2	S_3	
S_1	0	4	10	1	0	0	100
S_2	0	2	1	0	1	0	22
S_3	0	3	3	0	0	1	39
Z_j		0	0	0	0	0	0
$C_j - Z_j$		60	50	0	0	0	

Interpretation of the Initial Feasible Solution

To be noted first is that the values of each basic variable (variables that are in solution) is composed of a single 1 and the rest 0's. This is called a *unit vector*. Basic variables will have a unit vector. Moreover, '1' will appear in the same row that the variable appears in. The unit vector concept will help us in developing subsequent tableaus when we want to change the list of variables that are in solution.

The Z_j row in the quantity column indicates that the value of the objective function is 0.

The values in the $C_j - Z_j$ row indicate the net contribution of the variables if one unit of each variable is added into that solution. For example, the 60 in column X_1 indicates that bringing one unit of X_1 would increase the value of the objective function by \$60. The same is true for the X_2 column, i.e. bringing one unit of X_2 would increase the value of the objective function by \$50. On the other hand, since the slack variables are at their maximum, their values in the $C_j - Z_j$ row are all 0. According to what has been said before, we have the following rule in order to identify an optimal solution.

A simplex solution in a maximization problem is optimal if the $C_j - Z_j$ row consists of entirely zeros and negative numbers, i.e. there are no positive values in the row.

So for our case in the initial tableau, we have two positive values under the non-basic variables, which indicate that further improvement of the solution is possible. As a result, we go for the optimal solution by developing the second simplex tableau

Developing the Second Simplex Tableau

In the initial feasible solution, the slack variables form entirely the basic variables and the decision variables entirely form the non-basic variables. Since further improvement is possible, one of the decision variables will be brought into the solution and one of the current basic variables will be leaving the solution. ‘Which non-basic variable should be brought into the solution and which basic variable should leave the solution?’ is a major concern here.

In answering the first question, the non-basic variable that should enter the solution should be the one with the highest positive value in the $C_j - Z_j$ row since bringing that non-basic variable into the solution would make the highest contribution to the solution (objective function) and result in the largest profit potential. As a result, the variable with the highest value in the $C_j - Z_j$ row of the initial simplex tableau is the X_1 column with a value 60. Therefore, X_1 should enter the basis or the solution mix. The X_1 column is now called the *pivot column*. The numbers in this column (4 2 3) indicate the amount of the basic variables needed to get one unit of X_1 . For example, the number 4 indicates that 4 units of the first slack are needed to obtain one unit of X_1 .

Since X_1 has the highest profit potential, we need to make as much X_1 units as possible. The amount of X_1 that can be made depends on the values in the pivot column and the amount of slack available shown under the quantity column. By dividing the values in the pivot column by their respective values in the quantity column, we can identify the variable that is most limiting.

The values obtained by dividing will help us in determining the variable that should leave the solution.

C_j Basic V.	60 X_1	50 X_2	0 S_1	0 S_2	0 S_3	Quantity (Q)	Ratio = $\frac{Q}{Pv \text{ column}}$
S_1 0	4	10	1	0	0	100	$100/4 = 25$
S_2 0	2	1	0	1	0	22	$22/2 = 11$
S_3 0	3	3	0	0	1	39	$39/3 = 13$
Z_j	0	0	0	0	0	0	
$C_j - Z_j$	60	50	0	0	0		

In interpreting the ratios obtained by the division, if only the constraint was the first one, we could make 25 units of X_1 . But there are also other constraints. Therefore, the one with the smallest non-negative ratio is the most restrictive since it determines the amount of X_1 . In this particular case, there is only enough of the second constraint to make 11 units of X_1 . In making the 11 units of X_1 , the second resource (S_2) will be down to zero indicating that S_2 will leave out the solution mix. The row of the leaving variable is called the *pivot row*. The intersection of the pivot row and the pivot column is called the *pivot element*. As a rule,

The leaving variable is the one with the smallest non-negative ratio.

If a zero is obtained in the results of division, none of the corresponding variable is needed to obtain one unit of the entering variable. If a negative value is obtained in the division, bringing the entering variable into the solution would increase the amount of the basic variable. These values will not limit the amount of the entering variable. Therefore, there is no need to divide the quantity column by zero or negative.

Since we have determined the leaving and the entering variables and since the initial feasible solution can be improved further, we need to develop the second tableau in order to find the optimal solution. In developing the second tableau, we should compute for revised values of the constraint equations, the Z_j row and the $C_j - Z_j$ row and remember that the variables in solution will have a unit vector, with a value of 1 in the intersection of the column and the row of the basis, i.e. the pivot element.

To obtain a unit vector in the basis column, we perform elementary row operations resulting in new row values by either multiplying/dividing all the elements in a row by a constant or adding/subtracting the multiple of a row to or from another row.

In computing for the new row values from our initial simplex tableau, we first multiply the second constraint by $\frac{1}{2}$ obtaining the values as follows:

$$1X_1 + \frac{1}{2}X_2 + 0S_1 + \frac{1}{2}S_2 + 0S_3 = 11, \text{ which results in making the pivot element 1.}$$

Next, we multiply the above new row values by 4 and subtract it from the first constraint obtaining the following results:

$$0X_1 + 8X_2 + 1S_1 + 2S_2 + 0S_3 = 56.$$

Then, we multiply the new row values in the pivot row by 3 and subtract it from the third constraint resulting as follows:

$$0X_1 + \frac{3}{2}X_2 + 0S_1 - \frac{3}{2}S_2 + 1S_3 = 6.$$

Having these new row values, we develop the second simplex tableau as shown below.

Second Simplex Tableau

C _j		60	50	0	0	0	Quantity
Basic V.		X ₁	X ₂	S ₁	S ₂	S ₃	
S ₁	0	0	8	1	-2	0	56
X ₂	60	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	11
S ₃	0	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	1	6
Z _j		60	30	0	30	0	660
C _j -Z _j		0	20	0	-30	0	

Interpretation of the Second Simplex Tableau

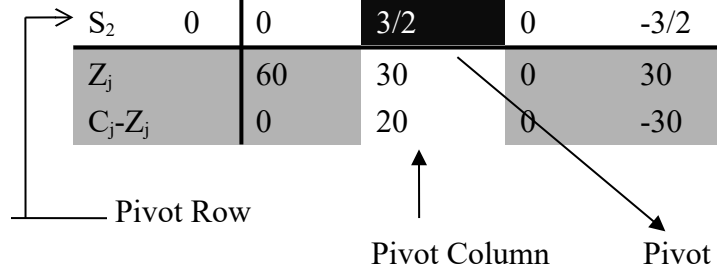
The profit obtained at this point of solution is \$660. In the $C_j - Z_j$ row, we search for the highest positive value. If there is, it means that we can further improve this solution. Therefore, we have a positive value in the $C_j - Z_j$ row which indicates that this is not the optimal solution. As a result, we go for the next tableau.

Developing the Third Tableau

Here, we select the entering and the leaving variables. The entering variable is the one with the highest $C_j - Z_j$ row value which is the X_2 column. This means that bringing one unit of X_2 into the solution increases profit by \$20. Therefore, the X_2 will be the entering variable designated as the pivot column.

To determine the leaving variable, we divide the values in the pivot column by their corresponding row values in the quantity column. The result obtained, as shown in the table below indicates that S_3 is the leaving variable with the smallest non-negative ratio. This means that S_3 is the most limiting resource for how much units of X_2 can be made.

C_j	60	50	0	0	0	Quantity	
Basic V.	X_1	X_2	S_1	S_2	S_3		
S_1 0	0	8	1	-2	0	56	$56/8 = 7$
X_2 60	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	11	$11/(1/2) = 22$
S_2 0	0	$\frac{3}{2}$	0	$-\frac{3}{2}$	1	6	$6/(3/2) = 4$
Z_j	60	30	0	30	0	660	
$C_j - Z_j$	0	20	0	-30	0		



After identifying the entering and the leaving variables, we perform elementary row operations as follows. To obtain a unit vector with 1 in the pivot element, we multiply the pivot row by $2/3$ resulting as follows:

$$0X_1 + 1X_2 + 0S_1 + 1S_2 + 2/3S_3 = 4$$

Then, multiply the above new row values by 8 and subtract it from the first constraint (row) resulting as follows:

$$0X_1 + 0X_2 + 1S_1 + 6S_2 + 16/3S_3 = 24.$$

For row 2, first multiply the new row values in the pivot row by $\frac{1}{2}$ and subtract it from the second row resulting as follows:

$$1X_1 + 0X_2 + 0S_1 + 1S_2 + 1/3S_3 = 9.$$

And the third tableau looks like the following.

Third Simplex Tableau

C_j	60	50	0	0	0	Quantity
Basic V.	X_1	X_2	S_1	S_2	S_3	
S_1 0	0	0	1	6	$-16/3$	24
X_1 60	1	0	0	1	$-1/3$	9
X_2 50	0	1	0	-1	$2/3$	4
Z_j	0	0	10	$40/3$	0	740
$C_j - Z_j$	0	0	-10	$-40/3$	0	

Interpreting the Third Tableau

All the values in the $C_j - Z_j$ row are zero and negative indicating that there can no be additional improvement. This makes the third tableau to contain the optimal solution with the following basic variables:

$S_1 = 24$, $X_1 = 9$, and $X_2 = 4$ producing a maximum profit of \$740.

This means that in order to achieve the maximum profit, the company should produce 9 units of X_1 and 4 units of X_2 leaving 24 hours of unused resource of the second constraint.

2.4.2. SIMPLEX MAXIMIZATION WITH MIXED CONSTRAINTS

(= AND $\geq$ CONSTRAINTS)

2.4.2.1. The Big M Method—introducing slack, surplus and artificial variables

- If any problem constraint has negative constants on the right side, multiply both sides by -1 to obtain a constraint with a non-negative constant. (If the constraint is an inequality, this will reverse the direction of the inequality.)
- Introduce a slack variable in each $\leq$ constraints.
- Introduce a surplus variable and an artificial variable in each $\geq$ constraint.
- Introduce an artificial variable in each '=' constraint.
- For each artificial variable A_i , add $-MA_i$ to the objective function in case of maximization and $+MA_i$ in case of minimization.
- Use the same constant M for all artificial variables.
- Form the simplex tableau for the modified problem.
- Solve the modified problem using the simplex method.
- Relate the solution of the modified problem to the original problem:
 - ✚ If the modified problem has no solution, then the original problem has no solution.
 - ✚ If any artificial variables are non-zero in the solution to the modified problem, then the original problem has no solution.

 **Note.** The artificial variables are introduced for the limited purpose of obtaining an initial solution and are required for the constraints of $\geq$ type or the constraints with '=' sign. It is not relevant whether the objective function is of the minimization or the maximization type.

Obviously, since artificial variables do not represent any quantity relating to the decision problem, they must be driven out of the system and must not remain in the final solution (and if at all they do, it represents a situation of infeasibility). This can be ensured by assigning an extremely high cost to them. Generally, a value M is assigned to each artificial variable, where M represents a number higher than any finite number. For this reason, the method of solving the problems where artificial variables are involved is termed as the '**Big M Method**'. When the problem is of the minimization nature, we assign in the objective function a coefficient of +M to each of the artificial variables. On the other hand, for the problems with the objective function of maximization type, each of the artificial variables introduced has a coefficient -M.

In case of equality constraints, a slack variable is not acceptable since these constraints require a precise amount. To overcome this problem, artificial variables are introduced which are added to

= and $\geq$ constraints. They have no physical representation for their purpose is to allow the use of simplex process. As a result, they should not appear in the final optimal solution. Assignment of coefficients for artificial variables depends on the type of the problem, whether it is a maximization or a minimization problem. *In case of maximization problem, a large negative contribution or coefficient in the objective function would ensure the removal of artificial variables from the optimal solution, commonly represented by $-M$.* For minimization problems, introducing a large positive coefficient would ensure the non-existence of artificial variables in the optimal solution, commonly represented by M .

In addition to this, a slack is not allowed in $\geq$ constraints since they can only happen an excess amount that is more than the minimum amount of a constraint. This excess amount is represented by a surplus variable and is subtracted from the constraint.

Example – Maximization Problems with Mixed Constraints

Assume the following maximization problem with mixed constraints.

$$\begin{array}{ll}\text{Max.} & Z = 6X_1 + 8X_2 \\ \text{Subject to} & X_2 \leq 4 \\ & X_1 + X_2 = 9 \\ & 6X_1 + 2X_2 \geq 24\end{array}$$

In solving for this problem, introducing a slack variable is not acceptable since they represent unused capacity and there is no unused capacity in = and $\geq$ constraints. Therefore, for $\geq$ constraints, we introduce a surplus variable and for both $\geq$ and = constraints we introduce artificial variables resulting as follows.

$$\begin{array}{ll}\text{Max} & Z = 6X_1 + 8X_2 + 0S_1 + 0S_3 - MA_2 - MA_3 \\ \text{Subject to} & X_2 + S_1 = 4 \\ & X_1 + X_2 + A_2 = 9 \\ & 6X_1 + 2X_2 - S_3 + A_3 = 24 \\ & X_1, X_2 \geq 0\end{array}$$

Wouldn't it be possible to set A_2 as A_1 and A_3 as A_2 in the above standardized LPM form?

Developing the Initial Simplex Tableau

In the initial tableau, we list the variable in the order of decision variable, slack/surplus variables and finally artificial variables. And for constraints, use artificial variables, and/or slack variables and/or surplus variables for the initial solution. Therefore the initial tableau for the problem is represented as follows.

Initial Simplex Tableau

C_j		6	8	0	0	-M	-M	Quantity
Basic V.		X_1	X_2	S_1	S_3	A_2	A_3	
S_1	0	0	1	1	0	0	0	4
A_2	-M	1	1	0	0	1	0	9
A_3	-M	6	2	0	-1	0	1	24

Z_j	-7M	-3M	0	+M	-M	-M	-33M
$C_j - Z_j$	6+7M	8+3M	0	-M	0	0	

Since this is a maximization problem, the entering variable is the one with the maximum $C_j - Z_j$ value. Therefore, the variable with the maximum $C_j - Z_j$ is X_1 , the decision variable. And the leaving variable is with the **minimum non-negative ratio** which is A_3 , the artificial variable. Since this artificial variable is not needed we remove it from the next tableau.

Developing the Second Simplex Tableau

After identifying the entering and the leaving variable, the usual elementary row operations are performed to obtain a unit vector in the entering variable column. The end result after performing the row operations is as follows which shows the second tableau.

Second Simplex Tableau

C_j		6	8	0	0	-M	Quantity
Basic V.		X_1	X_2	S_1	S_3	A_2	
S_1	0	0	1	1	0	0	4
A_2	-M	0	2/3	0	1/6	1	5
X_1	6	1	1/3	0	-1/6	0	4
Z_j		6	2-2/3M	0	-1-M/6	-M	24-5M
$C_j - Z_j$		0	6+2/3M	0	1+M/6	0	

Again here, we identify the leaving the entering variables, i.e. the pivot row and the pivot column respectively. The entering variable with the highest $C_j - Z_j$ row value is X_2 and the leaving variable with the smallest ratio is S_1 .

Developing the Third Simplex Tableau

Selecting the entering and leaving variables, we go for obtaining a unit vector in the pivot column by using the elementary row operations. We get the following third tableau.

Third Simplex Tableau

C_j		6	8	0	0	-M	Quantity
Basic V.		X_1	X_2	S_1	S_3	A_2	
X_2	8	0	1	1	0	0	4
A_2	-M	0	0	-2/3	1/6	1	7/3
X_1	6	1	0	-1/3	-1/6	0	8/3
Z_j		6	8	6+2/3M	-1-M/6	-M	48-7/3
$C_j - Z_j$		0	0	-6-2/3M	1+M/6	0	

Similarly, we identify the entering and the leaving variables which are S_3 and A_2 respectively representing the maximum $C_j - Z_j$ value and the minimum ratio.

Developing the Fourth Simplex Tableau

Perform elementary row operations usual to have a unit vector in the entering variable column and you will get the following fourth tableau.

Fourth Simplex Tableau

C_j		6	8	0	0	Quantity
Basic V.		X_1	X_2	S_1	S_3	
X_2	8	0	1	1	0	4
S_3	0	0	0	-4	1	14
X_1	6	1	0	-1	0	5
Z_j		6	8	2	0	62
$C_j - Z_j$		0	0	-2	0	

This tableau represents the final tableau since we have only zeros and negative values in the $C_j - Z_j$ row which indicates that it is the optimal solution. So we have the following results for each of the variables and the profit obtained.

$$X_1 = 5, X_2 = 4, S_3 = 14, \text{ and Profit} = 62$$

2.4.2.2. MINIMIZATION LINEAR PROGRAMMING PROBLEMS (Big M-method / Charnes Penalty Method)

The Big-M Method is a technique, which is used in removing artificial variables from the basis. In this method; we assign coefficients to artificial variables, undesirable from the objective function point of view. *If objective function Z is to be minimized, then a very large positive price (called **penalty**) is assigned to each artificial variable.* Similarly, if Z is to be maximized, then a very large negative cost (also called **penalty**) is assigned to each of these variables. Following are the characteristics of Big-M Method:

- High penalty cost (or profit) is assumed as M
- M as a coefficient is assigned to artificial variable A in the objective function Z .
- Big-M method can be applied to minimization as well as maximization problems with the following distinctions:
 - Minimization problems**
 - Assign **+M** as coefficient of artificial variable **A** in the objective function **Z** of the minimization problem.
 - Maximization problems:**

- Here $-M$ is assigned as coefficient of artificial variable A in the objective function Z of the maximization problem.
- d. Coefficient of S (slack/surplus) takes zero values in the objective function Z
- e. For minimization problem, the incoming variable corresponds to the ***smallest negative (or highest numerical value with negative sign)*** value of $C_j - Z_j$.
- f. The solution is optimal when there is no negative value of $C_j - Z_j$. (For minimization LPP case)

The various steps involved in using simplex method (Big M Method) for minimization problems are:

Step 1. Formulate the linear programming model, and express the mathematical model of L.P. problem in the standard form by introducing surplus and artificial variables in the left hand side of the constraints. Assign a 0 (zero) and $+M$ as coefficient for surplus and artificial variables respectively in the objective function. M is considered a very large number so as to finally drive out the artificial variables out of basic solution.

Step 2. Next, an initial solution is set up. Just to initiate the solution procedure, the initial basic feasible solution is obtained by assigning zero value to decision variables. This solution is now summarized in the initial simplex table. Complete the initial simplex table by adding two final rows Z , and $C_j - Z_j$. These two rows help us to know whether the current solution is optimum or not.

Step 3. Now; we test for optimality of the solution. If all the entries of $C_j - Z_j$ row are positive, then the solution is optimum. However, this situation may come after a number of iterations. But if at least one of the $C_j - Z_j$ values is less than zero, the current solution can be further improved by removing one basic variable from the basis and replacing it by some non-basic one.

Step 4. (i) Determine the variable to enter the basic solution. To do this, we identify the column with the ***smallest negative value (or highest numerical value with negative sign)*** in the $C_j - Z_j$ row of the table.

(ii) Next we determine the departing variable from the basic solution. If an artificial variable goes out of solution, then we discard it totally and even this variable may not form part of further iterations. Same procedure, as in maximization case, is employed to determine the departing variable.

Step 5. We update the new solution now. We evaluate the entries for next simplex table in exactly the same manner as was discussed earlier in the maximization case.

Step 6. Step (3—5) are repeated until an optimum solution is obtained.

So the following are the essential things to observe in solving for minimization problems:

-  The entering variable is the one with the ***smallest negative value (or highest numerical value with negative sign)*** in the $C_j - Z_j$ row while the leaving variable is the one with the smallest non-negative ratio.
-  The optimal solution is obtained when the $C_j - Z_j$ row contains entirely zeros and positive values.

Example: Assume the following minimization problem.

Min $Z = 7X_1 + 9X_2$
 Subject to $3X_1 + 6X_2 \geq 36$
 $8X_1 + 4X_2 \geq 64$
 $X_1, X_2 \geq 0$

We introduce both surplus and artificial variables into both constraints as follows.

Min $Z = 7X_1 + 9X_2 + 0S_1 + 0S_2 + MA_1 + MA_2$
 Subject to $3X_1 + 6X_2 - S_1 + A_1 = 36$
 $8X_1 + 4X_2 - S_2 + A_2 = 64$
 $X_1, X_2 \geq 0$

So the subsequent tableaus for this problem are shown below. To remind in these tableaus is in transforming from one tableau to another, we perform elementary row operations to obtain the unit vector in the pivot column for the entering variable into the solution.

Initial Simplex Tableau

C _j		7	9	0	0	M	M	Quantity	Ratio
Basic V.		X ₁	X ₂	S ₁	S ₂	A ₁	A ₂		
A ₁	M	3	6	-1	0	1	0	36	6 ←
A ₂	M	1	4	0	-1	0	1	64	16
Z _j		4M	10M	-M	-M	M	M	100M	
C _j -Z _j		7-4M	9-10M	M	M	0	0		

↑

Second Simplex Tableau

C _j		7	9	0	0	M	M	Quantity	Ratio
Basic V.		X ₁	X ₂	S ₁	S ₂	A ₁	A ₂		
X ₂	9	1/2	1	-1/6	0	1/6	0	6	12
A ₂	M	-1	0	2/3	-1	-2/3	1	-8	8 ←
Z _j		9/2-M	9	-3/2+2/3M	-M	3/2-2/3M	M	54+8M	
C _j -Z _j		5/2-M	0	3/2+2/3M	M	3/2+5M/3	0		

↑

Third Simplex Tableau

C _j		7	9	0	0	M	M	Quantity	Ratio
Basic V.		X ₁	X ₂	S ₁	S ₂	A ₁	A ₂		
X ₂	9	0	1	1/6	-1/2	-1/6	1/2	8	-16
X ₁	7	1	0	-2/3	1	2/3	-1	8	8 ←
Z _j		7	9	-19/6	5/12	19/6	-5/2	128	
C _j -Z _j		0	0	19/6	-5/12	M-19/6	M+5/2		

↑

Fourth Simplex Tableau

Basic V.	C _j	7	9	0	0	M	M	Quantity	Ratio
		X ₁	X ₂	S ₁	S ₂	A ₁	A ₂		
X ₂	9	1/2	1	-1/6	0	1/6	0	12	
S ₂	0	1	0	-2/3	1	2/3	-1	8	
Z _j		7	9	-19/6	5/12	19/6	-5/2	108	
C _j -Z _j		0	0	19/6	-5/12	M-19/6	M+5/2		

The third tableau represents a final tableau since it is the optimal solution with entirely zeros and non-negative values in the C_j-Z_j row.

Therefore, the optimal solution is: X₁ = 12 and S₂ = 8, and X₂ = 0 and value of objective function is 108

2.4.3. SPECIAL ISSUES

Several special situations which one may encounter during the application of simplex method are summarized below:

Non-feasible Solution/ Infeasibility

A situation with no feasible solution may exist if the problem was formulated incorrectly. Infeasibility comes about when there is **no solution** that satisfies all of the problem's constraints.

In the simplex method, an infeasible solution is indicated by looking at the final tableau. Whenever the optimality criteria is satisfied but still there exist an artificial variable in the basis or solution mix, this is the indication of infeasibility.

Example:

Minimization case

	C _j	5	8	0	0	M	
Basic V		X ₁	X ₂	S ₁	S ₂	A ₂	Q
X ₁	5	1	1	-2	3	0	200
X ₂	8	0	1	1	2	0	100
A ₂	M	0	0	0	-1	1	20
	Z _j	5	8	-2	31-M	M	1,800+200M
	C _j - Z _j	0	0	2	M-31	0	

Even though all $C_j - Z_j$ are positive or 0 (i.e. the criterion for an optimal solution in a minimization case), **no feasible solution** is possible because **an artificial variable (A_2)** remains in the solution mix.

Unbounded Solution

No finite solution may exist in problems that are not bounded. This means that a variable can be infinitely large without violating a constraint. In the simplex method, the condition of unboundedness will be discovered prior to reaching the final tableau. We will note the problem when trying to decide which variable to remove from the solution mix.

The procedure in unbounded solution is to divide each quantity column number by the corresponding pivot column number. The row with the smallest **positive** ratio is replaced. But if the entire ratios turn out to be **negative** or **undefined**, it indicates that the problem is unbounded.

👉 **Note:** A negative ratio means that increasing that variable would increase resources. A zero ratio means that increasing the variable would not use any resources.

Example:

Maximization case:

	C_j		6	9	0	0	
Basic V			X1	X2	S1	S2	Q
X_2	9		-1	1	2	0	30
S_2	0		-2	0	-1	1	10
Z_j			-9	9	18	0	270
$C_j - Z_j$			15	0	-18	0	

$30/-1 = -30$

The solution in the above case is not optimal because not all $C_j - Z_j$ entries are 0 or negative, as required in a maximization problem. The next variable to enter the solution should be X_1 . To determine which variable will leave the solution, we examine the ratios of the quantity column numbers to their corresponding numbers in the X_1 or pivot column. Since both pivot column numbers are negative, an unbounded solution is indicated.



Note: When unbounded solutions occur, no outgoing variable will exist.

Degeneracy /Tie for leaving basic variable (key row)/

If there is a tie for the smallest ratio, this is a signal that degeneracy exists. Degeneracy can occur right in the first (initial tableau). This normally happens when the *number of constraints is less than the number of variables* in the LP model. Such problems can be overcome by **trial and error method**. If this is resolved by a proper selection of the key element, degeneracy can be avoided. The main drawback to degeneracy is the increase in the computation, which reduces the efficiency of the simplex method considerably.

Example

C_j		5	8	2	0	0	0		
	SV	X_1	X_2	X_3	S_1	S_2	S_3	Q	RR
8	X_2	1/4	1	1	-2	0	0	10	10/1/4=40
0	S_2	4	0	1/3	-1	1	0	20	20/4=5
0	S_3	2	0	2	2/5	0	1	10	10/2=5
	Z_j	2	8	8	16	0	0	80	
	$C_j - Z_j$	3	0	-6	-16	0	0		

Tie for the smallest ratio indicates degeneracy.

Degeneracy could lead to a situation known as **cycling**, in which the simplex algorithm alternatives back and forth between the same non-optimal solutions, i.e., it puts a new variable in, then takes it out in the next tableau, puts it back in, and so on.

One simple way of dealing with the issue is to select either row (S_2 or S_3 in this case) arbitrary. If we are unlucky and cycling does occur, we simply go back and select the other row. However, the number of iterations required to arrive at the optimal solution can be minimized by adopting the following rule:

- Divide the coefficient of slack variables in the simplex table where degeneracy is detected by the corresponding positive numbers of the key column in the row, starting from left to right.
- The row which contains smallest ratio comparing from left to right column wise becomes the key row. For the above example:

|

Column		
Row	s_2	s_3
s_2	1	0
s_3	0	1

Divide the coefficients by the corresponding element of the key column, we obtain the following:

Column		
Row	s_2	s_3
s_2	$\frac{1}{4} = \frac{1}{4}$	$0/4 = 0$
s_3	$0/2 = 0$	$\frac{1}{2} = \frac{1}{2}$

Comparing the ratios from left to right column wise until they are not equal, the minimum ratio occurs for the second row (0). Therefore, s_3 is selected to leave the basis.



Note: When there is a tie between a slack and artificial variable to leave the basis, the preference shall be given to artificial variable to leave the basis and there is no need to apply the procedure for resolving such cases.

Two incoming variables / or Tie for entering variables/

In order to break this tie, the selection for the key column (entering variable) can be made arbitrary. However; the number of solution can be minimized by adopting the following rules:

1. **If there is a tie between two decision variables, then the selection can be made arbitrary.**
2. **If there is a tie between a decision variable and a slack (or surplus) variable, then select the decision variable to enter into basis first.**
3. **If there is a tie between slack or surplus variable, then selection can be made arbitrary.**

Example: If the equation is max Z:

C_j					Q
Basic V	x_1	x_2	s_1	s_3	
Z_j					
$C_j - Z_j$	5	2	5	0	



In such a case, X_1 is the entering variable

Multiple Optimum Solutions

This situation occurs when there can be infinite number of solutions possible for a given problem. This situation can be recognized in a simplex method when *one of the non-basic variables in the $C_j - Z_j$ row have a value of zero. This is to mean in the optimal solution when non basic variable (variable not in the basis) has a zero value in C-Z row, this is the indication of existence of multiple solution.* To obtain the other solution, we will make a non- basic variable with a zero C-Z value to enter in to the basis and solve in the same way as the one we did in the previous discussion.

Example: Maximization problem

C_j		3	2	0	0	Q
	BV	X_1	X_2	S_1	S_2	
2	X_2	3/2	1	1	0	6
0	S_2	1	0	1/2	1	3
	Z_j	3	2	2	0	12
	$C_j - Z_j$	0	0	-2	0	

$$\text{Max } Z = 3X_1 + 2X_2$$

$$X_1=0, X_2=6, S_2=3 \text{ and } \text{Max } Z=12 \text{ or: } X_1=3, X_2=3/2 \text{ and } \text{Max } Z=12$$

The $C_j - Z_j$ value of the non-basic variable (X_1) is 0. Thus, this shows the existence of alternative optimal solution. Can you identify more alternative optimal solutions?

CHAPTER THREE

3. TRANSPORTATION AND ASSIGNMENT MODEL

3.1. FORMULATING THE TRANSPORTATION MODEL

A transportation problem typically involves a set of sending locations, which are referred to as origins, and a set of receiving locations, which are referred to as destinations. In order to develop a model of a transportation problem, it is necessary to have the following information:

- a. Supply quantity (capacity) of each origin.
- b. Demand quantity of each destination.
- c. Unit transportation cost for each origin-destination route.

Assumptions

The transportation algorithm requires the assumption that:

- All goods be homogeneous, so that any origin is capable of supplying any destination, and
- Transportation costs are a direct linear function of the quantity shipped over any route.
- Though it will be modified later in our discussion, we shall add one additional requirement that the total quantity available is equal to the total demand.

Example

Let's consider an example. Harley's Sand and Gravel Pit has contracted to provide topsoil for three residential housing developments. Topsoil can be supplied from three different "farms" as follows:

<u>Farm</u>	<u>Weekly capacity (cubic yards)</u>
A	100

B	200
C	200

Demand for the topsoil generated by the construction projects is:

<u>Project</u>	<u>Weekly demand(cubic yards)</u>
1	50
2	150
3	300

The manager of the sand and gravel pit has estimated the cost per cubic yard to ship over each of the possible routes:

	<u>Cost per cubic yard to</u>		
<u>From</u>	<u>Project #1</u>	<u>Project #2</u>	<u>Project #3</u>
Farm A	Birr 4	Birr 2	Birr 8
Farm B	5	1	9
Farm C	7	6	3

This constitutes the information needed to solve the problem.

The next step is to arrange the information into a transportation table. This is shown in the following table.

Transportation table for Harley's sand and gravel

To:				
	From A	From B	From C	
From A	4	2	8	100
From B	5	1	9	200
From C	7	6	3	200



Note: It is possible to write this transportation problem in the form of LPM.

Nine (3x3) decision variables are there and the objective function is minimization.

LPM for this problem is:

$$Z \text{ min: } 4x_{11} + 2x_{12} + 8x_{13} + 5x_{21} + x_{22} + 9x_{23} + 7x_{31} + 6x_{32} + 3x_{33}$$

Subject to:

$$x_{11} + x_{12} + x_{13} \leq 100$$

$$x_{21} + x_{22} + x_{23} \leq 200$$

$$x_{31} + x_{32} + x_{33} \leq 200$$



Capacity/ Source constraint

$$x_{11} + x_{21} + x_{31} \leq 50$$

$$x_{12} + x_{22} + x_{32} \leq 150$$

$$x_{13} + x_{23} + x_{33} \leq 300$$



Demand/Destination constraint

This can be rewritten as:

$$Z_{\min}: 4x_{11} + 2x_{12} + 8x_{13} + 5x_{21} + x_{22} + 9x_{23} + 7x_{31} + 6x_{32} + 3x_{33}$$

Subject to:

$$x_{11} + x_{12} + x_{13} \leq 100$$

$$x_{21} + x_{22} + x_{23} \leq 200$$

$$x_{31} + x_{32} + x_{33} \leq 200$$

$$x_{11} + x_{21} + x_{31} \leq 50$$

$$x_{12} + x_{22} + x_{32} \leq 150$$

$$x_{13} + x_{23} + x_{33} \leq 300$$

$$x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}, x_{31}, x_{32}, x_{33} \geq 0$$

3.1.2. FINDING AN INITIAL FEASIBLE SOLUTION

The starting point of the transportation method is a feasible solution. For an assignment to be feasible, two conditions must be fulfilled:

- A feasible solution is one in which assignments are made in such a way that all supply and demand requirements are satisfied.
- The number of nonzero (occupied) cells should equal one less than the sum of the number of rows and the number of columns in a transportation table. In the case of a table with 3 rows and 3 columns, the number of occupied cells should be $3 + 3 - 1 = 5$ in order to be able to use the transportation algorithm. Sometimes, fewer occupied or completed cells appear in a solution. When that happens, the solution is referred to as a **degenerated solution**; such a solution requires modification in order to be able to determine if it is optimal. This topic will be covered later on.

A number of different approaches can be used to find an initial feasible solution. Three of these are described here:

- The northwest-corner method.
- An intuitive approach/Least cost method

3.1.2.1. The Northwest-Corner Method

The northwest corner method is a systematic approach for developing an initial feasible solution. Its chief advantages are that *it is simple to use and easy to understand*. Its chief drawback is that *it does not take transportation costs into account*. Consequently, such a solution may require much additional effort to obtain the optimal solution.

The northwest corner method gets its name because the starting point for the allocation process is the upper left-hand (Northwest) corner of the transportation table. For the Harley problem, this would be the cell that represents the route from Farm A to Project #1. The following set of principles guides the allocation:

- i. Begin with the upper left-hand cell, and allocate as many units as possible to that cell. This will be the smaller of the row supply and the column demand. Adjust the row and column quantities to reflect the allocation.
- ii. Remain in a row or column until its supply or demand is completely exhausted or satisfied, allocating the maximum number of units to each cell in turn, until all supply has been allocated (and all demand has been satisfied because we assume total supply and demand are equal).

Initial Feasible Solution for Harley using Northwest-corner method

To:				
	Project #1	Project #2	Project #3	Supply
From:				
Farm A	4 50 (first)	2 50 (second)	8	100
Farm B	5	1 100 (third)	9 100 (fourth)	200
Farm C	7	6	3 200 (last)	200
Demand	50	150	300	500

The total cost is found by multiplying the quantities in “completed” (i.e. nonempty) cells by the cell’s unit cost and, then, summing those amounts. Thus:

$$\text{Total cost} = 50(4) + 50(2) + 100(1) + 100(9) + 200(3) = \$1900$$

As noted earlier, the main drawback of the northwest-corner method is that it does not consider cell (route) costs in making the allocation. Consequently, if this allocation is optimal, that can be attributed **to chance** rather than the method used.

3.1.2.2. The Intuitive Approach

This approach, also known as the minimum-cost method, uses lowest cell cost as the basis for selecting routes. The procedure is as follows:

- i. Identify the cell that has the lowest unit cost. If there is a tie, select one arbitrarily. Allocate a quantity to this cell that is equal to the lower of the available supply for the row and the demand for the column.
- ii. Cross out the cells in the row or column that has been exhausted (or both, if both have been exhausted), and adjust the remaining row or column total accordingly.
- iii. Identify the cell with the lowest cost from the remaining cells. Allocate a quantity to this cell that is equal to the lower of the available supply of the row and the demand for the column.
- iv. Repeat steps (ii) and (iii) until all supply and demand have been exhausted.

The initial feasible solution for the Harley problem completed using the above steps is shown below.

Initial Feasible Solution for the Harley problem using the Intuitive approach

To:	Project #1	Project #2	Project #3	Supply
From:				
Farm A	50 ⁴	²	50 ⁸	100
Farm B	⁵	150 ¹	50 ⁹	200

		7	6	3	
Farm C			200		200
Demand	50	150	300	500	

We can easily verify that this is a feasible solution by checking to see that the row and column totals of the assigned cell quantities equal the supply and demand totals for the rows and columns. Now let us compute the total cost of this solution and compare it to that of the northwest corner solution.

$$\text{Total cost} = 50(4) + 50(8) + 150(1) + 50(9) + 200(3) = \$1800$$

Compared to the plan generated using the Northwest-corner method, this one has a total cost that is \$100 less. This is due to the fact that the previous one did not involve the use of cost information in allocating units.

3.1.2.3. Vogel's Approximation Method (VAM)

The third method for determining an initial solution, Vogel's Approximation Method (also called VAM), is based on the concept of **penalty cost or regret**. If a decision maker incorrectly chooses from several alternative courses of action, a penalty may be suffered (and the decision maker may regret the decision that was made). In transportation problem, the courses of action are the alternative routes and a wrong decision is allocating to a cell that does not contain the lowest cost.

This method is preferred over the other two methods because the initial feasible solution obtained with VAM is either optimal or very close to the optimal. With VAM the basis of allocation is unit cost penalty i.e. that column or row which has the **highest unit cost penalty** (difference between the **highest and the next highest cost**) is selected first for allocation and the subsequent allocations in cells are also done keeping in view the highest unit cost penalty.

Steps in VAM

1. Construct the cost, requirement, and availability matrix i.e. cost matrix with column and row information.
2. Compute a penalty for each row and column in the transportation table. The penalty is merely the **difference between the highest cost and the next highest cost element** in that particular row or column.
3. Identify the row and column with the **largest penalty**. In this identified row (column), choose the cell which has the **smallest cost** and allocate the maximum possible quantity to this cell. Delete the row (column) in which capacity (demand) is exhausted. When there is a tie for penalty, select one arbitrarily. After allocation, cross that row or column and disregard it from further consideration.
4. Repeat steps 1 to 3 for the reduced table until the entire capabilities are used to fill the requirement at different warehouses.
5. From step 4 we will get initial feasible solution. Now for initial feasible solution find the total cost.

The solution for our problem is as follows:

To: From:					Penalty		
					Hc	NHC	Hc- NHC
	4	2	8				
Farm A				100			
	5	1	9		8	4	4
Farm B		150		200			
		1					
	7	6	3				
Farm C				200			
					9	5	4
							selected

7	6	1	
HC	7	6	9
NHC	5	2	8
Penalty	2	4	1
HC: Highest cost			
NHC: Next Highest cost			

First allocation

Table 2

Penalty					Hc	NHC	Hc- NHC
To:							
From:							
	4	2		8			
	5	1		9			
		150					
	7	6		3	8	4	4
Farm C			200	200			2
	9	5	4				

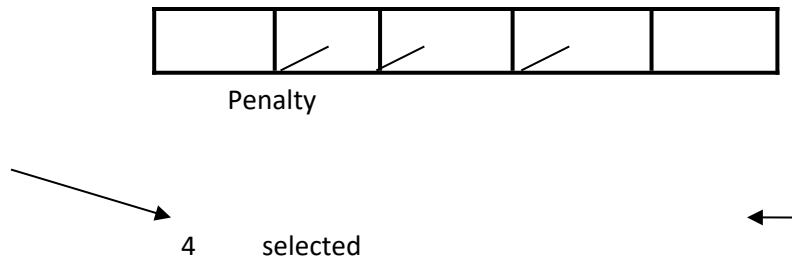
7 3 ← 4 selected



Penalty 2 - 1 Second allocation

Table 3

To:					
From:					
	4	2		8	3
Farm A	50				100 50
	5	1		9	
Farm B		150			200 50
	7	6		3	
Farm C			20		200



4

Penalty 1 - 1

Table 4

Since there is no penalty for the remaining cells, we allocate for these cells according to their cost.

To: From:					
	4	2	8		4
Farm A	50		50	100	
Farm B	5	1	50	9	5
		15			
Farm C	7	6	3	200	
			200	200	
					Fourth allocation

Fifth allocation

Therefore the final allocation will be:

To: From:	Plant 1	Plant 2	Plant 3	Plant 4	Plant 5
Farm A	4 50	2	8 50		100
Farm B	5	1 15	50		200
Farm C	7	6	3 200		200
	50	150	200		

4

5

Total cost: $1 \times 150 + 3 \times 200 + 4 \times 50 + 8 \times 50 + 9 \times 50 = \text{Birr } 1800$

4.1.3. Evaluating a Solution for Optimality

The test for optimality for a feasible solution involves a cost evaluation of *empty cells* (i.e., routes to which no units have been allocated) to see if an improved solution is possible. We shall consider two methods for cell evaluation:

- The Stepping-stone method
- The MODI method

The stepping-stone method involves a good deal of more effort than the **MODI** method. However, *it provides an intuitive understanding of the evaluation process*. Moreover, when a solution is not optimal, the distribution plan must be revised by reallocating units into and out of various cells, and **only the stepping-stone method can be used for the reallocation**.

3.1.3.1. The Stepping-stone method

The Stepping-stone method involves tracing a series of closed paths in the transportation table, using one such path for each empty cell. The path represents a shift of one unit into an empty cell, and it enables the manager or analyst to answer a “what-if” question: *What impact on total cost would there be if one unit were shifted into an unused route? The result is a cost change per unit shifted into a cell. If the shift would result in a cost savings, the stepping-stone path also can be used to determine the maximum number of units that can be shifted into the empty cell*, as well as modifications to other completed cells needed to compensate for the shift into the previously unused cell.

Reconsider the initial feasible solution we found using the northwest-corner method. Only the unoccupied cells need to be evaluated because the question at this point is not how many units to allocate to a particular route but only if converting a cell from zero units to nonzero (a positive integer) would decrease or increase total costs. The **unoccupied cells** are A-3, B-1, C-1, and C-2. They must be evaluated one at a time, but in no particular order.

Rules for tracing Stepping-stone paths:

- i. **All unoccupied cells must be evaluated.** Evaluate cells one at a time.
- ii. Except for the cell being evaluated, only add or subtract in occupied cells. (It is permissible to skip over unoccupied cells to find an occupied cell from which the path can continue.)

- iii. A path will consist of only horizontal and vertical moves, starting and ending with the empty cell that is being evaluated.
- iv. Alter + and – signs, beginning with a + sign in the cell being evaluated.



Note that it is not necessary to actually alter the quantities in the various cells to reflect the one-unit change, the + and – signs suffice.

The general implication of the plus and minus signs is that cells with + signs mean one unit would be added, cells with a – sign indicate one unit would be subtracted. The net impact of such a one-unit shift can be determined by adding the cell costs with signs attached and noting the resulting value. Thus, for cell B-1, we have a net change of +2 (i.e., $5+2-4-1$) which means that for each unit shifted into cell B-1, the total cost would increase by \$2.

Computed in similar way, the evaluations of cells C-1, A-3, and C-2 result in +10, -2, and +11 respectively. The negative value for cell A-3 indicates an improved solution is possible: For each unit we can shift into that cell, the total cost will decrease by \$2. The following table shows how empty cell C-1 can be evaluated using the **Stepping Stone Method**.

Evaluation path for cell C-1

To:	Project #1	Project #2	Project #3	Supply
From:	Project #1	Project #2	Project #3	Supply
Farm A	50 —	50 ← +	8 2	100
Farm B	5 ↓	100 ↑	9 100 + ↑	200
Farm C	7 +	6 —	3 - 200	200
Demand	50	150	300	500

3.1.3.2. The MODI method (*Occupied Cell Evaluation*)

The MODI (Modified Distribution) method of evaluating a transportation solution for optimality involves the use of index numbers that are established for the rows and columns. These are based on the unit costs of the **occupied cells**. **The index numbers can be used to obtain the cell evaluations for empty cells without the use of stepping-stone paths.**

There is one index number for each column and one for each row. These can be conveniently displayed along the left and upper edges of a matrix. The index numbers are determined in such a way that for any occupied cell, the sum of the row index and the column index equal the cell's unit transportation cost:

Row index	+	Column index	= Cell cost
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The index numbers are determined sequentially in a manner dictated by the position of occupied cells. **The process always begins by assigning a value of zero as the index number of row 1.**

The method will be illustrated by developing index numbers for the initial feasible solution for the Harley problem generated by the northwest-corner method. We begin assigning a value of zero for row 1. Once a row index has been established, it will enable us to compute column index numbers for all occupied cells in that row. Similarly, once a column index number has been determined, index numbers for all rows corresponding to occupied cells in that column can be determined. The complete set of row and column index numbers is shown in the following table.

Cell evaluations for Northwest Corner solution for the Harley Problem using the MODI method

			k_1	k_2	k_3	
			+4	+2	+10	
	To:		Project #1	Project #2	Project #3	Supply
	From:					
r_1	0	Farm A	50 4	50 2	-2 8	100
r_2	-1	Farm B	5 +2	100 1	9 100	200
r_3	-7	Farm C	7 +10	6 +11	3 200	200

Demand	50	150	300	500

The cell evaluations (improvement potentials) for each of the unoccupied cells are determined using the relationship:

Cell evaluation = Cell cost	–	Row index	–	Column index
e_{ij}	=	c_{ij}	–	$r_i - k_j$

For example, the cell evaluations for A-3 is $8 - 0 - 10 = -2$. Similarly, the evaluation for B-1 is +2, for C-1, +10, and for C-2, it is +11. Note that they agree with the values we computed earlier using the stepping-stone method.

When cell evaluations are positive or zero, an optimal solution has been found. If one or more is negative, the cell with the largest negative should be brought into solution because that route has the largest potential for improvement per unit. In this case, we found that cell A-3 had an evaluation of –2, which represented an improvement potential of \$2 per unit. Hence, an improved solution is possible.

3.1.3.3. DEVELOPING AN IMPROVED SOLUTION

Developing an improved solution to a transportation problem requires focusing on the unoccupied cell that has the largest negative cell evaluation. Improving the solution involves reallocating quantities in the transportation table. More specifically, we want to take advantage of the improvement potential of cell A-3 by transferring as many units as possible into that cell. The stepping-stone path for that cell is necessary for determining how many units can be reallocated while retaining the balance of supply and demand for the table. The stepping-stone path also reveals which cells must have quantity changes and both the magnitude and direction of changes. The + signs in the path indicate units to be added, the – signs indicate units to be subtracted. The limit on subtraction is the smallest quantity in a negative position along the cell path. With each iteration (new solution), it is necessary to evaluate the empty cells to see if further improvement is possible. This requires use of either the MODI or the Stepping-stone method. Both will yield the same values.

After reallocating the units using the stepping stone method, the empty cells will be A-2, B-1, C-1 and C-2. Suppose we use the MODI method for evaluation. After assigning all the row and column indices, we find the cell evaluations to be as follows:

Unoccupied cell $C_{ij}-U_i-V_j= E_{ij}$

Cell A-2: $2 - 0 - 0 = +2$

Cell B-1: $5 - 1 - 4 = 0$

Cell C-1: $7 - (-5) - 4 = +8$

Cell C-2: $6 - (-5) - 0 = +11$

Because none of these numbers is negative, this is an optimal solution. You may recall that this was the same solution obtained using the intuitive method for the initial feasible solution. At that point, it was determined that the total cost for the distribution plan was \$1800.

Optimal Solution for the Harley problem

 V_j (Column indexes)

U_i (Row indexes)			+4		0		+8			
		To:	Project #1		Project #2		Project #3		Supply	
		From:								
		0	Farm A	50	4	+2	2	50	8	100
		+1	Farm B	0	5		1	150	9	200
			+8		+11					
		-5	Farm C		7		6	200	3	200
			Demand	50		150		300		500

3.1.6. SPECIAL ISSUES

There are a number of special issues in relation to the transportation model. They are:

- i. Determining if there are alternate optimal solutions.
- ii. Defining and handling degeneracy (too few occupied cells to permit evaluation of a solution).
- iii. Avoiding unacceptable or prohibited route assignments.
- iv. Dealing with problems in which supply and demand are not equal.
- v. Solving maximization problems.

Alternate Optimal Solutions

Sometimes, transportation problems have multiple optimal solutions. In such instances, it can be useful for a manager to be aware of alternate solutions, because this gives the manager an option of bringing non-quantitative considerations into the decision.

In the case of the transportation problem, the existence of an *alternate solution is signaled by an empty cell's evaluation equal to zero*. In fact, you may have noted that cell B-1 had an evaluation equal to zero in the final solution of the Harley problem. We can find out what that alternate solution is by reallocating the maximum number of units possible around the stepping-stone path for that cell. After the cell evaluation of, and reallocating units to cell B-1, we find the following table to be an alternate optimal solution.

Alternate optimal solution for the Harley problem

To:				
	Project #1	Project #2	Project #3	Supply
From:				
Farm A	4	2	8	100
Farm B	5	1	9	200
Farm C	7	6	3	200

Demand	50	150	300	500

Degeneracy

A solution is degenerate if the number of occupied cells is less than the number of rows (m) plus the number of columns (n) minus one (1) or $(m+n-1)$. i.e., there are too few occupied cells to enable all the empty cells to be evaluated. In the case of the stepping-stone method, this means that there will be at least one empty cell for which an evaluation path cannot be constructed. *For the MODI method, it means that it will be impossible to determine all of the row and column index numbers.* Obviously, some modification has to be made to determine if such a degenerate solution is optimal. The modification is to **treat some of the empty cells as occupied cells**. This is accomplished **by placing a delta (Δ)** in one of the empty cells. The delta represents an extremely small quantity (e.g., 0.001 unit); it is so small that supply and demand for the row and column involved will be unaffected even without modifying other quantities in the row or column, and so small that total cost will not change.

The purpose of the delta is to enable evaluation of the remaining empty cells. The choice of location for the delta can be somewhat **tricky**: some empty cells may be unsuitable if they do not enable evaluations of the remaining empty cells. Moreover, the **delta cannot be placed in a cell which later turns out to be in a negative position of a cell path involved in reallocation** because delta will be the “smallest quantity in a negative position” and shifting that minute quantity around the cell path will leave the solution virtually unchanged. Consequently, a certain amount of trial and error may be necessary before a satisfactory location can be identified for delta.

The technique can be demonstrated for the degenerate alternate solution of the Harley problem. Suppose that after some experimentation, cell A-1 has been selected for the location of delta. The resulting index numbers generated using MODI and the improvement potential for empty cells based on delta in cell A-1 are shown in the following table. This confirms that the solution is optimal

Harley Alternate Solution Modified for Degeneracy

		+4	0	+8	
To:		Project #1	Project #2	Project #3	Supply
From:					
		4	2	8	
		Δ	+	100	
0	Farm A				100
		5	1	9	
				0	

+1	Farm B	50	150		200
-5	Farm C	7 (+)	6 (+11)	3 (200)	200
	Demand	50	150	300	500

Unacceptable Routes

In some cases, an origin-destination combination may be unacceptable. This may be due to weather factors, equipment breakdowns, labor problems, or skill requirements that either prohibit, or make undesirable, certain combinations (routes).

Suppose that in the Harley problem route A-3 was suddenly unavailable because of recent flooding. In order to prevent that route from appearing in the final solution (as it originally did), the manager could assign a unit cost to that cell that was large enough to make that route uneconomical and, hence, prohibit its occurrence. **One rule of thumb would be to assign a cost that is 10 times the largest cost in the table** (or a very big $+M$). Then, this revised problem could be solved using either of the methods we have discussed earlier. Note that the prohibited route may appear in a non-optimal solution, but it will be eliminated by the time the optimal solution is reached.

Alternatively, to handle unacceptable routes in a minimization problem, infeasible arcs must carry an extremely high cost, denoted M , to keep them out of the solution. Thus, if we have a route (arc) from an origin to a destination that for some reason cannot be used, we simply assign this arc a cost per unit of M , and it will not enter the solution. Unacceptable arcs would be assigned a profit per unit of $-M$ in a maximization problem.

Dear learner, would you find the optimal solution if route A-3 is not available? How do you compare the cost with the original cost?

Unequal Supply and Demand

Up to this point, examples have involved cases in which supply and demand were equal. As you might guess, there are situations in which the two are not equal. When such a situation is encountered, it is necessary to **modify the original problem so that supply and demand are equal. This is accomplished by adding either a dummy column or a dummy row**; a dummy row is added if supply is less than demand and a dummy column is added if demand is less than supply. The dummy is assigned unit costs of zero for

each cell, and it is given a supply (if a row) or a demand (if a column) equal to the difference between supply and demand. Quantities in dummy routes in the optimal solution are not shipped. Rather, they serve to indicate which supplier will hold the excess supply, and how much, or which destination will not receive its total demand, and how much it will be short.

Let's consider an example. Suppose that Farm C in the Harley problem has experienced an equipment breakdown, and it will be able to supply only 120 cubic yards of topsoil for a period of time. Therefore, total supply will be 80 units less than total demand. This will require adding a dummy origin with a supply of 80 units. The final solution is shown in the following table. We interpret the solution indicating that Project #3 will be short by 80 units per week until the equipment is repaired. Note, though, that this analysis has considered only transportation costs, and that other factors, such as shortage costs or schedules of the projects, may dictate some other course of action.

If the intuitive approach (Least Cost Method) is used to obtain the initial feasible solution when a dummy is involved, make assignments to the dummy last. Hence, *begin by assigning units to the cell with the lowest nonzero cost, then the next lowest nonzero cost, and so on.* For the Harley problem this would mean that units would be assigned first to cell B-2 because its cost of \$1 is the lowest nonzero cell cost.

Solution using the Dummy Origin

To:				
	Project #1	Project #2	Project #3	Supply
From:				
Farm A	50 4	2	50 8	100
Farm B	5	150 1	50 9	200
Farm C	7	6	120 3	120
Dummy	0	0	80 0	80

Demand	50	150	300	500
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Maximization

Some transportation type problems concern profits or revenues rather than costs. In such cases, the objective is to maximize rather than to minimize. Such problems can be handled by adding one additional step at the start: **Identify the cell with the largest profit and subtract all the other cell profits from that value. Then replace the cell profits with the resulting values.** These values reflect the opportunity costs that would be incurred by using routes with unit profits that are less than the largest unit profit. Replace the original unit profits with these opportunity cost solution. This will be identical to maximizing the total profit. For example, suppose in the Harley problem, the cell values had been unit profits instead of unit costs. Cell B-3 had the largest dollar value: \$9. Hence, each cell's dollar amount would be subtracted from 9. For cell A-1, the resulting opportunity cost would have been $9 - 4 = 5$ and so on. Cell B-3 would have an opportunity cost of 0 making it the most desirable route.

The remainder of the steps for developing an initial feasible solution, evaluation of empty cells, and reallocation are identical to those used for cost minimization. When the optimal distribution plan has been identified, use the original cell values (i.e., profits) to compute the total profit for that plan.

Alternatively, the transportation method also can be used to solve maximization problems by only modification necessary involves the selection of an incoming cell. Instead of picking the cell with the smallest or most negative e_{ij} value, we pick that cell for which e_{ij} is largest. That is, we pick the cell that will cause the largest increase per unit in the objective function. If $e_{ij} \leq 0$ for all unoccupied cells, we stop; the maximization solution has been reached.

3.2. ASSIGNMENT MODEL

There are many situations where the assignment of people or machines and so on, may be called for. Assignment of workers to machines, clerks to various counters, salesmen to different sales areas, service crews to different districts, are typical examples of these. *The assignment is a problem because people posses varying abilities for performing different jobs and, therefore, the costs of performing the jobs by different people are different.* Obviously, if all persons could do a job in the same time or at the same cost then it would not matter who of them is assigned the job. *Thus, in assignment problem, the question is how should the assignment be made in order that the total cost involved is minimized* (or the total value is maximized when pay-offs are given in terms of, say, profits).

A typical assignment problem follows:

Example

A production supervisor is considering how he should assign the four jobs that are performed, to four of the workers working under him. He want to assign the jobs to the workers such that the aggregate time to perform the job in the least. Based on the previous experience, he has the information on the time taken by the four workers in performing these jobs, as given in below

Worker	Job(in Hrs to perform)			
	A	B	C	D
1	45	40	51	67
2	57	42	63	55
3	49	52	48	64
4	41	45	60	55

Example

A computer centre has got four Expert Programmers. Centre needs four application programmers to be developed. The head of dept., estimate the **computer required by the respective experts** to develop the application programs as follows. Make appropriate selection of experts.

Program Expert	Computers required			
	A	B	C	D
1	120	100	80	90
2	80	90	110	70
3	110	140	120	100
4	90	90	80	90

3.2.1. THE ASSIGNMENT PROBLEM (A.P)

Definition:

1. The Assignment Problem (A.P) is a special case of Transportation Problem (T.P) under the condition that the number of origins is equal to the number of destinations,

i.e. $m=n$

Hence assignment is made on the basis of 1:1

- Number of jobs equal to number of machines or persons.
- Each man or machine is loaded with one and only one job.
- Each man or machine is independently capable of handling any of the jobs being presented.

- Loading criteria must be clearly specified such as “ minimizing operating time or “ maximizing profit”, or “ minimizing production cost” or minimizing production cycle time e.t.c

2. The Assignment Problem (A.P.) is a special case of Transportation Problem (T.P.) in which the number of sources and destinations are the same, and the objective is to assign the given job (task) to most appropriate machine (person) so as to optimize the objective like minimizing cost.

- **Cost Vector (C_{ij})**

Cost vector (C_{ij}) is the cost of assigning i^{th} job (Task) to j^{th} machine (person),

- **Assignment Vector (X_{ij}) is defined as follows**

$X_{ij} = 0$; If i^{th} job (Task) is not assigned by j^{th} machine (person) or

$X_{ij} = 1$; If i^{th} job (Task) is assigned to j^{th} machine (person).

- **Cost Matrix (C_{ij})**

Assignment problem is stated in the form of ($n \times n$) matrix. This is called cost matrix. This is illustrated as given below.

C_{ij} = cost of assigning i^{th} job to j^{th} machine

(Symbols j = Job (Task) M = Machine (person))

General Assignment Problem Table

J \ M	1	2	3	...	k	...	N
1	C_{11}	C_{12}	C_{13}	...	C_{1k}	...	C_{1n}
2	C_{21}	C_{22}	C_{23}	...	C_{2k}	...	C_{2n}
3	C_{31}	C_{32}	C_{33}	...	C_{3k}	...	C_{3n}
...	...	...	...	...	...	...	...
K	C_{k1}	C_{k2}	C_{k3}	...	C_{kk}	...	C_{kn}
...	...	...	...	...	...	...	...
N	C_{n1}	C_{n2}	C_{n3}	...	C_{nk}	...	C_{nn}

- **Effective Matrix**

A cost Matrix in A.P. is called an “Effectiveness Matrix” when there is at least one zero in each row and column. Following is an example of Effectiveness Matrix.

	1	2	3	4
1	0	1	3	4
2	5	0	6	0
3	7	8	0	9
4	0	4	3	8

3. Mathematical Modeling of an A.P.

Let there be 'n' jobs in a manufacturing plant. Let there be 'n' machines to process the product. A job i (i = 1, 2... n) when processed in a machine j (j = 1, 2... n), it is assumed to incur a cost of Cij. The assignment is made in such a way that one job is associated with one machine (see assumption). Hence we have the following:

$$\sum_{i=1}^n \sum_{j=1}^n x_{ij} = 1 \quad \sum_{i=1}^n \sum_{j=1}^n x_{ij} = 1$$

Where: - x_{ij} is the assignment- vector discussed earlier

C_{ij} is the cost vector associated with the assignment vector x_{ij} .

Total assignment cost (Z) is, therefore, given by the following:

$$Z = \sum_{i=1}^n \sum_{j=1}^n C_{ij} * x_{ij}$$

Thus the mathematical formulation of assignment Problem is given as follows:

Minimize

$$Z = \sum_{i=1}^n \sum_{j=1}^n C_{ij} * x_{ij}$$

Subject to

$$\sum_{j=1}^n X_{ij} = 1$$

$$i = 1, 2, 3, \dots, n$$

$$\sum_{i=1}^n X_{ij} = 1$$

$$j = 1, 2, 3, \dots, n$$

This special structure of A.P. makes solution much easy compared to the conventional T.P.

Remark:

1. It may be noted that assignment problem is a variation of transportation problem with two characteristics (i) the cost matrix is square matrix, and (ii) the optimum solution for the problem would always be such that there would be only one assignment in a given row or column of the cost matrix.
2. In assignment problem if a constant is added or subtracted from every element of any row or column in the given cost matrix then an assignment that minimizes the total cost in one matrix also minimizes the total cost in the other matrix.

3.2.3. HUNGARIAN ASSIGNMENT METHOD (HAM)

A method, designed specially to handle the assignment problems in an efficient way, called the Hungarian Assignment Method, is available, which is *based on the concept of opportunity cost*. It is shown in figure 6.1 and discussed here. For a typical balanced assignment problem involving a certain number of persons and an equal number of jobs, and with an objective function of the minimization type, the method is applied as listed in the following steps:

Step 1. Locate the ***smallest cost element in each row*** of the cost table. Now subtract this smallest from each element in that row. As a result, there shall be at least one zero in each row of this new table, called the Reduced Cost Table (Row Reduction).

Step 2. In the reduced cost table obtained, ***consider each column and locate the smallest element in it. Subtract the smallest value from every other entry in the column.*** As a consequence of

this action, there would be at least one zero in each of the rows and columns of the second reduced cost table (Column Reduction).

Step 3. Draw the minimum number of horizontal and vertical lines (not diagonal ones) that are required to cover the entire 'zero' elements. If the number of lines drawn is equal to n (the number of rows/columns) the solution is optimal, and proceeds to step 6. If the number of lines drawn is smallest than n , go to step 4.

Step 4. Select the smallest uncovered (by the lines) cost element. *Subtract this element from all uncovered elements including it-self and add this element to each value located at the intersection of any lines.* The cost elements through which only one line passes remain unaltered.

Step 5. Repeat steps 3 and 4 until an optimal solution is obtained.

Step 6. Given the optimal solution, ***make the job assignments as indicated by the 'zero' elements.*** This done as follows:

- Locate a **row** which only 'zero' element. Assign the job corresponding to this element to its corresponding person. *Cross out the zero's, if any, in the column corresponding to the element*, which is indicative of the fact that the particular job and person are no more available.
- Repeat (a) for each of such rows which contain only one zero. Similarly, perform the same operation in respect of each column containing only one 'zero' element, crossing out the zero(s), if any, in the row in which the element lies.
- If there is no row or column with only a single 'zero' element left, then select a row/column arbitrarily and choose one of the jobs (or persons) and make the assignment. Now cross the remaining zeros in the column and row in respect of which the assignment is made.
- Repeat steps (a) through (c) until all assignments are made.
- Determine the total cost with reference to the original cost table.

Example

Solve the assignment problem given in Illustrative Example 1 for optimal solution using HAM. The information is reproduced in the following table

Time Taken (in minutes) by 4 workers

Worker	Job			
	A	B	C	D
1	45	40	51	67
2	57	42	63	55

3	49	52	48	64
4	41	45	60	55

The solution to this problem is given here in a step-wise manner.

Step 1: The minimum value of each row is subtracted from all elements in the row. It is shown in the reduced cost table, also called opportunity cost table, given as follows.

Reduced Cost Table 1

Worker	Job			
	A	B	C	D
1	5	0	11	27
2	15	0	21	13
3	1	4	0	16
4	0	4	19	14

Step 2: For each column of this table, the minimum value is subtracted from all the other values. Obviously, the columns that contain a zero would remain unaffected by this operation. Here only the fourth column values would change. The table below shows this.

Reduced Cost Table 2

Worker	Job			
	A	B	C	D
1	5	0	11	14
2	15	0	21	0
3	1	4	0	3
4	0	4	19	1

Step 3: Draw the minimum number of lines covering all zeros. As a general rule, we should first cover those rows/columns which contain larger number of zeros. The above table is reproduced in the next table and the lines are drawn.

Reduced Cost Table3

Worker	Job			
	A	B	C	D
1	5	0	11	14
2	15	0	21	0
3	1	4	0	3
4	0	4	19	1

Step 4: Since the number of lines drawn is equal to 4(=n), the optimal solution is obtained. The assignments are made after scanning the rows and columns for unit zeros. Assignments made are shown with squares, as shown in the following table.

Assignment of Jobs

Worker	Job			
	A	B	C	D
1	5	0 ☐	11	14
2	15	0 X	21	0 ☐
3	1	4	0 ☐	3
4	0 ☐	4	19	1

Assignments are made in the following order. Rows 1, 3, and 4 contain only one zero each. So assign 1-B, 3-C and 4-A. Since worker 1 has been assigned job B, only worker 2 and job D are left for assignment. The final pattern of assignments is 1-B, 2-D, 3-C, and 4-A, involving a total time of $40+55+48+41=184$ minutes. This is the optimal solution to the problem-the same as obtained by enumeration and transportation methods.

Example

Using the following cost matrix, determine (a) optimal assignment, and (b) the cost of assignments.

Reduced Cost Table 1

Machinist	Job				
	1	2	3	4	5
A	10	3	3	2	8

B	9	7	8	2	7
C	7	5	6	2	4
D	3	5	8	2	4
E	9	10	9	6	10

Iteration 1: Obtain row reductions.

Reduced Cost Table 1

Machinist	Job				
	1	2	3	4	5
A	8	1	1	0	6
B	7	5	6	0	5
C	5	3	4	0	2
D	1	3	6	0	2
E	3	4	3	0	4

Iteration 2: Obtain column reductions and draw the minimum number of lines to cover all zeros.

Reduced Cost Table2

Machinist	Job				
	1	2	3	4	5
A	7	0	0	0	4
B	6	4	5	0	3
C	4	2	3	0	0
D	0	2	5	0	0
E	2	3	2	0	2

Since the number of lines covering all zeros is less than the number of columns/rows, we modify the Table 6.13. The least of the uncovered cell values is 2. This value would be subtracted from each of the uncovered values and added to each value lying at the intersection of lines (corresponding to cells A-4, D-4, A-5 and D-5). Accordingly, the new table would appear as shown as follows.

Iteration 3

Machinist	Job				
	1	2	3	4	5
A	7-----	0----- ☐	0 X-----	2-----	6-----
B	4-----	2-----	3-----	0----- ☐	3-----
C	2-----	0 X-----	1-----	0 X-----	0----- ☐
D	0----- ☐	2-----	5-----	2-----	2-----
E	0 X-----	1-----	0----- ☐	0 X-----	2-----

The optimal assignments can be made as the least number of lines covering all zeros in Table 6.14 equals 5. Considering rows and columns, the assignments can be made in the following order:

- Select the second row. Assign machinist B to job 4. Cross out zeros at cells C-4 and E-4.
- Consider row 4, Assign machinist D to job 1. Cancel the zero at cell E-1.
- Since there is a single zero in the row, put machinist E to job 3 and cross out the zero at A-3.
- There being only a single left in each of the first and third rows, we assign job 2 to machinist A and job 5 to C.

The total cost associated with the optimal machinist-job assignment pattern A-2, B-4, C-5, D-1 and E-3 is $3+2+4+3+9 = 21$

3.2.5. Special Issues

When we solve assignment problems, there are cases which are treated differently from the usual way.

Unbalanced Assignment Problems

The Hungarian Method of solving an assignment problem requires that the number of columns should be equal to the number of rows. They are equal, the problem is balanced problem, and when not, it is called an unbalanced problem. Thus, where there are 5 workers and 4 machines, or when there are 4 workers and 6 machines, for instance, we have unbalanced situations in which one-to-one match is not possible. In case the machines are in excess, the excess machine(s) would remain idle and so is the case when men are in excess- the number of excess people would get an assignment.

In such situations, dummy column(s)/row(s), whichever is smaller in number, are inserted with zeros as the cost elements. For example, when the given cost matrix is of the dimension 4×5 , a dummy row would

be included. In each column in respect of this row, a 'zero' would be placed. After this operation of introducing dummy columns/rows, the problem is solved in the usual manner.

Example:

A company has 4 machines to do 3 jobs. Each job can be assigned to one and only one machine. The cost of each job on each machine is given below. Determine the job assignments which will minimize the total cost.

Job	Machine			
	W	X	Y	Z
A	18	24	28	32
B	8	13	17	18
C	10	15	19	22

Constrained/Prohibited/ Assignment Problems

It happens sometimes that a worker cannot perform a certain job or is not to be assigned a particular job. To cope with this situation, the cost of performing that job by such person is taken to be extremely large (which is written as M). Then the solution to the assignment problem proceeds in the manner discussed earlier. The effect of assigning prohibitive cost to such person-job combinations is that they do not figure in the final solution.

Example:

You are given the information about the cost of performing different jobs by different persons. The job-person marking X indicates that the individual involved cannot perform the particular job. Using this information, state (i) the optimal assignment of jobs, and (ii) the cost of such assignment,

person	Job				
	J1	J2	J3	J4	J5
P1	27	18	X	20	21
P2	31	24	21	12	17
P3	20	17	20	X	16

P4	22	28	20	16	27
-----------	----	----	----	----	----

Solution: - Balancing the problem not assigning a high cost to the pairings P1-J3 and P3-J4, we have the cost given in the table below.

person	Job					
		J1	J2	J3	J4	J5
	P1	27	18	M	20	21
	P2	31	24	21	12	17
	P3	20	17	20	M	16
	P4	22	28	20	16	27
	P5 dummy	0	0	0	0	0

Now we can derive the reduced cost table as shown in table shown below. Note that the cells with prohibited assignments continue to be shown with the cost element M, since **M is defined to be extremely large so that subtraction or addition of value does not practically affect it.** To test optimality, lines are drawn to cover all zeros. Since the number of lines covering all zeros is less than n, we select the lowest uncovered cell, **which equals 4**. With this value, we can obtain the revised reduced cost table, shown in the final table.

Reduced Matrix

person	Job					
		J1	J2	J3	J4	J5
	P1	9	0	M	2	3
	P2	19	12	9	0	5
	P3	4	1	4	M	0
	P4	6	12	4	0	11
	P5 dummy	0	0	0	0	0

Reduced Matrix

person	Job					
		J1	J2	J3	J4	J5
	P1	9	0	M	2	3
	P2	19	12	9	0	5
	P3	4	1	4	M	0
	P4	6	12	4	0	11
	P5 dummy	0	0	0	0	0

Assignment of Job

person	Job								
		J1	J2	J3	J4	J5			
	P1	9	0	☐	M	2		3	
	P2	15	8		9	0	☐	1	
	P3	4	1		4	M		0	☐
	P4	2	8		0	☐	0	X	7
	P5 dummy	0	☐	0	X		0	X	

The number of lines covering zeros is equal to 5(=n), hence the optimal assignment can be made. The assignment is: P1-J2, P2-J4, P3-J5, P4-J3, while job J1 would remain unassigned. This assignment pattern would cost $18+12+16+20=66$ in the aggregate.

Unique Vs Multiple Optimal Solutions

In the process of making assignments, it was stated earlier that we select a row/column with only a single zero to make an assignment. However, a situation may wherein the various rows and columns, where assignment are yet to be done, have **all multiple zeros. In such cases, we get multiple optimal solutions to the given problem.** In any of the problems discussed so far, we have not experienced such a situation. Hence, each one of them has had a unique optimal solution. When a problem has a unique optimal solution, it means that no other solution to the problem exists which yields the same objective function value (cost, time, profit e. t. c) as the one obtained from the optimal solution derived. On the other hand in a problem with multiple optimal solutions, there exists more than one solution which all is optimal and equally attractive. Consider the following example.

Example:

Solve the following assignment problem and obtain the minimum cost at which all the jobs can be performed.

Worker	Job (cost in '00 Br.)				
	1	2	3	4	5
A	25	18	32	20	21
B	34	25	21	12	17
C	20	17	20	32	16
D	20	28	20	16	27

Solution: This problem is unbalanced since number of jobs is 5 while the number of workers is 4. We first balance it by introducing a dummy worker E, as shown in table below.

Balancing the Assignment Problem

Worker	Job (cost in '00 Br.)				
	1	2	3	4	5
A	25	18	32	20	21
B	34	25	21	12	17
C	20	17	20	32	16
D	20	28	20	16	27
E	0	0	0	0	0

Step 1: Obtain reduced cost values by subtracting the minimum value in each row from every cell in the row. This is given in Table below.

Reduced Cost 1

Worker	Job (cost in '00 Br.)				
	1	2	3	4	5
A	7	0	14	2	3

B	22	13	9	0	5
C	4	1	4	16	0
D	4	12	4	0	11
E	0	0	0	0	0

Since there is at least one zero in each row and column, we test it for optimality. Accordingly, lines are drawn. All zeros are covered by 4 lines, which is less than 5 (the order of the given matrix). Hence, we proceed to improve the solution. The least uncovered value is 4. Subtracting from every uncovered value and adding it to every value lying at the intersection of lines, we get the revised values as shown below.

Reduce cost 2 and Assignment

Worker	Job (cost in '00 Br.)				
	1	2	3	4	5
A	7	0 ☐	14	6	3
B	18	9	5	0 ☐	1
C	4	1	4	20	0 ☐
D	0 ☐	8	0	0 X	7
E	0 X	0 X	0 ☐	4	0 X

The solution given in the reduced cost 2 table is optimal since the number of lines covering zeros matches with the order of the matrix. We can, therefore, proceed to make assignments. To begin with, since each of the columns has multiple zeros, we cannot start making assignments considering columns and have, therefore, to look through rows. The first row has a single zero. Thus, we make assignment A-2 and cross out zero at E-2.

Further, the second and the third rows have one zero each. We make assignments B-4 and C-5, and cross out zeros at D-4 and E-5. Now, both the rows left two zeros each and so have both the columns. This indicates existence of multiple optimal solutions. To obtain the solutions, we select zeros arbitrarily and proceed as discussed below.

- Select the zero at D-1, make assignment and cross out zeros at D-3 and E-1 (as both, worker D and job 1, are not available any more). Next, assign worker E to job 3, corresponding to the only zero left. Evidently, selecting the zero at E-3 initially would have the effect of making same assignments.

- ii. Select the zero at D-3, make assignment and cross at D-1 and E-3. Next, make assignment at the only zero left at E-1. Obviously, selecting the zero at E-1 making assignment in the first place would lead to the same assignments.

To conclude, the problem has two optimal solutions as given below.

Solution 1			Solution 2		
Worker	Job	('00 Br.) Cost	Worker	Job	('00 Br.) cost
A	2	18	A	2	18
B	4	12	B	4	12
C	5	16	C	5	16
D	1	20	D	3	20
Job left	3		Job left	1	
	Total	66		Total	66

Maximization Case in Assignment Problem

In some situations, the assignment problem may call for maximization of profit, revenue, etc as the objective. For example, we may be faced with the problem of assignment of salesmen in different regions in which they can display different qualities in making sales (reflected in amounts of sales executed by them). Obviously, assignment would be made in such a way that the total expected revenue is maximized.

For dealing with a maximization problem, we first change it into an equivalent minimization problem. This is achieved by subtraction each of the elements of the given pay-off matrix from a constant (value) k . Thus, we may simply put a negative sign before each of the pay-off values (which is equivalent to subtracting each value from zero). *Usually, the largest value of all values in the given matrix is located and then each one of the values is subtract from it (the largest value is taken so as to avoid the appearance of negative signs). Then the problem is solved the same way as a minimization problem is.*

Example:

A company plans to assign 5 salesmen to 5 districts in which it operates. Estimates of sales revenue in thousands of birr for each salesman in different districts are given in the following table. In your opinion, what should be the placement of the salesmen if the objective is to maximize the expected sales revenue?

	District
--	----------

Salesman	D1	D2	D3	D4	D5
S1	40	46	48	36	48
S2	48	32	36	29	44
S3	49	35	41	38	45
S4	30	46	49	44	44
S5	37	41	48	43	47

Solution: Since it is a maximization problem, we would first subtract each of the entries in the table from the largest one, which equals 49 here. The resultant data are given in Table below.

Opportunity Loss Matrix

Salesman	District				
	D1	D2	D3	D4	D5
S1	9	3	1	13	1
S2	1	17	13	20	5
S3	0	14	8	11	4
S4	19	3	0	5	5
S5	12	8	1	6	2

Now, we will proceed as usual.

Step 1: Subtract *smallest* value in each row from every value in the row. The resulting values are given in table below.

Reduced cost Table 1

Salesman	District				
	D1	D2	D3	D4	D5
S1	8	2	0	12	0
S2	0	16	12	19	4
S3	0	14	8	11	4
S4	19	3	0	5	5
S5	11	7	0	5	1

Step 2: Subtract *smallest/minimum* value in each column in reduced cost table 1 from each value in the column. Test for optimality by drawing lines to cover zeros. These are shown in table below (in the Reduced cost Table 2)

Reduced cost Table 2

Salesman	District				
	D1	D2	D3	D4	D5
S1	8	0	0	7	0
S2	0	14	12	14	4
S3	0	12	8	6	4
S4	19	1	0	0	5
S5	11	5	0	0	1

Since the number of lines covering all zeros is fewer than n, we select uncovered cell value, which equals 4. With this, we can modify the table as given in the Reduced Cost Table 3.

Steps 4, 5, 6: Find improved solution. Test for optimality and make assignments.

Reduced cost Table 3

Salesman	District				
	D1	D2	D3	D4	D5
S1	12	0 ☐	0 X	7	0 X
S2	0 ☐	10	8	10	0 X
S3	0 X	8	4	2	0 ☐
S4	23	1	0 ☐	0 X	5
S5	15	5	0 X	0 ☐	1

More than one optimal assignment is possible in this case because of the existence of multiple zeros in different rows and columns. The possible assignments are:

S1-D2, S2-D5, S3-D1, S4-D3, S5-D4; or

S1-D2, S2-D1, S3-D5, S4-D3, S5-D5; or

S1-D2, S2-D1, S3-D1, S4-D4, S5-D3; or

S1-D2, S2-D1, S3-D5, S4-D4, S5-D

Each of these assignment patterns would lead to expected aggregated sales equal to 231 thousand birr.

CHAPTER THREE

3. Post optimality Analysis



Dear learner, in the previous chapters we discussed about how to solve LP problems using either graphic or simplex approach which is a base for this section. This unit deals with the other version of the simplex model, dual and the sensitivity of the optimal solution to changes in different parameters.

Objectives:

Up on completing this chapter, the learners would be able to:

- Explain what post optimality analysis is
- Describe the dual and primal forms of LP problems
- Conduct sensitivity analysis on different parameters

3.1. DUALITY

Section objective:

Up on completion of this sub section the learner will be able to:

- Define duality
- Describe the importance of duality
- Explain how to convert dual in to primal and the vice versa
- Formulate dual when the constraints are mixed
- Elucidate how to read the solution for primal from dual's solution and vice versa.
- Analyze the impact of addition of a new product on the decision

The term 'dual' in a general sense implies two or double. Every linear programming problem can have two forms. The original formulation of a problem is referred to as its **Primal form**. The other form is referred to as its dual LP problem or in short **dual form**. In general, however, it is immaterial which of the two problems called primal or dual, since the dual of the dual is primal.

The dual involves setting up and solving an LP problem that is almost a 'mirror image' of an LP problem that has been formulated. Both in its formulation and solution, the dual is the flip flop version of the primal.

In the context of LP, duality implies that each LP problem can be analyzed in two different ways, but having equivalent solution. For example, consider the problem of production planning. By using the primal LP problem, the production manager attempts to optimize resource allocation by determining quantities for each product to be produced that will maximize profit. But through dual LP problem approach, he attempts to achieve production plan that optimizes resource allocation so that each product is produced at that quantity such that its marginal opportunity cost equals its marginal return. Thus, the main focus of dual is to find for each resource its best marginal value or shadow price. This value reflects the scarcity of resources, i.e., the maximum additional prices to be paid to obtain one additional unit of the resources to maximize profit under the resource constraints. If resource is not completely used, i.e., there is slack, then its marginal value is zero.

The shadow price is also defined as the rate of change in the optimal objective function value with the respect to the unit change in the availability of a resource. Precisely for any constraint, we have,

Shadow Price = change in the optimal objective function value

unit change in the availability of a resource

Analysis of the dual can also enable a manager to evaluate the potential impact of a new product, and it can be used to determine the marginal values of resources (i.e. constraints). Relative to a product, a manager would want to know what impact adding a new product would have on the solution quantities and the profit; relative to resources, a manager can refer to a dual solution to determine how much profit one unit of each resource equivalent to. Whereas the primal gives solution results in terms of the amount of profit gained from producing products, the dual provides information on the value of the constrained resources in achieving that profit.

?

Often the manager is less concerned about profit than about the use of resource in dual.

Do you agree? Dear learner, would you discuss with your class mates.

3.1.1. Formulating the Dual

There are two important forms of primal and dual problems, namely the symmetrical (or canonical) and the standard form.

Symmetrical form

Suppose the primal LP problem is given in the form

Maximize $Zx = c_1x_1 + c_2x_2 + \dots + c_nx_n$
Subject to

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \leq b_2$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \leq b_m$$

$$x_1, x_2, \dots, x_n \geq 0$$

and then the corresponding dual LP problem is defined as:

$$\text{Minimize } Z_y = b_1y_1 + b_2y_2 + \dots + b_my_m$$

Subject to

$$a_{11}y_1 + a_{21}y_2 + \dots + a_{m1}y_m \leq c_1$$

$$a_{12}y_1 + a_{22}y_2 + \dots + a_{m2}y_m \leq c_2$$

$$\vdots$$

$$a_{1n}y_1 + a_{2n}y_2 + \dots + a_{mn}y_m \leq c_n$$

$$y_1, y_2, \dots, y_m \geq 0$$

The above pair of LP problems can be expressed in the general LP model form as

Primal	Dual
$\text{Max } Z_x = \sum_{j=1}^n c_j x_j$	$\text{Min } Z_y = \sum_{i=1}^m b_i y_i$
Subject to constraints	Subject to constraints

$$\begin{array}{ccc}
 & \downarrow & \\
 \sum_{j=1}^n a_{ij}x_j \leq b_i & & \sum_{i=1}^m a_{ji}y_i \geq c_j \\
 \uparrow & & \uparrow \\
 & a_{ij} = a_{ji} & \\
 \text{and } x_j \geq 0; j = 1, 2, \dots, n & & \text{and } y_i \geq 0; i = 1, 2, \dots, m
 \end{array}$$

The following rules which guide the formulation of the dual problem will give you the summary of the general relationship between primal and dual LP problems.

1. If the primal's objective is to minimize, the dual's will be to maximize; and the vice versa
2. The coefficients of the primal's objective function become the RHS values for the dual's constraints.
3. The primal's RHS values become the coefficients of the dual's objective function.
4. The coefficients of the first "row" of the primal's constraints become the coefficients of the first "column" of the dual's constraint
5. The $\leq$ constraints become $\geq$ and the vice versa.

Consider this Primal problem:

$$\begin{aligned}
 &\text{Minimize} && 40x_1 + 44x_2 + 48x_3 \\
 &\text{Subject to:} && 1x_1 + 2x_2 + 3x_3 \geq 20 \\
 &&& 4x_1 + 4x_2 + 4x_3 \geq 30 \\
 &&& x_1, x_2, x_3 \geq 0
 \end{aligned}$$

The dual of this problem is:

$$\begin{aligned}
 &\text{Maximise } Z = && 20y_1 + 30y_2 \\
 &\text{Subject to:} && \\
 &&& 1y_1 + 4y_2 \leq 40 \\
 &&& 2y_1 + 4y_2 \leq 44 \\
 &&& 3y_1 + 4y_2 \leq 48 \\
 &&& y_1, y_2 \geq 0
 \end{aligned}$$

The following table shows how the primal problem is transformed into its dual.

	Primal			Dual		
Objective function and right-hand side values	Minimize	$40x_1 + 44x_2 + 48x_3$		Maximize	$20y_1 + 30y_2$	
	Subject to			Subject to		
	1		≥ 20	1		≤ 40
	2		≥ 30	2		≤ 44
				3		≤ 48
	Primal			Dual		
Constraint coefficients	1	$1x_1 + 2x_2 + 3x_3$	$\geq$	1	$1y_1 + 4y_2$	$\leq$
	2	$4x_1 + 4x_2 + 4x_3$	$\geq$	2	$2y_1 + 4y_2$	$\leq$
				3	$3y_1 + 4y_2$	$\leq$

We can see from the table that the original objective was to minimize, whereas the objective of the dual is to maximize. In addition, the coefficients of the primal's objective function become the right-hand side values for the dual's constraints, whereas the primal's right-hand side values become the coefficients of the dual's objective function.

Note that the primal has three decision variables and two constraints; whereas the dual has two decision variables and three constraints.

The constraint coefficients of the primal are constraint coefficients of the dual, except that the coefficients of the first "row" of the primal become the coefficients of the first "column" of the dual, and the coefficients of the second "row" of the primal become the coefficients of the second "column" of the dual.

When the primal problem is a maximization problem with all $\leq$ constraints, the dual is a minimization problem with all $\geq$ constraints.

3.1.2. Formulating the Dual when the Primal has Mixed Constraints

In order to transform a primal problem into its dual, it is easier if all constraints in a maximization problem are of the $\leq$ variety, and in a minimization problem, every constraint is of the $\geq$ variety.

To change the direction of a constraint, multiply both sides of the constraints by -1. For example,

$$-1(2x_1 + 3x_2 \geq 18) \text{ is } -2x_1 - 3x_2 \leq -18$$

If a constraint is an equality, it must be replaced with two constraints, one with a $\leq$ sign and the other with a $\geq$ sign. For instance,

$$4x_1 + 5x_2 = 20$$

Will be replaced by;

$$4x_1 + 5x_2 \leq 20$$

$$4x_1 + 5x_2 \geq 20$$

Then one of these must be multiplied by -1, depending on whether the primal is maximization or a minimization problem.

EXAMPLE:

Formulate the dual of this LP model.

$$\text{Maximize } z = 50x^1 + 80x^2$$

Subject to:

$$C^1 \quad 3x^1 + 5x^2 \leq 45$$

$$C^2 \quad 4x^1 + 2x^2 \geq 16$$

$$C^3 \quad 6x^1 + 6x^2 = 30$$

$$x^1, x^2 \geq 0$$

SOLUTION

Since the problem is a max problem, put all the constraints in to the $\leq$ form. Subsequently, C^2 and C^3 will be first adjusted in to $\leq$ constraints.

- C^2 will be multiplied by -1:

$$-1(4x^1 + 2x^2 \geq 16) \text{ becomes } -4x^1 - 2x^2 \leq -16$$

- C^3 is equality, and must be restated as two separate constraints. Thus, it becomes:

$$6x^1 + 6x^2 \leq 30 \text{ and } 6x^1 + 6x^2 \geq 30 \text{ Then the second of these must be multiplied by -1.}$$

$$-1(6x^1 + 6x^2 \geq 30) \text{ becomes } -6x^1 - 6x^2 \leq -30$$

After making the above adjustments, rewrite the LP model again.

$$\text{Maximize } z = 50x^1 + 80x^2$$

Subject to:

$$C^1 \quad 3x^1 + 5x^2 \leq 45$$

$$C^2 \quad -4x^1 - 2x^2 \leq -16$$

$$C^3 \quad 6x^1 + 6x^2 \leq 30$$

$$C^4 \quad -6x^1 - 6x^2 \geq -30$$

$$x^1, x^2 \geq 0$$

? Dear learner, would you guess the reason why the RHS quantity consists of negative values and do you think that it contradicts with rules of LP?

The dual of the above problem will be:

$$\text{Minimize } 45y_1 - 16y_2 + 30y_3 - 30y_4$$

Subject to

$$C^1 \quad 3y_1 - 4y_2 + 6y_3 - 6y_4 \geq 50$$

$$C^2 \quad 5y_1 - 2y_2 + 6y_3 - 6y_4 \geq 80$$

$$y_1, y_2, y_3, y_4 \geq 0$$

3.1.3. Comparison of the Primal and Dual Simplex Solutions

Cross -referencing the values in the primal and dual final simplex tableaus is shown as follows.

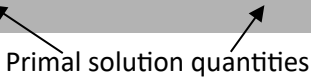
PRIMAL	HOW LEBELED/ WHERE FOUND IN THE PRIMAL	CORRESPONDENCE IN THE DUAL
-Decision variable	 $x^1, x^2, x^3, \dots$	$s_1, s_2, s_3, \dots$
- slack variable	$s_1, s_2, s_3, \dots$	$y_1, y_2, y_3, \dots$
- shadow prices	z row under slack column	Quantity column in decision variable rows.
- solution quantities	Quantity column	C-Z row under slack and decision variable columns

Example:

To show that the flip-flopping of values between the primal and the dual carries over to their final simplex tableaus, let us look at the following tables. The first table contains the final tableau for the dual and the second one contains the final tableau for the primal.

Final tableau of Dual solution to the Microcomputer problem

C		100	22	39	0	0	M	M	Quantity	
	Basis	y_1	y_2	y_3	s_1	s_2	a_1	a_2		
y_3	39	16/3	0	1	1/3	-2/3	-1/3	2/3	40/3	← Primal shadow prices
y_2	22	-6	1	0	-1	1	1	-1	10	← Primal shadow prices
Z		76	22	39	-9	-4	9	4	740	
C-Z		24	0	0	9	4	9-M	4-M		


 Primal solution quantities

Final tableau of Primal solution to the Microcomputer problem:

C		60	50	0	0	0	Quantity
	Basis	x_1	x_2	s_1	s_2	s_3	
s_1	0	0	0	1	6	-16/3	24
x_1	60	1	0	0	1	-1/3	9
x_2	50	0	1	0	-1	2/3	4
Z		60	50	0	10	40/3	740
C-Z		0	0	0	-10	-40/3	

The primary concern with a simplex solution is often three fold:

1. Which variables are in solution?
2. How much of each variable is in the optimal solution?

3. What are the shadow prices for the constraints?

Let's consider how we can obtain the answers to these questions from the dual solution. Notice that the solution quantities of the dual are equal to the shadow prices of the primal (i.e. $40/3$ and 10). Next, notice that the values of the solution quantities of the primal (i.e., 24, 9, and 4) can be found in the bottom row of the dual. Now, in the primal solution, s_1 equals 24. In the dual, the 24 appears in the y_1 column. The implication is that a slack variable in the primal solution becomes a real variable (i.e., a decision variable) in the dual. The reverse is also true: A real variable in the primal solution becomes a slack variable in the dual. Therefore, in the primal solution we have $x_1 = 9$ and $x_2 = 4$; in the dual, 9 appears under s_1 in the bottom row, and 4 appears under s_2 . Thus, we can read the solution to the primal problem from the bottom row of the dual: In the first three columns of the dual, which equate to slack variables of the primal, we can see that the first slack equals 24 and the other two are zero. Under the dual's slack columns, we can read the primal values of the decision variables. In essence, then, the variables of the primal problem become the constraints of the dual problem, and vice versa.

3.1.4. Economic Interpretation of the Dual

Suppose the microcomputer firm has been approached by a representative of a department store chain that wants the firm to make computers that will be sold under the store's brand name. The microcomputer company has only a limited capacity for producing computers, and therefore, must, decide whether to produce its own computers or produce computers for the department store.

For convenience, the original problem is repeated here:

x_1 = number of Model I

x_2 = number of Model II's

Maximize $60x_1 + 50x_2$

Subject to: Assembly time: $4x_1 + 10x_2 \leq 100$

Inspection time: $2x_1 + 1x_2 \leq 22$

Storage space: $3x_1 + 3x_2 \leq 39$

$$x_1, x_2 \geq 0$$

The manager of the firm would reason in the following way. For each unit of Model I that the firm sacrifices to produce computers for the department store, it will gain 4 hours of assembly time, 2 hours of inspection time, and 3 cubic feet of storage space, which can be applied to the store computers. However, it will also give up a unit profit of \$60. Therefore, in order for the firm to realistically consider the store's offer, the amounts of scarce resources that will be given up must produce a return to the firm that is at least equal to the foregone profit. Hence, the value of 4 assembly hours + 2 inspection hours + 3 cubic feet of storage space $\geq$ \$60. By similar reasoning, giving up one unit of Model II will require that the value received by giving up 10 assembly hours + 1 inspection hour + 3 cubic feet of storage must equal or exceed the Model II profit of \$50 per unit. These, then, become the constraints of the dual problem. Thus:

Value Received	Resources	Minimum Profit
<u>Per Unit of</u>	<u>Freed Up</u>	<u>Required</u>
Model I	$4y_1 + 2y_2 + 3y_3$	\$60
Model II	$10y_1 + 1y_2 + 3y_3$	\$50

Hence the constraints of the dual refer to the value of capacity (i.e., the scarce resources). The formulation indicates that in order to switch from making units of Models I and II to making computers for the department store, the value received from that switch must be at least equal to the profit foregone on the microcomputer models. The variables y_1 , y_2 and y_3 are the marginal values of scarce resources (assembly time, inspection time, and storage space). Solving the dual will tell us the imputed values of the resources given our optimal solution.

Naturally, the department store would want to minimize the use of the scarce resources, because the computer firm almost certainly would base its charges on the amount of resources required. Consequently, the objective function for the dual problem focuses on minimizing the use of the scarce resources. Thus:

$$\text{Minimize: } 100y_1 + 22y_2 + 39y_3$$

Looking at the optimal solution to the dual of the microcomputer problem, we can see the marginal values of y_2 and y_3 in the Quantity column, but not the value of y_1 . This is because the optimal solution to the primal did not completely use up all of the assembly capacity. Consequently, no amount of either x_1 or x_2 would need to be given up to obtain one free hour of assembly time. Thus, the marginal value of one hour of assembly time is \$0.

Finally, the optimal dual solution always yields the same value of the objective function as the primal optimal. In this case, it is 740. The interpretation is that the imputed value of the resources that are required for the optimal solution equals the amount of profit that the optimal solution would produce.

3.1.5. Adding another Variable

The concepts introduced in the preceding section also can be applied to evaluating the feasibility of adding another decision variable to the problem. For example, suppose the microcomputer firm is considering a third model of microcomputer that will yield a profit of Birr 70 per unit, and this will need resource requirements of 8 hours of assembly time, 4 hours of inspection time, and 5 cubic feet of storage space per unit. The dual constraint for this model would be:

$$8y_1 + 4y_2 + 5y_3 \geq 70$$

In order to determine if this new variable would come into the optimal (primal) solution, we can substitute the dual solution of $y_1 = 0$, $y_2 = 10$, and $y_3 = 40/3$ (i.e., the shadow prices, or marginal values of the resources) into this constraint to see if it would be satisfied. Thus, we find, $8(0) + 4(10) + 5(40/3) = 106.67$. Because this amount is greater than the new dual constraint, the original solution remains optimal. Hence, the new variable (x_3) would not come into solution.

Conversely, if the dual constraint has not been satisfied, the new variable would have come into solution. In this instance, the marginal value of scarce resources that would be required for the new model exceeds the marginal contribution to profit that the new model would provide. That is, Birr106.67 > Birr70.

Note: The value of this approach is that it is not necessary to rework the entire problem in order to test the potential impact that adding a new decision variable would have on the optimal solution.

3.1.6. Managerial Significance of Duality

The importance of the dual LP problem in terms of the information which it provides about the value of the resources. The economic analysis is concerned with deciding whether or not to secure more resource and how much to pay for these additional resources. The significance of the study dual is as follows;

- The dual variables provide the decision maker a basis for deciding how much to pay for additional unit of resources.
- The maximum amount that should be paid for one additional unit of a resource is called its shadow price (also called simplex multiplier).
- The total marginal value of the resources equals the optimal objective function value. The dual variables equal the marginal value of resources (shadow prices).

Note: The value of i_{th} dual variable is the rate at which the primal objective function value will increase as the i_{th} resource increases, assuming that all other data is unchanged.

3.1.7. Advantages of Duality

- It is advantageous to solve the dual of a primal having less number of constraints, because the number of constraints usually equals the number of iterations required to solve the problem.
- It avoids the necessity of adding surplus or artificial variables and solves the problem quickly.
- The dual variables provide an important economic interpretation of the final solution of an LP problem.
- It is quite useful when investigating change in the parameters of an LP problem (called Sensitivity analysis)

Activity

Convert the following in to dual and solve it using simplex approach

1. Min $Z=4x_1+x_2$

Subject to:

$$3x_1+x_2=3$$

$$4x_1+3x_2\geq 6$$

$$x_1+2x_2\leq 3$$

$$x_1, x_2\geq 0$$

b. Min $Z=2x_1+8x_2$

Subject to:

$$5x_1+x_2\geq 10$$

$$2x_1+2x_2\geq 14$$

$$x_1+4x_2\geq 14$$

$$x_1, x_2\geq 0$$

3.2. Sensitivity Analysis

★ Overview

Dear learners, this section take you beyond the optimal solution to analyze the impact of change in the parameters on the optimal solutions. Assumptions may not always hold. If this is the case, it should be tested. This section will do this purpose.

Under this section, we will discuss about sensitivity analysis which involves an examination of the potential impact of changes in any of the parameters (i.e. numerical values) of a problem (e.g., a change in the amount of scarce resource available).

When LP models are formulated, the input data of the model (also called parameters) such as: (i) profit (or cost) contribution (C_j) per unit of decision variable, ii) availability of resources(b_i), and (iii) consumption of resources per unit of decision variables (a_{ij}), implicitly assumed constant and known with certainty during a planning period. However, rarely does a manager know all of these parameters exactly. In reality the model parameters are simply estimates (or best “guesses”) that are subject to change. For this reason it is of interest to the manager to see what effect a change in a parameter will have in the solution to the model. Changes may be either to the reactions to anticipated uncertainties in the parameters or reactions to the information gained from the dual. The analysis of the parameter changes and their effects on the model solution is known as sensitivity analysis.

? Dear learner, discuss with your group members about why sensitivity analysis is conducted?

Section Objective

Up on completion of this section, the learner will be able to:

- Define sensitivity analysis
- Describe the significance of sensitivity analysis
- Conduct sensitivity analysis on Two parameters; namely
 - Coefficient of OF- when the variable is in the solution mix and not
 - RHS

3.2. Sensitivity Analysis

As it is indicated above, sensitivity analysis is the study of sensitivity of the optimal solution of an LP problem due to discrete variations (changes) in its parameters. The degree of sensitivity of the solution due to these changes can range from no change to all to a substantial change in the optimal solution of the given LP problem.

Sensitivity analysis carries the LP analysis beyond the determination of the optimal solution and begins with the final simplex tableau. Its purpose is to explore how changes in any of the parameters of a problem, such as the coefficients of the constraints, coefficients of the objective function, or the right hand side values; would affect the solution. For this, instead of resolving the entire problem as a new problem with new parameters, we may consider the original optimal solution as an initial solution for the purpose of knowing the ranges, both lower and upper, within which a parameter may assume a value.

For instance, a manager may want to consider obtaining an additional amount of scarce resource that might be available, in which case the manager would want to know the answers to questions such as:

- b. Will an increase in the right hand side of this constraint affect the objective function?
- c. If so, how much of the resource can be used?
- d. Given the answer to the preceding question, what will be the revised optimal solution?

Conversely, the decision maker might be facing a situation in which the amount of scarce resource available is less than the original amount, in which case the issues would be:

- a. Will the decreased level of the resource have an impact on the value of the objective function?
- b. If yes, how much of an impact will it have, and what will be the revised optimal solution?

The process of studying the sensitivity of the optimal solution of an LP problem is also called *post optimality analysis* because it is done after an optimal solution, assuming a given set of parameters, has been obtained for the model.

The discussion here covers only two types of changes: changes in the right hand side levels of the constraints, and changes in the coefficients of the objective function as change in the coefficient of constraints is beyond the coverage of this material.

To demonstrate sensitivity analysis, the microcomputer example will again be used.

Maximize	$60x_1 + 50x_2$
Subject to	
Assembly time	$4x_1 + 10x_2 \leq 100$
Inspection time	$2x_1 + 1x_2 \leq 22$
Storage space	$3x_1 + 3x_2 \leq 39$
	$x_1, x_2 \geq 0$

The most obvious way to ascertain the effect of a change in the parameter of the model is to make the change in the original model, resolve the model, and compare the solution result with the original solution. However, resolving a problem can be very time consuming and as it will

be demonstrated below, it is unnecessary. In most cases the effect of changes in the model can be determined directly from the final simplex tableau.

3.2.1. A Change in the RHS of a constraint

The first step in determining how a change in the RHS (Right Hand Side) of a constraint (e.g., the amount of scarce resource that is available for use) would influence the optimal solution is to examine the shadow prices in the final simplex tableau. Shadow prices are the values in the Z row in the slack columns.

The final tableau for the microcomputer is shown in the following table since sensitivity analysis starts from final tableau.

	Cj	60	50	0	0	0	Quantity
Basis		x_1	x_2	s_1	s_2	s_3	
s_1	0	0	0	1	6	-16/3	24
x_1	60	1	0	0	1	-1/3	9
x_2	50	0	1	0	-1	2/3	4
Z		60	50	0	10	40/3	740

C-Z	0	0	0	-10	-40/3
-----	---	---	---	-----	-------



 Negatives of

A shadow price is a marginal value; it indicates the impact that a one-unit change in the amount of a constraint would have on the values of the objective function. As we can see from the table, the shadow prices are \$0 for s_1 (Assembly time), \$10 for s_2 (Inspection time), and \$40/3 for s_3 (Storage space). These tell us that an increase in the amount of assembly time would have no effect on profit; if inspection time is increased by one hour, it will increase the profit by \$10, and if storage space is increased by 1 cubic foot, profit would increase by \$40/3. The reverse also holds. If we decrease them by such respective amounts, the decrease in profit will take the same figure.

What shadow prices do not tell us is the extents to which the level of a scarce resource can be changed and still have the same impact per unit. The ability to use additional amounts of a resource will disappear at some point because of the fixed amounts of the other constraints. Therefore, we need to determine the range over which we can change the right hand side quantities and still have the same shadow price. This is called the *range of feasibility*, or the right hand side range.

The key to computing the range of feasibility for the constraint lies in each slack column of the final simplex tableau. To compute the range for each constraint, the entries in the associated slack column must be divided into the values in the Quantity column. For example, for the storage column values, the resulting ratios are:

$$\frac{24}{1/3} \qquad \frac{9}{1/4} \qquad \frac{4}{2/3}$$

Here, the smallest positive ratio indicates how much the constraint level can be decreased before it reaches the lower limit of its range of feasibility, and the smallest negative ratio (i.e. the negative ratio closest to 0) indicates how much the storage constraint can be increased

Before it reaches its upper limit of feasibility. The general rule applies when computing the upper and lower limits on the range of feasibility for maximization problem is:

Allowable decrease: The smallest positive ratio

Allowable increase: Negative ratio closest to zero.

Note: The reverse rule will be true for minimization problems.

Alternative Method

Assume q_i symbolically represent quantity in the original model, and Δ stands for change.

Max.	$Z = 60X_1 + 50X_2$
Subject to	$4X_1 + 10X_2 \leq 100$
	$2X_1 + X_2 \leq 22$
	$3X_1 + 3X_2 \leq 39$

If we consider the same example as in the previous example, i.e., change in $q_3 = 39$. If q_3

changes by some amount, the new q'_3 equals the original $q_3 + \Delta$. The Δ changes form a separate column identical to the s_1 column in the final simplex tableau. Therefore, to determine the Δ change, we need only observe the slack (s_i) column corresponding to the model constraint quantity

(q_i) being changed.

Final simplex tableau

	Cj	60	50	0	0	0	Quantity
Basis		x_1	x_2	s_1	s_2	s_3	
s_1	0	0	0	1	6	-16/3	$24 - 16/3 \Delta$
x_1	60	1	0	0	1	-1/3	$9 - 1/3 \Delta$

x_2	50	0	1	0	-1	$\frac{2}{3}$	$4 + \frac{2}{3} \Delta$
Z		60	50	0	10	$\frac{40}{3}$	740
C-Z		0	0	0	-10	$-\frac{40}{3}$	

Recall that the requirement of simplex method is that the quantity values can not be negative. If any q_1 value becomes negative, the solution no longer be feasible.

Thus the inequalities

$$24 - 16\Delta/3 \geq 0$$

$$9 - \Delta/3 \geq 0$$

$$4 + 2\Delta/3 \geq 0$$

are solved for Δ :

$$24 - 16\Delta/3 \geq 0 \quad 9 - \Delta/3 \geq 0 \quad 4 + 2\Delta/3 \geq 0$$

$$\frac{72 - 16\Delta}{3} \geq 0 \quad \frac{27 - \Delta}{3} \geq 0 \quad \frac{12 + 2\Delta}{3} \geq 0$$

$$3 \quad 3 \quad 3$$

$$72 - 16\Delta \geq 0 \quad 27 - \Delta \geq 0 \quad 12 + 2\Delta \geq 0$$

$$72 \geq 16\Delta \quad 27 \geq \Delta \quad 2\Delta \geq -12$$

$$4.5 \geq \Delta \quad \Delta \leq 27 \quad \Delta \geq -6$$

$$\Delta \leq 4.5$$

Since the new $q_3 = 39 + \Delta$, then $\Delta = q_3 - 39$. These values are substituted to the above inequalities to result in:

$\Delta \leq 4.5$	$\Delta \leq 27$	$\Delta \geq -6$
$q_3 - 39 \leq 4.5$	$q_3 - 39 \leq 27$	$q_3 - 39 \geq -6$
$q_3 \leq 43.5$	$q_3 \leq 66$	$q_3 \geq 33$

Summarizing these inequalities, we have:

$$33 \leq 43.5 \leq q_3 \leq 66$$

The value 66 can be eliminated since q_3 must be less than 43.5; thus

$$33 \leq q_3 \leq 43.5$$

As long as q_3 remains in this range, the present basic solution variables remain positive and feasible. However, the quantity value of those basic variables may change.

? Dear learner, can you determine the range within which q_1 & q_2 vary with out causing the basic solution mix change?

Once we have computed these limits, we proceed to assess the impact that a contemplated change would have on the optimal solution.

Example

The manager in the microcomputer problem is contemplating a change in the level of the storage constraint – an increase of 3 cubic feet. Determine the revised optimal solution for the change.

Solution

First, note that an increase of 3 cubic feet is within the range. Then, the effect of an increase of three cubic feet is computed in the following way:

Basis	s_3	Current solution	Change	Revised solution
s_1	$-16/3$	24	$3(-16/3) = -16$	8
x_1	$-1/3$	9	$3(-1/3) = -1$	8
x_2	$2/3$	4	$3(2/3) = +2$	6
Z	$40/3$	740	$3(40/3) = +40$	780

For an increase in storage space of 8 cubic feet, we take the upper limit of 4.5 cubic feet for the computation. The resulting revised figures, then, would be $s_1 = 0$, $x_1 = 7.5$, $x_2 = 7$, and Profit = 800.  **Note**, however, that beyond the upper limit, s_3 would come into solution, replacing s_1 , which would no longer be slack. The amount of slack would be $8 - 4.5 = 3.5$ cubic feet. Consequently, the revised solution would be $s_3 = 3.5$, $x_1 = 7.5$, $x_2 = 7$, and Profit = 800.

3.2.2. A Change in an Objective Function Coefficient

It is useful to know how much the contribution of a given decision variable to the objective function can change *without* changing the optimal solution. There are two cases: Changes for a variable that is not currently in the solution mix, and Changes for a variable that is currently in the solution mix.

If a variable is not currently in the solution in a maximization problem, its objective function coefficient would have to increase by an amount that exceeds the C-Z value for that variable in the final tableau in order for it to end up as a basic variable in the optimal solution. The range over which a non-basic variable's objective function coefficient can change without causing that variable to enter the solution mix is called its **range of insignificance**.

Range of Insignificance: the range over which a non-basic variable's objective function coefficient can change without causing that variable to enter the solution mix.

Maximize $3x^1 + 2x^2 + 5x^3$

Subject to:

C1 $x^1 + 2x^2 + 2x^3 \leq 18$

C2 $3x^1 + 2x^2 + 6x^3 \leq 12$

C3 $2x^1 + 3x^2 + 4x^3 \leq 12$

$x^1, x^2, x^3 \geq 0$

C	3	2	5	0	0	0	
Basis	x^1	x^2	x^3	s^1	s^2	s^3	Quantity
s^1 0	0	0	0	1	1/5	-4/5	4/5
x^1 3	1	0	2	0	3/5	-2/5	12/5
	0	1	0	0	-2/5	3/5	12/5

x^2	2						
Z	3	2	6	0	1	0	12
C-Z	0	0	-1	0	-1	0	

In the above tableau we see that x^3 is a non- basic variable. To determine the range over which the objective function coefficient of x^3 would change without changing the optimal solution, recall how a variable in a max problem enters the solution mix. Because x^3 is not in solution, its objective function coefficient would need to increase in order for it to come in to solution (i.e., to make its c-z value positive). The amount of increase must be greater than the absolute value of its c-z value, which is $|-1|$. Therefore, its objective function coefficient must increase by more than 1. Hence, the range of insignificance for x^3 is 6 or less (i.e, its Z value). In general the rule is: **Range of insignificance for a non- basic variable is $\leq$ the Z value of the variable**. The allowable increase is the absolute value of its own C- Z value.

Range of Optimality: the range over which objective function coefficient of a variable that is in solution can change without changing the optimal values of the decision variables. Note, however, that such a change would change the optimal value of the objective function.

For variables that are in solution, the determination of the range of optimality (the range over which the objective function coefficient of a variable that is in the solution can change without changing the optimal values of the decision variables) requires a different approach. Divide the values in row C-Z by the corresponding row values of the variable in question and follow the following rule for both maximization and minimization problems.

Allowable increase: **The smallest positive ratio of C-Z value and the variable's substitution rate.**

Allowable decrease: **the smallest negative ratio of C-Z value and the**

 **Note:** If there is no positive ratio, it means that there is no upper limit on that variable's objective function coefficient.

Example

Determine the range of optimality for the decision variables in the microcomputer problem.

Solution

The final simplex tableau for that problem is repeated here for convenience.

C_j		60	50	0	0	0	
	Basis	x_1	x_2	s_1	s_2	s_3	Quantity
s_1	0	0	0	1	6	$-16/3$	24
x_1	60	1	0	0	1	$-1/3$	9
x_2	50	0	1	0	-1	$2/3$	4
Z		60	50	0	10	$40/3$	740
C-Z		0	0	0	-10	$-40/3$	

For x_1 we find:

Column	x_1	x_2	s_1	s_2	s_3
<u>C-Z value</u>	$0/1 = 0$	$0/0 = \text{Undefined}$	$0/0 = \text{Undefined}$	$-10/1 = -10$	$-40/3 = +40$
x_1 row value				$-1/3$	

The smallest positive ration is +40. Therefore, the coefficient of x_1 can be increased by Birr 40 without changing the optimal solution. The upper end of its range of optimality is this amount added to its current (original) value. Thus, its upper end is (Birr60 +Birr40) = Birr100. Also, the smallest negative ratio is -10; therefore, the x_1 coefficient can be decreased by as much as Birr10 from its current value, making the lower end of the range equal to (Birr60 - Birr10)= Birr 50.

For x_2 we find:

Column	x_1	x_2	s_1	s_2	s_3
C-Z value	$0/0 = \text{Undefined}$	$0/1 = 0$	$0/0 = \text{Undefined}$	$-10/-1 = +10$	$-40/3 = -20$
x_2 row value					$\frac{2}{3}$

The smallest positive ratio is +10. This tells us that the x_2 coefficient in the objective function could be increased by Birr10 to $(\text{Birr}50 + \text{Birr}10) = \text{Birr} 60$. The smallest negative ratio is -20, which tells us that the x_2 coefficient could be decreased by Birr20 to $(\text{Birr}50 - \text{Birr}20) = \text{Birr}30$. Hence, the range of optimality for the objective function coefficient of x_2 is Birr30 to Birr 60.

This can also be done in the same way as change in the constraints by using the concept of change starting from the final tableau. That is, the range of c_j that will maintain the optimal solution can be determined directly from the optimal simplex tableau.

C_j		$60 + \Delta$	50	0	0	0	
	Basis	x_1	x_2	s_1	s_2	s_3	Quantity
s_1	0	0	0	1	6	$-16/3$	24
x_1	$60 + \Delta$	1	0	0	1	$-1/3$	9
x_2	50	0	1	0	-1	$2/3$	4
Z		$60 + \Delta$	50	0	$\Delta + 10$	$(40 - \Delta)/3$	740
C-Z		0	0	0	$-10 - \Delta$	$(-40 + \Delta)/3$	

Consider a Δ change for c_1 . This will change the c_1 value from $c_1 = 60$ to $c_1 = 60 + \Delta$. This new value is not only included on the top c_j row, but also in the left-hand c_j column. Since $60 + \Delta$ is in left-hand column, it becomes a multiple of the column values when the new Z_j row values and the subsequent $C_j - Z_j$ row values.

The solution becomes optimal as long as the $C_j - Z_j$ row values remain negative. If $C_j - Z_j$ becomes positive, the product mix will change, if it becomes zero, there will be an alternative solution. Thus, for the solution to remain uniquely optimal,

$$-10 - \Delta < 0$$

and

$$(-40 + \Delta)/3 < 0$$

Both of these inequalities must be solved for Δ .

$$-10 - \Delta < 0 \quad \text{and} \quad (-40 + \Delta)/3 < 0$$

$$-\Delta < 10(-40 + \Delta) < 0$$

$$\Delta > -10$$

$$\Delta < 40$$

Now recall that $c_1 = 60 + \Delta$; therefore, $\Delta = c_1 - 60$. Now substituting $c_1 - 60$ for Δ in the above inequalities, we get:

$$\Delta > -10$$

$$\Delta < 40$$

$$c_1 - 60 > -10$$

$$c_1 - 60 < 40$$

$$c_1 > 50$$

$$c_1 < 100$$

Therefore, the range of values for C_1 over which the solution basis remain optimal (although the value of the objective function may change) is:

$$50 \leq C_1 \leq 100$$

? Dear learner, would you try to determine the range over which coefficient of X_2 in the objective function change without changing the solution mix in the same way as above?

Summary

- Every linear programming problem can have two forms. The original formulation of a problem is referred to as its **Primal form**. The other form is referred to as its dual LP problem or in short **dual form**.
- The main focus of dual is to find for each resource its best marginal value or shadow price. This value reflects the scarcity of resources, i.e., the maximum additional prices to be paid to obtain one additional unit of the resources to maximize profit under the resource constraints.
- If the primal's objective is to minimize, the dual's will be to maximize; and the vice versa; the coefficient's of the primal's objective function become the RHS values for the dual's constraints; the primal's RHS values become the coefficients of the dual's objective function; the coefficients of the first "row" of the primal's constraints become the

coefficients of the first "column" of the dual's constraint; the $\leq$

Activity

1. ABC Metal crafters company makes brass tray and buckets. The number of trays (x_1) and buckets (x_2) that can be produced daily is constrained by the availability of brass and labor, as reflected in the following LP model.

$$\text{Maximize } Z = 6x_1 + 10x_2$$

Subject to:

$$x_1 + 4x_2 \leq 90 \text{ (brass, lb)}$$

$$2x_1 + 2x_2 \leq 960 \text{ (labor, hrs)}$$

$$x_1, x_2 \geq 0$$

The final optimal simplex table for this model is as follows:

		6	10	0	0	
Basic variables	cj	x ₁	x ₂	s ₁	s ₂	quantity
x ₂ 10	0	1	1/3	-1/6	20	
x ₁ 6	1	0	-1/3	2/3	10	
Zj		6	10	4/3	7/3	260
Cj - Zj		0	0	-4/3	-7/3	

- Formulate the dual of this problem.
 - Define the dual variables and explain what they mean.
 - What do the dual variable equal?
 - resolve the problem for its dual and see the differences in the solution if any
2. A Manufacturing firm produces electric motors for washing machines and vacuum cleaners. The firm has resource constraints for production time, steel and wire. The LP model determining the number of washing machine motors (x_1) and vacuum cleaner motors (x_2) to produce has been formulated as follows.

$$\text{Maximize } Z = 70x_1 + 80x_2$$

Subject to:

$$2x_1 + x_2 \leq 19 \text{ (production, hrs)}$$

$$x_1 + x_2 \leq 14 \text{ (steel, lb)}$$

$$x_1 + 2x_2 \leq 20 \text{ (wire, ft)}$$

$$x_1, x_2 \geq 0$$

The final simplex table for this model is as follows.

		70	80	0	0		
Basic variables	cj	x ₁	x ₂	s ₁	s ₂	s ₃	quantity
x ₁	70	1	0	2/3	0	-1/3	6
s ₂	0	-1/3	1	1	0	0	1
x ₂	0	0	1	-1/3	0	2/3	7
Zj		70	80	20	0	30	980
Cj - Zj		0	0	-20	0	-30	

- Formulate the dual of this problem.
- Define the dual variables and explain what they mean.
- What do the dual variable equal?
- resolve the problem for its dual and see the differences in the solution if any

e. Interpret the $C_j - Z_j$ values in the above table

f. allocate the profit among the resources to show how much each resource contributes to the total profit.

3. The evergreen products firm produces three types of pressed paneling from pine and spruce. The three types of paneling are wide(x_1), medium (x_2) and small(x_3). Each sheet must be cut and pressed. The resource requirements are given in the following LP formulation.

$$\text{Maximize } Z = 4x_1 + 10x_2 + 8x_3$$

Subject to:

$$5x_1 + 4x_2 + 4x_3 \leq 200 \text{ (pine, lb)}$$

$$2x_1 + 5x_2 + 2x_3 \leq 160 \text{ (spruce, lb)}$$

$$x_1 + x_2 + 2x_3 \leq 50 \text{ (cutting, hrs)}$$

$$2x_1 + 4x_2 + 2x_3 \leq 80 \text{ (pressing, hr)}$$

$$x_1, x_2, x_3 \geq 0$$

The final simplex table for this model is as follows.

x_3	8	1/3	1	0	0	0	2/3	1/6	20	
x_2	10	1/3	1	0	0	0	-1/3	1/3	10	
Z_j			6	10	8	0	0	2	2	260
$C_j - Z_j$			0	0	0	0	0	-2	-2	

- What is the marginal value of an additional pound of spruce? Over what range is this value valid?
- What is the marginal value of additional hours of cutting? Over what range is this value valid?
- Given a choice between securing more hours and or more pressing hours, which should the management select?
- If the amount of the spruce available to the firm were decreased from 160 to 100 pounds, would this reduction affect the solution? Why? If it is what is the new solution?
- What unit profit would have to be made from wide paneling before management consider producing it?
- Management is considering changing the profit of small paneling from Birr 8 to Birr 13. would this change affect the solution?

4. A manufacturing firm produces four products. Each product requires material and machine processing. The linear programming model formulated to determine the number of product 1 (x_1)

Product 2 (x_2) product 3 (x_3), and product 4 (x_4), to produce is as follows.

Maximize $Z = 2x_1 + 8x_2 + 6x_4$ (profit, Birr)

Subject to:

$$2x_1 + x_2 + 4x_3 + 2x_4 \leq 200 \text{ (material, lb)}$$

$$x_1 + 2x_2 + 2x_3 + x_4 \leq 160 \text{ (machine processing, hrs)}$$

$$x_1, x_2, x_3, x_4 \geq 0$$

The final simplex table for this model is as follows.

		2	8	10	6	0	0	
Basic variables	cj	x ₁	x ₂	x ₃	x ₄	s ₁	s ₂	quantity
x ₃	6	1	0	2	1	2/3	-1/3	80
x ₂	8	0	1	0	0	-1/3	2/3	40
Zj		6	8	12	6	4/3	10/3	800
Cj - Zj		-4	0	-2	0	-4/3	-10/3	

a. What is the marginal value of an additional pound of material? Over what range is this value valid?

b. What is the marginal value of additional hours of processing time? Over what range is this

CHAPTER VII

NETWORK MODELS

Network diagram is a practical representation of the various events and activities.

Network modeling: Techniques that enable complex projects to be scheduled, taking into account the precedence of each activity.

7.1. PROJECTS

Definition: A *project* is a set of proposals for the investment of resources in to a clearly identified set of actions that are expected to produce future benefits that exceeds costs.

Examples of a project:

- *New business establishment*
- *Expansion of existing business*
- *Launching new product*
- *Cost reduction project*
- *Total Quality of Management (TQM), etc*

Characteristics of a project:

1. *The duration of a project lasts long*
2. *A project is complex in nature involving many interrelated activities and participants*
3. *Delay in completion time may be very costly*
4. *Project activities are sequential*

7.2. NETWORK CONVENTIONS

1. The project can be sub-divided into a set of predictable independent activities each of which has a clear beginning and ending.
2. Each activity can be sequenced as to its predecessors or successors.
3. The network is not cyclical

i.e: Each activity is executed once and only once during the life of the project.

4. Activity times may be estimated either as a **single point estimate (CPM)** or as a **3- point estimate (PERT)**
 - **CPM-** Critical Path Method
 - **PERT-** Program Evaluation and Review Technique
5. The duration of the activities is independent of each other.

7.3. EVENT ANALYSIS

Events: Events of the network represent **project milestone**, such as *the start* or *the completion* of an activity (task) or activities, and occur at a particular instant of time at which some specific part of the project has been or is to be achieved.

Event: An event is a point in time that marks the beginning or ending of activity.

Events are commonly represented by **circles (nodes)** in the network diagram.

The events can be further classified into the following two categories (types):

i. Merge event

An event which represents the **joint completion** of more than one activity is known as **merge event**.

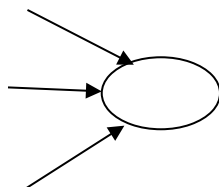


Fig: merge event

ii. Burst event

An event which represents the *initiation (beginning)* of more than one activity is known as *burst event*.

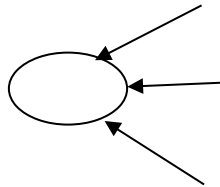


Fig: Burst event

Events in the network diagram are identified by numbers. Each event should be identified by a number higher than that allotted to its immediately preceding event to indicate progress of work. The numbering of events in the network diagram must *start* from *left* (start of the project) to *right* (completion of the project) and *top* to the *bottom*. Care should be taken that there is no duplication in the numbering.

7.4. ACTIVITY ANALYSIS

Activity is a time-consuming job or task that is a key subpart of the total project.

Activities of the net work represent project operations or tasks to be conducted. As such each activity except dummy consumes time and resources and incurs costs. An arrow is commonly used to represent an activity with its head indicating the direction of progress in the project.

Activities are identified by the numbers of their starting (tail or initial) event and ending (head or terminals) event. An arrow (i, j) extended between two events; the *tail event* i represents the start of the activity and the *head event* j , represents the completion of the activity.

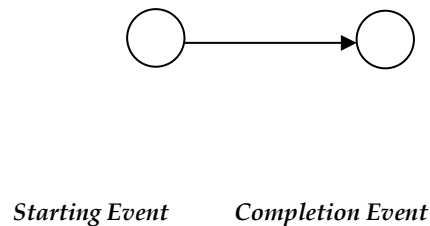


Fig: Activity -Node relationship in network diagram.

Activities can be further classified into the following three categories:

i. Predecessor activity

Predecessor activity is an activity which must be completed before one or more other activities start.

I

ii. Successor activity

Successor activity is an activity which started immediately after one or more of other activities are completed.

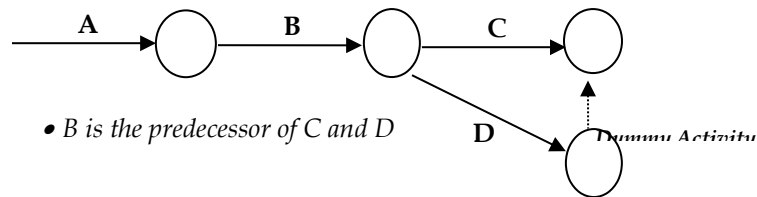
iii. Dummying activity

Dummying activity is an activity which does not consume either any resource or time.

A ***dummy activity*** in the network is added only to represent the given precedence relationships among activities of the project and is needed when

- a) two or more parallel activities in a project have same head and tail events, or
- b) two or more activities have some (but not all) of their immediate predecessor activities in common.

A ***dummy activity*** is depicted by ***dotted line*** in the network diagram.



7.5. RULES OF NETWORK CONSTRUCTION

Following are some of the rules which have to be followed while constructing a network:

1. In network diagram, arrows represent activities and circles the events. The length of an arrow is of no significance.
2. Each activity should be represented by one arrow and must start and end in a circle called event. The tail of an activity represents the start and head the completion of work.

3. *The event numbered 1 denotes start of the project and is called initial event. All activities emerging (or taking off) from event 1 should not be preceded by any other activity or activities. Event carrying the highest number denotes the completion events. A network should have only one initial event and only one terminal event.*
4. *The general rule of numbering the event is that the number at an activity's head should always be larger than that at its tail. That is, events should be numbered such that for each **activity (i, j)**, $i < j$.*
5. *An activity must be uniquely identified by its starting and completion event which implies that*
 - a. *an event number should not get repeated or duplicated*
 - b. *two activities should not be identified by the same completion event*
 - c. *activities must be represented either by their symbols or by the corresponding ordered pair of starting-completion events.*
6. *The logical sequence (or interrelationship) between activities must follow the following rules :*
 - a. *An event cannot occur until all the incoming activities into it have been completed.*
 - b. *An activity cannot start unless all the preceding activities, on which it depends, have been completed.*
 - c. *Though a dummy activity does not consume either any resource or time, even then it has to follow the rules 6(a) and 6(b).*

7.5. ERRORS AND DUMMIES IN NETWORK

i. Looping and Dangling

Looping (cycling) and dangling are considered as faults in a network. Therefore, these must be avoided.

1. Looping

A case of endless loop in a network which is also known as *looping* is shown in figure below, where activities *A*, *B*, and *C* form a cycle.

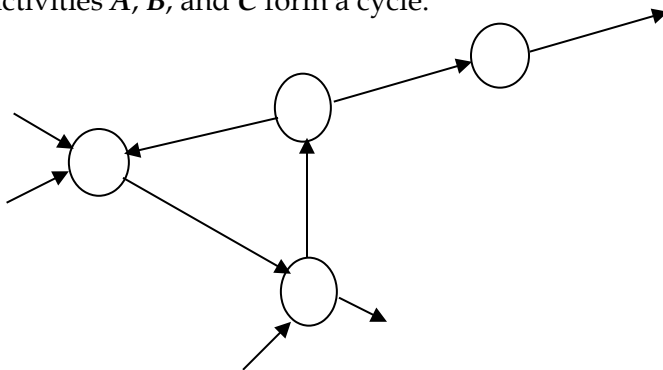


Fig: Looping in network diagram

In this case it is difficult to number three events associated with activity *A*, *B*, and *C* so as to satisfy *rule 6* of constructing the network.

2. Dangling

A case of disconnect activity before the completion of all activities which is also known as *dangling*.

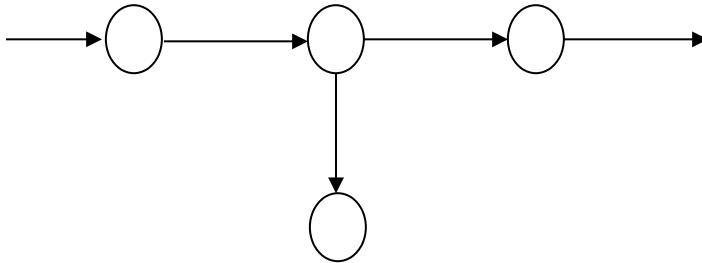


Fig: Dangling in network diagram

In this case, activity C does not give any result as per the rules of the network. The dangling may be avoided by adopting *rule 5* of constructing the network.

i. Dummy or Redundant Activity.

The following is the one of the cases in which the use of dummy activity may help in drawing the network correctly as per the various rules

When two or more parallel activities in a project have the same head and tail events:

i.e: two events are connected with more than one arrow.

In a), activities *B* and *C* have a common predecessor activity *A*. At the same time, they have activity *D* as a common successor. To get the correct network a dummy activity for the ending event of *B* to show that *D* may not start before both *B* and *C* are completed is shown in figure b)

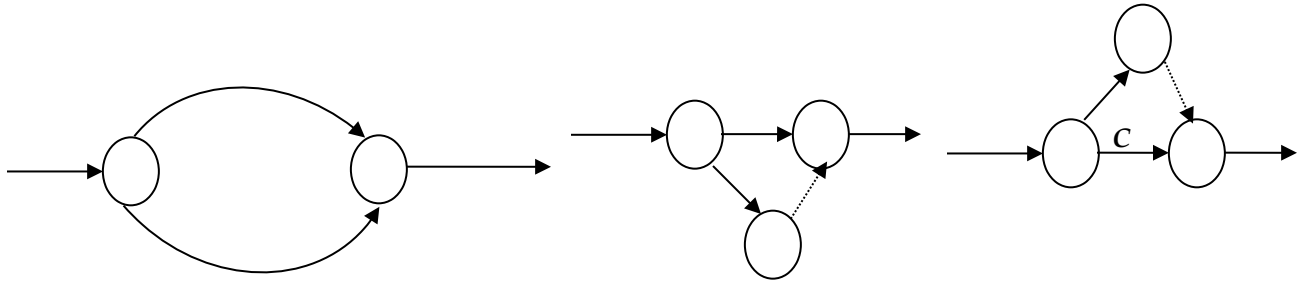


Fig: a) Parallel Activities

Fig: b) Dummy Activity

Example

1. An assembly is to be made from two parts *X* and *Y*. Both parts must be turned on a lathe and *Y* need not be polished. The sequence of activities together with their predecessors is given below:

<i>Activity</i>	<i>Description</i>	<i>Predecessor Activity</i>
A	Open work order	—
B	Get material for X	A
C	Get material for Y	A
D	Turn X on Lathe	B
E	Turn Y on Lathe	B, C

F	Polish Y	E
G	Assemble X and Y	D, F
H	Pack	G

Draw a network diagram for the project.

Solution

The network diagram for the project is show below:

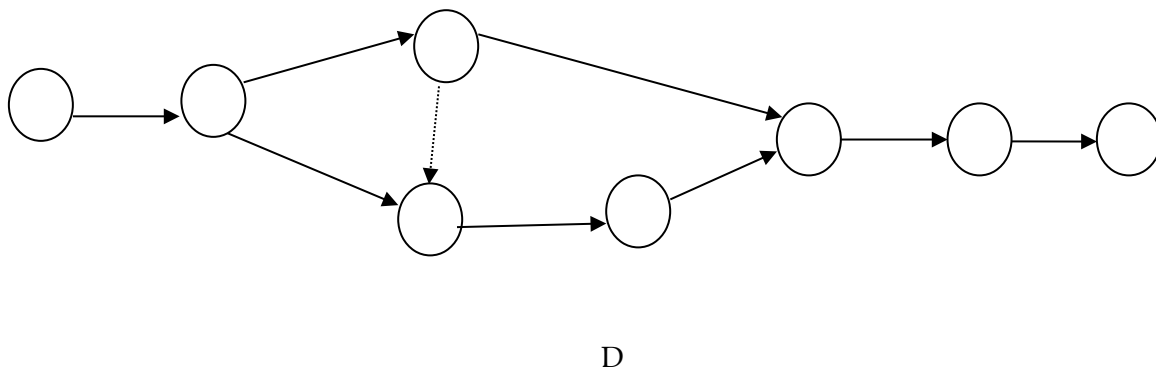


Fig: Network Diagram

2. Listed in the table are the activities and sequencing necessary for a maintenance job on the heat exchangers in a refinery.

<i>Activity</i>	<i>Description</i>	<i>Predecessor Activity</i>
A	Dismantle pipe connections	-
B	Dismantle header, closure, and floating front	A
C	Remove tube bundle	B
D	Clean bolts	B
E	Clean header and floating head front	B
F	Clean tube bundle	C
G	Clean shell	C
H	Replace tube bundle	F, G
I	Prepare shell pressure test	D, E, H
J	Prepare tube pressure test and reassemble	I

Required:

Draw a network diagram for the project

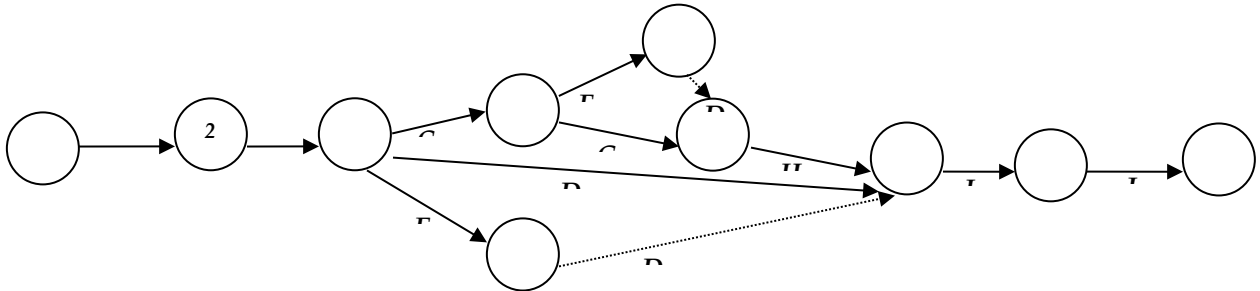


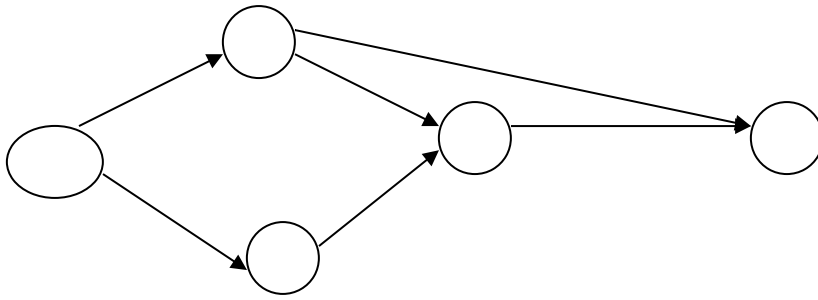
Fig: Network Diagram

3. The following table gives the activities in a construction project:

<u>Activity</u>	<u>Immediate Predecessor</u>
A	-
B	-
C	-
D	A
E	C
F	A
G	D, B, E

Required: Draw an arrow diagram for the project

Solution:



Exercise:

Listed in the table are the activities & sequencing necessary for the completion of a recruitment procedure for management trainees (MT) in a firm.

<u>Activity</u>	<u>Predecessor Activity</u>
A	-
B	A
C	A
D	C
E	B

C

D, E

G

H

F

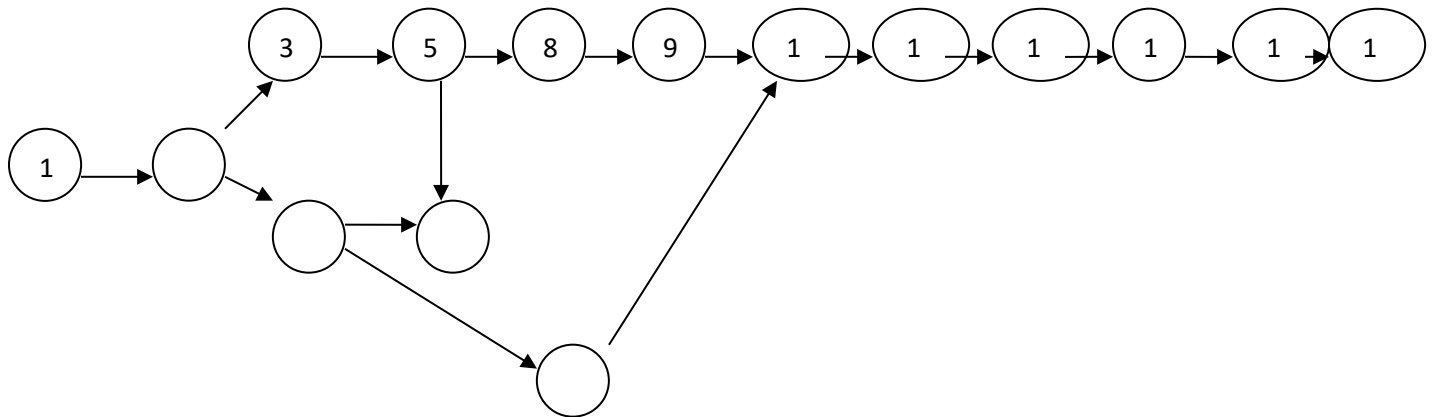
I, J

K

L

M, N.

Draw a network diagram for the project Ans:



F IG Net work diagram

7.5 CRITICAL PATH ANALYSIS

The objective of critical path analysis is to estimate the total project duration & to assign starting & finishing times to all activities involved in the project. This helps in checking actual progress against the scheduled duration of the project.

The duration of individual activities may be uniquely determined (in case of CPM) or may involve the three time estimates (in case of PERT) out of which the expected duration of an activity is computing. Having done this, the following factors should be known to prepare project scheduling

- i. Total completion time of the project
- ii. Earliest & latest start time of each activity
- iii. Float for each activity

i.e. the amount of time by which the completion of an activity can be delayed without delaying the total project completion time.

- iv. Critical activities & critical path.

Consider the following notation for the purpose of calculating various times of event & activities

- A. Earliest occurrence time of event i E_i it is the earliest time at which an event can occur without affecting the total project time.
- B. Latest occurrence time of event i L_i

It is the latest time at which an event can occur without affecting the total project time

- C. Earliest start time for activity $(i, j) = ES_{ij}$

It is the time at which the activity can start without affecting the total project time

D. Latest start time for activity $(i, j) = LS_{ij}$

It is the latest possible time by which an activity must start without affecting the total project time

E. Earliest finish time for activity $(i, j) = EF_{ij}$

It is the earliest possible time at which an activity can finish without affecting the total project time.

F. Latest finish time for activity $(i, j) = LF_{ij}$

It is the latest time by which an activity must get completed without delaying the project completion.

G. Duration of activity $(i, j) = t_{ij}$

Expected time estimate

In a network diagram, there should only be one initial event & one final event, while other events are numbered consecutively with integers $1, 2, \dots, n$ such that $i < j$, for any two event i & j connected by an activity which starts at i & finish at j

For calculating the above mentioned times, there are two methods; namely;

- 1) Forward pass method &
- 2) Backward pass method

7.5.1 Forward Pass Method

(For Earliest Event Time)

In this method's calculation begins from the initial event 1, proceeds through the network visiting events in an increasing order of event no & ends at the final event, say N . At each event we calculate earliest occurrence event time (E) & earliest start & finish time for each activity

that begins at that event. When calculations end at the final event N, its earliest occurrence time gives the earliest possible completion time of the entire project.

The method may be summarized as follows:

- 1) Set the earliest occurrence time of initial event 1 to zero, i.e.

$$E_1 = 0, i = 1$$

- 2) Calculate earliest start time for each activity that begins at event I ($i=1$) This is equal to earliest occurrence time of event, i (tail event). That is

$$ES_{ij} = E_i, \text{ for LL activities } (i, j) \text{ starting at event } i.$$

- 3) Calculate the earliest finish time of each activity that begins at event i. This is equal to the earliest start time of the activity plus the duration of the activity. That is,

$$EF_{ij} = ES_{ij} + t_{ij}$$

$$= E_i + t_{ij} \text{ for all activities } (i, j) \text{ beginning at event } i$$

- 4) Proceed to the next event, say j, $j > i$
- 5) Calculate the earliest occurrence time for the event j. This is the maximum of the earliest finish times of all activities ending into that event, i.e.

$$E_j = \max \{EF_{ij}\}$$

$$= \max \{E_i + t_{ij}\} \text{ for all immediate predecessor}$$

Activities

- 6) If $j = N$ (final event), then earliest finish time for the project, i.e. the earliest occurrence time E_N for the final event is given by

$$E_N = \max \{EF_{ij}\}, \text{ for all terminal activities.}$$

$$= \max \{E_{N-1} + t_{ij}\} \text{ \& stop the method.}$$

7.5.2 Backward Pass Method

(For Latest Allowable Event Time)

In this method calculations begin from final event N, proceed through the network visiting events in the decreasing order of event no & end at the initial event 1. At each event, we calculate the latest occurrence event time (L) for the corresponding event, latest finish & start time for each activity that is terminating at the event, such that the earliest finish time for the project remains the same. The method may be summarized as follows.

1. Set the latest occurrence of last event N equal to its earliest occurrence time (known from forward pass method), that is

$$L_N = E_N, \quad j = N$$

2. Calculate latest finish time of each activity which ends at event j. This is equal to latest occurrence time of final event N. That is $LF_{ij} = L_j$, for all activities (i,j) ending at event j
3. Calculate the latest start times of all activities ending at j. It is obtained by subtracting the duration of the activity from the latest finish time of the activity, That is,

$$LF_{ij} = L_j$$

$$LS_{ij} = LF_{ij} - t_{ij}$$

$$= L_j - t_{ij}, \text{ for each activity (i,j) ending at}$$

Event, j

4. Proceed backward to the event in the sequence that decreases j by 1.
5. Calculate the latest occurrence time of event i (i<j). This is the minimum of the latest start time of all activities starting from that event. That is,

$$L_i = \min \{LS_{ij}\}$$

$$= \min \{L_j - t_{ij}\}$$

6. If j = 1 (initial event), then the latest finish time for project, i.e. latest occurrence time L1 for the initial event is given by.

$L_i = \min [LS_{ij}]$ for all immediate successor activities

$= \min \{ L_j - t_{ij} \}$ & stop the method

7.5.3. Float (or slack) of an Activity and Event

The *float* of an activity is the amount of time by which it is possible to delay its completion time without affecting the total project completion time.

1. Event Float

The *float* (also sometimes called '*slack*') of an event is the difference between its latest time (L_i) and its earliest time (E_i). That is

$$\text{Event Float} = L_i - E_i$$

It is a measure of how much later than expected a particular event could occur without delaying the completion of the entire project.

2. Activity Float

It is the float (or slack) in the activity time estimates.

Types of activity floats:

a. Total Float

It is the difference between the time available to perform the activity (*measured from the earliest start time to the latest finish time*) and the expected completion time of the activity. That is, for activity (i,j), the total float is given by

$$\begin{aligned} \text{Total Float (TF}_{ij}) &= (L_j - E_i) - t_{ij} \\ &= LS_{ij} - ES_{ij} \end{aligned}$$

b. Free Float

For calculating the total float, only a particular activity was considered with respect to its tail and head event times or by considering latest start and finish time of an activity with respect to its earliest start and finish time. However, it may be needed to know that how much an activity's scheduling time can be increased or decreased without changing its immediate successor activities.

The time by which the completion of an activity can be delayed beyond its *earliest finish time* without affecting the earliest start time of a succeeding activity is called *free float*.

$$\begin{aligned} \text{Free Float (FF}_{ij}) &= (E_j - E_i) - t_{ij} \\ &= \text{Min} \left\{ ES_{ij}, \text{for all immediate successors of (i,j)} - EF_{ij} \right\} \end{aligned}$$

c. Independent float

Remarks:

1. Since latest event time is always greater than or equal to the earliest event time (i.e $L_i \geq E_i$), $TF_{ij} \geq FF_{ij}$.

This implies the value of free float may range from zero to total float and can never exceed total float value.

2. The calculation of various floats can help the decision-maker in identifying the underutilized resources, flexibility in the total schedule and possibilities of redeployment of resources.

3. Once the float of an activity is disturbed, float of all other activities of the project is changed and should be recalculated.

4. If total float value is

a. Negative (i.e $L-E < 0$),

-resources are not adequate and activity may not finish in time

-Induce extra resources or certain activities (also called critical activities) need crashing to reduce negative float value.

b. Zero (i.e $L-E = 0$)

-Resources are just sufficient to complete the activity. Any delay in activity execution will necessarily increase the project cost.

c. Positive (i.e $L-E > 0$)

-Resources are surplus

-Surplus resources can be deployed elsewhere or execution of the activity can be delayed.

7.6 CRITICAL PATH

Path A path is unbroken chain of activities that connects from the start node to the end node (terminal node)

Certain activities in a network diagram of a project are called critical activities because delay in their execution will cause further delay in the project completion time. Thus, all activities having zero total float value are identified as critical activities.

The critical path is the continuous chain of critical activities in a network diagram. It is the longest path starting from first to the last event & is shown by the thick line or double lines in the network diagram.

The length of the critical path is the sum of the individual times of the critical activities living on it & defines the minimum time required to complete the project.

The critical path on a network diagram can be identified as:

- i. For all activities (i,j) lying on the critical path the E -values & L-values for tail & head events are equal

That is, $E_j = L_j$ & $E_i = L_i$

- ii. On critical path $E_j - E_i = L_j - L_i = L_{ij}$

Finding the critical path is important for directing decision-makers attention & effort to critical activities where improvement will pay the largest dividend.

Example

1. An established company has decided to add a new product to its line. It will buy the product from manufacturing concern, package it & sell it to a number of distributors selected on a geographical basis. Market research has identified the volume expected & the size of sales force required. The steps

<u>Activity</u>	<u>Description</u>	<u>Time (Week)</u>
A	Organize sales office	6
B	Hire Salesman	4
C	Train salesman	7
D	Select advertising agency	2
E	Plan advertising campaign	4
F	Conduct	10
G	Design package	2
H	Setup packaging facilities	10

I	Package initial Stocks	6
J	Order Stock from manufacture	13
K	Select distributors	9
L	Sell to distributors	3
M	Ship stocks	5

Required:

- Indicate the critical path the critical activities & the total project time
- For each non-critical activity, find the total float

A. the critical path in the network diagram above has been shown by thick lines by joining all those events where two values E_i & L_j are equal

the critical path of the project is:

1-2-5-6-9-10 &

the critical activities are A,B,C,L&M.

the total project time is 25 weeks

B. For each non-critical activity, the total float is:

Activity (i,j)	Duration (t _{ij})	Earliest time		Last time		Float	
		Start (E _i)	Finish (E _i +t _{ij})	Start (L _j -t _{ij})	Finish (L _j)	Total [(L _j -t _{ij})E _j]	Free (E _j -E _i)-t _{ij}
1-3	2	0	2	2	4	2	0

1-4	13	0	13	1	14	1	0
2-6	9	6	15	8	17	2	
2-7	2	6	8	9	11	3	0
3-4	10	2	12	4	14	2	
4-9	6	13	19	14	20	1	
7-8	4	8	13	11	15	3	
8-10	10	12	22	15	25	3	

Remark:

- What does the critical path really mean?
 - The critical path shows the shortest time in which a project can be completed. Even though the critical path is the longest path, it represents the shortest time it takes to complete a project.
 - There can be more than one critical path on a project. Project managers should closely monitor performance of activities on the critical path to avoid late project completion. If there is more than one critical path, project managers must keep their eyes on all of them
- C. A small maintenance project consists of 5 activities which take place according to the following conditions:

<u>Activity</u>	<u>Immediate predecessors</u>	<u>Estimated Activity Time in Week</u>
A	-	3

B	-	5
C	-	5
D	A	2
E	B	4

Required:

- Draw the network diagram for the following activities.
- Find the critical path & the critical activities & the critical path duration
- Find the total float for each activity

b. Path 1: 1-2-4 length $3+2= 5$ weeks

Path 2:1-4 Length =6 weeks

Path 3: 1-3-4 Length = $5+4= 9$ weeks.

The critical path of the project is 1-3-4

The critical activities are B& E

The critical path duration 9 weeks

Activity (i,j)	Duration (tij)	Earliest start (Ei)	Time Finish (Ei+tij)	Last time		Total Float (Li-tij)-Ej
				Start (Lj-tij)	Finish Lj	
1-2	3	0	3	4	7	4
1-3	5	0	5	0	5	0

1-4	6	0	6	3	9	3
2-4	2	3	5	7	9	4
3-4	4	3	9	5	9	0

Exercise

1. A research & development department is developing a new power supply for a console television set. It has broken the job down into the following form:

Job	Immediate predecessors	Expected tome (days)
A	-	5
B	A	7
C	B	2
D	B	3
E	C	1
F	D	2
G	C	1
H	E,F	3
I	G,H	10

- a. Draw a critical path scheduling arrow diagram, identifying jobs by letters & associating times with each. Indicate the critical path.
- b. What is the minimum time for completion of the project?
- c. Find the total float of each job.

2. Based on the following network diagram a) identify the critical path. What is its length b) Find the total float for each task.

7.7 PERT NETWORKS

Program Evaluation and Review Technique (PERT)- is a network analysis technique used to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates. PERT applies the critical path method to a weighted average duration estimate.

If the duration of activities in a project is uncertain than activity scheduling calculations are done by using the expected value of the durations. However, such expected duration estimated may not be given an accurate answer. Thus, rather than estimating directly the expected completion time of an activity, three values are considered in PERT.

The three time estimates are:

1. Optimistic time (to or a):
 - This is the shortest possible time to perform an activity, assuming that everything goes well.
2. Pessimistic time (tp or b):

This is the maximum (longest) time that is required to perform an activity, under extremely bad conditions. However, such conditions do not include acts of God like earthquakes, flood, etc.

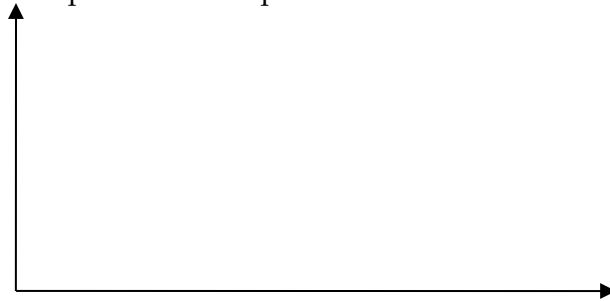
3. Most likely time (tm or m):
 - This is the most realistic time to complete the activity. Statistically, it is the modal value of duration of the activity

i.e consider the time estimate under ht normal situation

PERT uses probabilistic time estimates duration estimates based on using optimistic most likely & pessimistic estimates of activity durations instead of one specific or discrete duration estimate. Like the CPM, PERT is based on a project network diagram.

To use the PERT method, we have to calculate the expected activity time.

The most likely time (t_m) estimate may not coincide with the midpoint $(t_o + t_p)/2$ & may occur to its left or its right as shown in fig below. Because of these properties, it is justified to assume that the duration of each activity may follow Beta (B) distribution with unmodel point occurring at t_m & it's and points at t_o & t_p



In Beta distribution, the mid-point $(t_o + t_p)/2$ is assumed to weight half as much as the most likely point (t_m). Thus the expected or mean (t_e) value of the activity duration can be approximated as the arithmetic mean of $(t_o + t_p)/2$ & $2t_m$.

Example

1. A project is represented by a network shown below & has the following data

Task:	A	B	C	D	E	F	G	H	I
Optimistic time	5	18	26	16	15	6	7	7	3
Pessimistic time:	10	22	40	20	25	12	12	9	5
Most Likely time:	8	20	33	18	20	9	10	8	4

Determine the following

- Expected task time & their variance
- The earliest & latest expected times to reach each event
- The critical path

Activity	To	tp	tm	$t_e = (t_o + 4t_m + t_p) / 6$	$S^2 = [(t_p - t_o) / 6]^2$
1-2	5	10	8	7.8	0.694
1-3	18	22	20	20.0	0.444
1-4	26	40	33	33.0	5.444
2-5	16	20	18	18.0	0.444
2-6	15	25	20	20.0	2.778
3-6	6	12	9	9.0	1.000
4-7	7	12	10	9.8	0.694
5-7	7	9	8	8.0	0.111
6-7	3	5	4	4.0	0.111

Forward Pass Method

Exercise:

A small project is composed of 7 activities whose time estimates are listed in the table below. Activities are identified by their beginning (i)& ending (j) node numbers.

<u>Activity</u> (i,j)	<u>Estimated Duration (in weeks)</u>		
	Optimistic	Most likely	Pessimistic
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	6	15

- Draw the project network
- Find the expected duration& variance for each activity. What is the expected project length?
- Calculate the variance & standard deviation of the project length.

UNIT FIVE

DECISION THEORY/ANALYSIS

Dear learner, in the previous units dealing with LP, models were formulated and solved in order to aid the manager in making decision. The solutions to the models were represented by values for the decision variables. However, these LP models are formulated under the assumption that ***certainty*** existed. In actual practice, however, many decision-making situations occur under conditions of ***uncertainty***. For example, the demand for a product may be not 100 units next week, but 50 or 200 units, depending on the market (which is uncertain).

5.1. Characteristics of Decision Theory

Decision theory problems are characterized by the following:

1. **List of alternatives:** are a set of mutually exclusive and collectively exhaustive **decisions** that are available to the decision maker (sometimes, not always, one of these alternatives will be to “do nothing”).)
2. **States of nature:** - the set of possible **future conditions**, or **events**, beyond the control of the decision maker, that will be the primary determinants of the eventual consequence of the decision. The states of nature, like the list of alternatives, must be mutually exclusive and collectively exhaustive.
3. **Payoffs:** - the payoffs might be profits, revenues, costs, or other measures of value. Usually the measures are financial. Usually payoffs are estimated values. The more accurate these estimates, the more useful they will be for decision-making purposes and the more likely, it is that the decision maker will choose an appropriate alternative. The number of payoffs depends on the number of alternative/state of nature combination.
4. **Degree of certainty:** - the **approach** often used by a decision maker depends on the degree of certainty that exists. There can be different degrees of certainty. One extreme is complete certainty and the other is complete uncertainty. The latter exists when the likelihood of the various states of nature are unknown. Between these two extremes is risk (probabilities are known for the states of nature). Knowledge of the likelihood of each of the states of nature can play an important role in selecting a course of action.
5. **Decision criteria:** - the **decision maker's attitudes** toward the decision as well as the degree of certainty that surrounds a decision. Example; maximize the expected payoffs.

5.2. THE PAYOFF TABLE

A payoff table is a **device** a decision maker can use to summarize and organize information relevant to a particular decision. It includes a list of alternatives, the possible future states of nature, and the payoffs associated with each of the alternative/state of nature combinations. If probabilities for the states of nature

are available, these can also be listed. The general format of the table is illustrated below:

States of nature

		S1	S2	S3
Alternatives	A1	V11	V12	V13
		V21	V22	V23
	A2	V31	V32	V33

A3

where:

A_i = the i th alternative

S_j = the j th states of nature

V_{ij} = the value or payoff that will be realized if alternative i is chosen and event j

occurs.

Decision situations can be categorized in to three classes: Situation of certainty, Situations where probabilities cannot be assigned to future occurrences and Situations where probabilities can be assigned to future occurrences. In this chapter, we will discuss each of these classes of decision situations separately.

5.3. DECISION MAKING UNDER CERTAINTY

The simplest of all circumstances occurs when decision-making takes place in an environment of complete certainty. When a decision is made under conditions of complete certainty, the attention of the decision maker is focused on the column in the payoff table that corresponds to the state of nature that will

occur. The decision maker then selects the alternative that would yield the best payoff, given that state of nature.

EXAMPLE

The following payoff table provides data about profits of the various states of nature/alternative combination.

	S1	S2	S3
A1	4	16	12
A2	5	6	10
A2	-1	4	15
A3			

If we know that S2 will occurs the decision maker then can focus on the first raw of the payoff table. Because alternative A1 has the largest profit (16), it would be selected.

5.4. DECISION MAKING UNDER COMPLETE UNCERTAINTY (Without probabilities)

Under complete uncertainty, the decision maker either is unable to estimate the probabilities for the occurrence of the different state of nature, or else he or she lacks confidence in available estimates of probabilities, and for that reason, probabilities are not included in the analysis.

A decision-making situation includes several components- the decision themselves and the actual event that may occur future, known as state of nature. At the time the decision is made, the decision maker is uncertain which state of nature will occur in the future, and has no control over them.

Decisions made under these circumstances are at the opposite end of the spectrum from the certainty case just mentioned. Once the decision has been organized in to a payoff table, several criteria are available making the actual

decision. There are several approaches (criteria) to decision making under complete uncertainty. Some of these discussed in this section include: maximax, maximin, minimax regret, and equal likelihood.

5.4.1. MAXIMAX

With the maximax criterion, the decision maker selects the decision that will result in the maximum of the maximum payoffs. (In fact this is how this criterion derives its name- maximum of maximum). The maximax is **very optimistic**. The decision maker assumes that the most favorable state of nature for each decision alternative will occur. For example, the investor would optimistically assume that good economic conditions will prevail in the future. The best payoff for each alternative is identified, and the alternative with the maximum of these is the designated decision.

For the previous problem:

	S1	S2	S3	Row Maximum
A1	4	16	12	16*maximum
	5	6	10	10
A2	-1	4	15	15
A3				

Decision: A1 will be chosen.

☞ **Note:** If the pay off table consists of costs instead of profits, the opposite selection would be indicated: The minimum of minimum costs. For the subsequent decision criteria we encounter, the same logic in the case of costs can be used.

5.4.2. Maximin Criteria

This approach is the opposite of the previous one, i.e. it is **pessimistic**. This strategy is a conservative one; it consists of identifying the worst (minimum) payoff for each alternative, and, then, selecting the alternative that has the best (maximum) of the worst payoffs. In effect, the decision maker is setting a floor on the potential payoff by selecting maximum of the minimum; the actual payoff cannot be less than this amount. It involves selecting best of the worst.

For the previous problem:

	S1	S2	S3	Row minimum
A1	4	16	12	4
	5	6	10	5*maximum
A2	-1	4	15	-1

A3

Decision: A2 will be chosen.

☞ **Note:** If it were cost, the conservative approach would be to select the maximum cost for each decision and select the minimum of these costs.

5.4.3. MINIMAX REGRET or savage

Both the maximax and maximin strategies can be criticized because they focus only on a single, extreme payoff and exclude the other payoffs. Thus, the maximax strategy ignores the possibility that an alternative with a slightly smaller payoff might offer a better overall choice. For example, consider this payoff table:

	S1	S2	S3	Row Max.
A1	-5	16	-10	16*max
	15	15	15	15
A2	15	15	15	15

A3

A similar example could be constructed to demonstrate comparable weaknesses of the maximin criterion, which is also due to the failure to consider all payoffs.

An approach that does take all payoffs in to consideration is Minimax regret. In order to use this approach, it is necessary to develop an **opportunity loss table**. The opportunity loss reflects the difference between each payoff and the best possible payoff in a column (i.e., given a state of nature). Hence, opportunity loss amounts are found by identifying the best payoff in a column and, then, subtracting each of the other values in the column from that payoff. Therefore, this decision avoids the greatest regret by selecting the decision alternative that minimizes the maximum regret.

EXAMPLE:

	S1	S2	S3
A1	4	16	12
	5	6	10
A2	-1	4	15

A3

opportunity loss table:

	S1	S2	S3	
	$5-4=1$	$16-16=0$	$15-12=3$	A1
	$5-5=0$	$16-6=10$	$15-10=5$	
	$5-(-1)=6$	$16-4=12$	$15-15=0$	A2

A3

The values in an opportunity loss table can be viewed as potential “regrets” that might be suffered as the result of choosing various alternatives. A decision

maker could select an alternative in such a way as to minimize the maximum possible regret. This requires identifying the maximum opportunity loss in each row and, then, choosing the alternative that would yield the best (minimum) of those regrets.

S1	S2	S3	Max. Loss	
5-4=1	16-16=0	15-12=3	3*minimum	A1
5-5=0	16-6=10	15-10=5	10	
5-(-1)=6	16-4=12	15-15=0	12	A2
A3				

Decision: A will be chosen.

Although this approach makes use of more information than either Maximin or Maximax, it still ignores some information, and, therefore, can lead to a poor decision.

EXAMPLE:

Opportunity loss table

	S1	S2	S3	S4	Max. Loss
A1	0	0	0	24	24
	15	15	15	0	15*minimum
A2	15	15	15	0	15*minimum

A3

5.4.4. PRINCIPLE OF INSUFFICIENT REASON/ Equal likelihood/ Laplace

The Minimax regret criterion's weakness is the inability to factor row differences. Hence, sometimes the minimax regret strategy will lead to a poor decision because it ignores certain information.

The principle of insufficient reason offers a method that incorporates more of the information. It treats the states of nature as if each were equally likely, and it focuses on the average payoff for each row, selecting the alternative that has the highest row average.

EXAMPLE

	S1	S2	S3	S4	S5	Row Average
A1	28	28	28	28	4	23.2*maximum
	5	5	5	5	28	9.6
A2	5	5	5	5	28	9.6

A3

Decision: A1 is selected

The basis for the criterion of insufficient reason is that under complete uncertainty, the decision maker should not focus on either high or low payoffs, but should treat all payoffs (actually, all states of nature), as if they were equally likely. Averaging row payoffs accomplishes this.

5.5. DECISION MAKING UNDER RISK (WITH PROBABILITIES)

Dear learner, the decision making criteria just presented were based on the assumption that no information regarding the likelihood of the states of the nature

was available. Thus, no probabilities of occurrence were assigned to the states of nature, except in the case of the equal likely hood criterion.

It is often possible for the decision maker to know enough about the future state of nature to assign probabilities to their occurrences. The term risk is often used in conjunction with partial uncertainty, presence of probabilities for the occurrence of various states of nature. The probabilities may be subjective estimates from managers or from experts in a particular field, or they may reflect historical frequencies. If they are reasonably correct, they provide the decision maker with additional information that can dramatically improve the decision making process.

Given that probabilities can be assigned, several decision criteria are available to aid the decision maker. Some of these are discussed below.

5.5.1. EXPECTED MONETARY VALUE (EMV)

The EMV approach provides the decision maker with a value which represents an average payoff for each alternative. The best alternative is, then, the one that has the highest EMV. The average or expected payoff of each alternative is a weighted average:

$$EMV_i = \sum_{j=1}^k P_i \cdot V_{ij}$$

Where:

EMV_i = the EMV for the i th alternative

P_i = the probability of the i th state of nature

V_{ij} = the estimated payoff for alternative i under state of nature j .

Note: the sum of the probabilities for all states of nature must be 1.

EXAMPLE:

Probability 0.20 0.50 0.30

	S1	S2	S3	Expected payoff
A1	4	16	12	12.40*maximum
A2	5	6	10	7
A3	-1	4	15	6.30

Decision: A1 will be chosen.

☞ Note that it does not necessarily follow that the decision maker will receive a payoff equal to the expected monetary value of a chosen alternative. Similarly, the expected payoffs for either of the other alternatives do not equal any payoffs in those rows. What, then, is the interpretation of the expected payoff? Simply a long-run average amount; the approximate average amount one could reasonably anticipate for a large number of identical situations.

5.5.2. Expected Opportunity Loss (EOL)

The table of opportunity loss is used rather than a table of payoffs. Hence, the opportunity losses for each alternative are weighted by the probabilities of their respective state of nature to compute a long run average opportunity loss, and the alternative with the smallest expected loss is selected as the best choice.

$$\text{EOL (A1)} = 0.20(1) + 0.50(0) + 0.30(3) = 1.10 \text{ *minimum}$$

$$\text{EOL (A2)} = 0.20(0) + 0.50(10) + 0.30(5) = 6.50$$

$$\text{EOL (A3)} = 0.20(6) + 0.50(12) + 0.30(0) = 7.20$$

☞ **Note:** The EOL approach resulted in the same alternative as the EMV approach

(Maximizing the payoffs is equivalent to minimizing the opportunity losses).

5.5.3. Expected Value of Perfect Information (EVPI)

It can sometimes be useful for a decision maker to determine the potential benefit of knowing for certain which state of nature is going to prevail. The EVPI is the measure of the difference between the certain payoffs that could be realized under a condition involving risk.

If the decision maker knows that S1 will occur, A2 would be chosen with a payoff of \$5. Similarly for S2 \$16 (for A1) and for S3, \$15 (with A3) would be chosen.

Hence, the expected payoff under certainty (EPC) would be:

$$\text{EPC} = 0.20(5) + 0.50(16) + 0.30(15) = 13.50$$

The difference between this figure and the expected payoff under risk (i.e., the EMV) is the expected value of perfect information. Thus:

$$\text{EVPI} = \text{EPC} - \text{EMV}$$

$$= 13.50 - 12.40 = 1.10$$

☞ **Note:** The EVPI is exactly equal to the EOL. The EOL indicates the expected opportunity loss due to imperfect information, which is another way of saying the expected payoff that could be achieved by having perfect information.

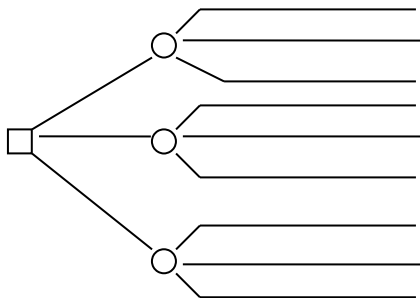
☞ **Note:** The expected value approach is particularly useful for decision making when a number of similar decisions must be made; it is a long-run approach. For one-shot decisions, especially major ones, other methods (perhaps, maximax or maximin) may be preferable. In addition, non-monetary factors, although not

included in a payoff table, may be of considerable importance. Unfortunately, there is no convenient way to include them in an expected value analysis.

5.6. DECISION TREES

Decision trees some times are used by decision makers to obtain a visual portrayal of decision alternatives and their possible consequences. The term gets its name from the tree-like appearance of the diagram.

Decision tree format:



Decision tree, like probably tree is composed of squares, circles, and lines:

- The squares indicate decision points
- Circles represent chance events(circles and squares are called nodes)
 - The lines (branches) emanating from squares represent alternatives.
 - The lines from circles represent states of nature
- The tree is read from right to left.

It should be noted that although decision trees represent an alternative approach to payoff tables, they are not commonly used for problems that involve a single decision. Rather, their greatest benefit lies in portraying sequential decisions (i.e.,

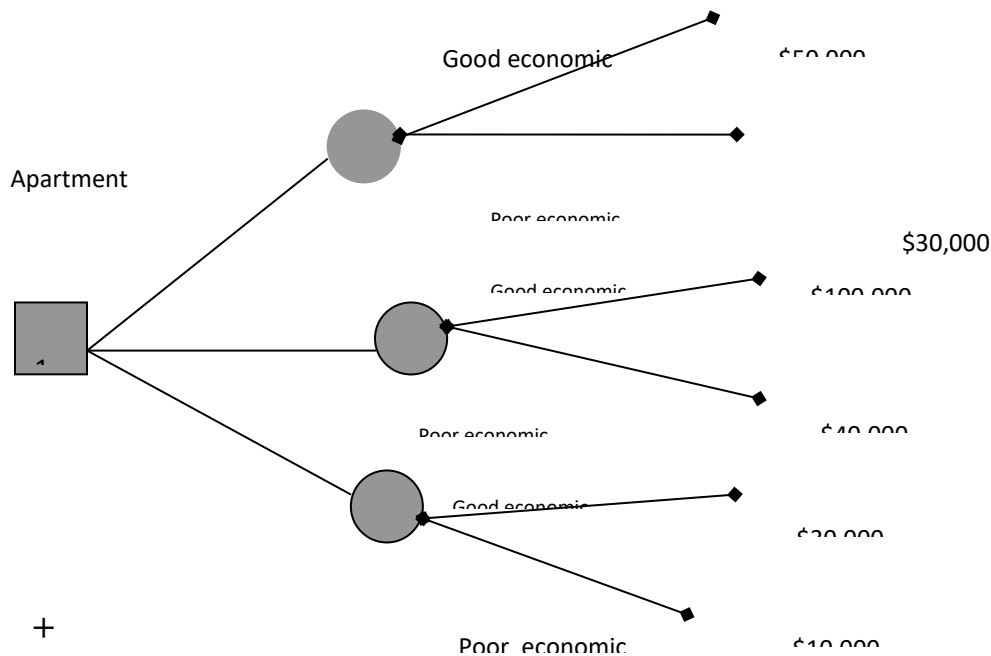
a series of chronological decisions). In the case of a single decision, constructing a decision tree can be cumbersome and time consuming.

Example

Pay off table for Real Estate investment

State of Nature_____		
	Good economic	poor economic
Decision	conditions	condition
(Purchase)	0.6	0.4
Apartment building	50,000	30,000
Office building	100,000	-40,000
Warehouse	30,000	10,000

The decision tree for the above example will be:



The circles (○) and squares (□) in the above figure are referred to as **nodes**. The squares are decision nodes, and the **branches** emanating from a decision node reflect the alternative decisions possible at that point. For example, in the above figure, node 1 signifies a decision to purchase an apartment building, Office building or warehouse. The circles are probability nodes, and the branches emanating from them indicate the state of nature that can occur: good economic conditions or poor economic conditions. The decision tree represents the sequence of events in a decision situation. First, one of the three decision choices is selected at node 1. Depending on the branch selected, the decision maker arrives at probability node 2, 3, or 4, where one of the states of nature will prevail, resulting in one of six possible payoffs.

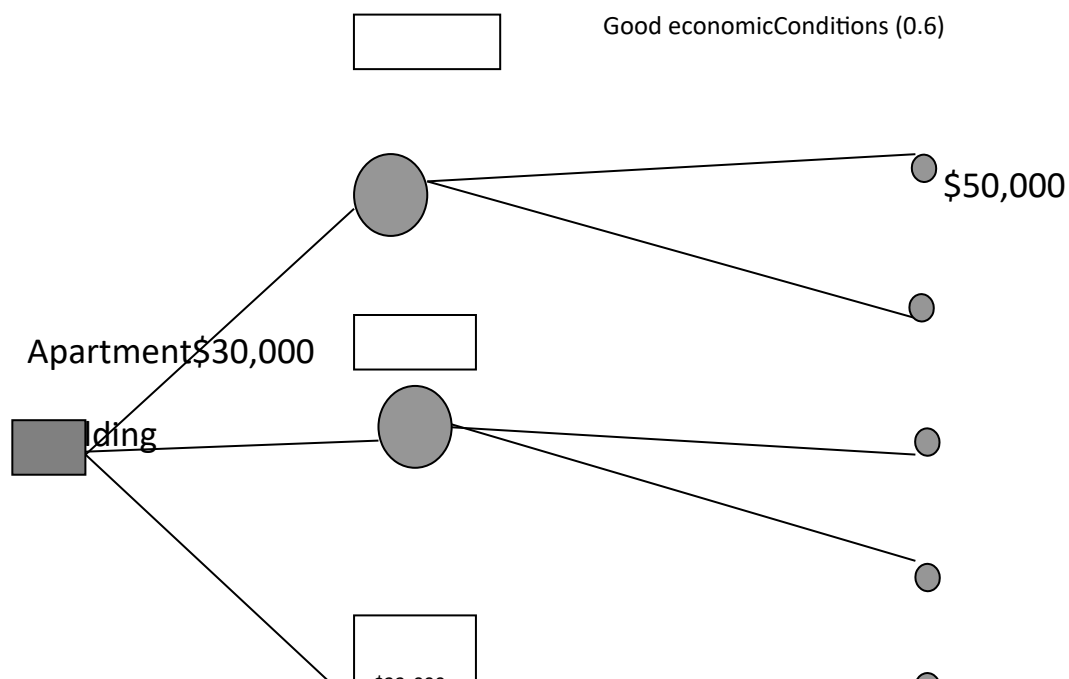
Determining the best decision using a decision tree involves computing the expected value at each probability node. This is accomplished by starting with the final outcomes (payoffs) and working backward through the decision tree toward node 1. First, the expected value of the payoffs is computed at each probability node.

$$EV(\text{node 2}) = .60(\$ 50,000) + .40(\$ 30,000) = \$42,000$$

$$EV(\text{node 3}) = .60(\$100,000) + .40(\$-40,000) = \$44,000$$

$$EV(\text{node 4}) = .60(\$ 30,000) + .40(\$ 10,000) = \$22,000$$

These values are now shown as the **expected payoffs** from each of the three branches emanating from node 1 in figure below. Each of these three expected values at nodes 2, 3, and 4 is the outcome of a possible decision that can occur at node 1. Moving toward node 1, we select the branch that comes from the probability node with the highest expected payoff. In figure below, the branch corresponding to the highest payoff, \$44,000 is from node 1 to node 3. This branch represents the decision to purchase the office building. The decision to purchase the office building, with an expected payoff of \$44,000, is the same result we achieved earlier using the expected value criterion. In fact, when only one decision is to be made (i.e., there is not a series of decisions), the decision tree will always yield the same decision and expected payoff as the expected value criterion. As a result, in these decision situations the decision tree is not very useful. However, when a sequence or series of decisions is required, the decision tree can be very useful.



Poor economicConditions(0.4)

Good economicConditions (0.6)

\$100,000

Purchase

Poor economicConditions -\$40,000

Good economic

Warehouse

Conditions (0.6)

\$30,000

Poor economic

Conditions(0.4)

\$10,000

Activity

1. The owner of the Burger Doodle Restaurant is considering two ways to expand operations: opening a drive-up window or serving breakfast. The increase in profits resulting from these proposed expansions depends on whether a competitor opens a franchise down the street. The possible Profits from each expansion in

operations given both future competitive situations are shown in the following payoff table.

Competitor

<u>Decision</u>	<u>open</u>	<u>Not Open</u>
Drive-up window	\$-6,000	\$20,000
Breakfast	4,000	8,000

Select the best decision using the following decision criteria.

- a) Maximax
- b) Maximin
- c) Equally likely
- d) Minimax regret

2. Consider the following payoff table for three alternatives, A,B, and C, under two future states of the economy, good and bad.

Economic Conditions

<u>Investment</u>	<u>Good</u>	<u>Bad</u>
A	\$ 70,000	\$ 25,000
B	120,000	-60,000
C	40, 000	40,000

Determine the decision using the following decision criteria.

- a) Maximax
- b) Maximin
- c) Minimax regret
- d) Equal likelihood

3. An investor is considering investing in stock, real estate, or bonds under uncertain economic conditions. The payoff table of returns for the investor's decision situation is shown below

Investment	Economic Conditions		
	Good	Stable	Poor
Stocks	\$ 5,000	\$ 7,000	\$ 3,000
Real estate	-2,000	10,000	6,000
Bonds	4,000	4,000	4,000

Determine the best investment using the following decision criteria.

- a) Equal likelihood
- b) Maximin
- c) Maximax
- d) Minimax regret

